

Chapter 10. Positive End-Expiratory Pressure

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Positive End-Expiratory Pressure: Introduction

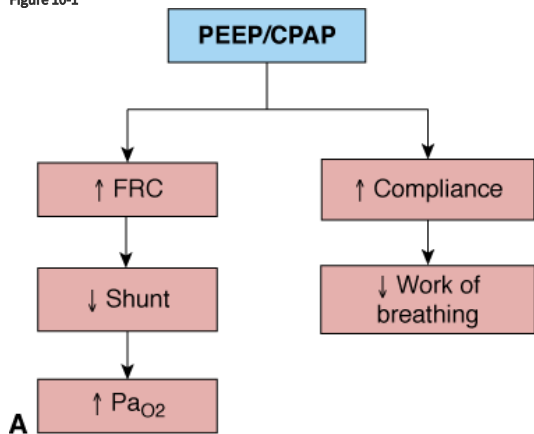
Positive end-expiratory pressure (PEEP) is not a ventilator mode itself, but rather an adjunctive treatment that can be combined with all forms of mechanical ventilation, both controlled and assisted,¹⁻⁷ or applied to spontaneous breathing throughout the entire respiratory cycle, so-called continuous positive airway pressure (CPAP).⁸⁻¹⁰ Following the pioneering work of Poulton and Oxon¹¹ and Barach and associates¹² who demonstrated in the mid-1930s that application of positive pressure to the airway can effectively treat patients with pulmonary edema, several pathological conditions were proved to benefit from PEEP, which is today considered by intensive care unit physicians as one of the most powerful treatments available for acute respiratory failure (ARF).¹³

Since publication of the second edition of this book in 2006, several relevant studies have been published on PEEP and CPAP, primarily concerning lung injury and cardiogenic pulmonary edema. To update the chapter with this new information, we have condensed material that was covered in greater depth in the second edition.

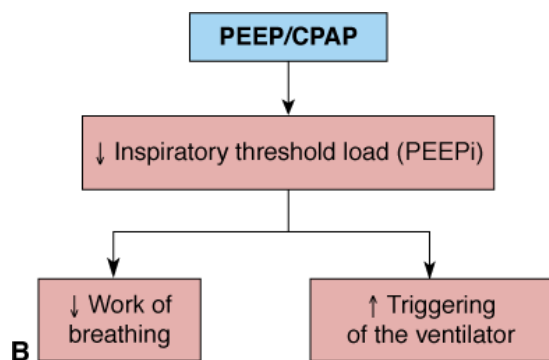
Overview of Pertinent Pathophysiology

The reversal of hypoxemia caused by intrapulmonary shunt and venous admixture requires interventions that recruit more aerated lung units for ventilation. In patients with an acute reduction of lung volume secondary to lung edema and/or atelectasis, PEEP can improve arterial oxygenation^{1,8} by increasing functional residual capacity (FRC),¹⁴⁻¹⁹ reducing venous admixture,²⁰⁻²⁴ shifting tidal volume (V_T) to a more compliant portion of the pressure-volume curve,²⁵ preventing the loss of compliance during mechanical ventilation,^{5,26} reducing intratidal alveolar opening and closing²⁷ and the work of breathing.²⁸ Figure 10-1A summarizes the rationale for PEEP in patients with ARF secondary to acute lung volume reduction.

Figure 10-1



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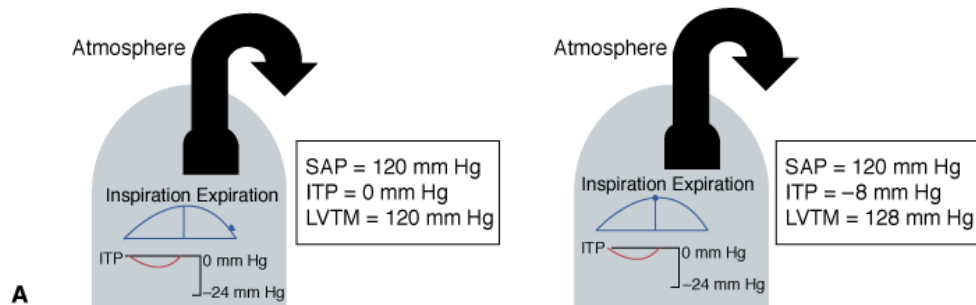
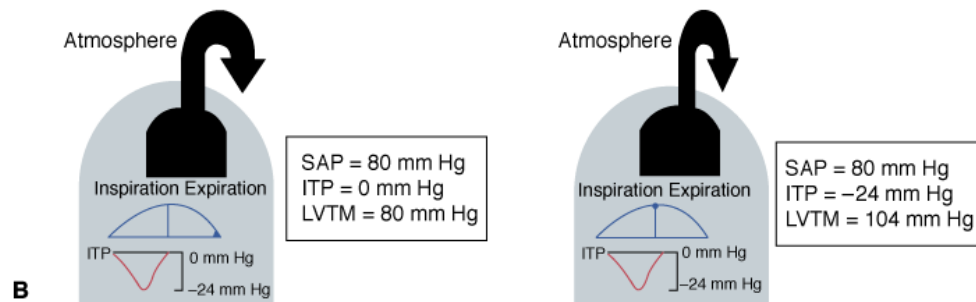
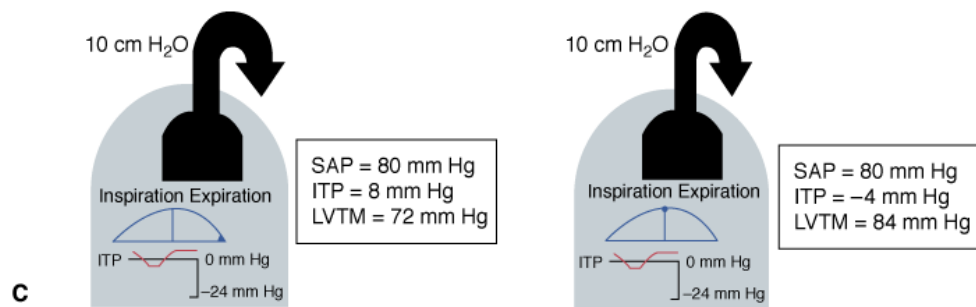
Respiratory effects of PEEP/CPAP in acute respiratory failure secondary to (A) lung volume reduction caused by edema and/or atelectasis and (B) airway obstruction and expiratory flow limitation. *FRC*, functional residual capacity; PaO_2 , partial pressure of arterial oxygen; *PEEP*, positive end-expiratory pressure.

Although there is general consensus about the potential of PEEP in treating patients with hypoxemic ARF secondary to lung edema and/or atelectasis, several aspects are controversial. For many years, it has been recognized that the actions of PEEP on gas exchange and pulmonary mechanics are variable,^{29,30} depending on the type and severity of lung injury, which may be sustained by different pathways^{31,32} and pathophysiologic mechanisms.^{31,33} As with any treatment, PEEP is not free of side effects.³⁴ In patients with acute respiratory distress syndrome (ARDS), PEEP may recruit nonaerated regions, but also distend normally aerated regions,^{35,36} contributing to barotrauma through increase in end-inspiratory plateau pressure.^{22,37–40} High levels of PEEP also have been shown to augment the physiologic dead space,^{24,41,42} and worsen gas exchange³⁵ and tissue perfusion.^{43–45} Potential extrapulmonary side effects of PEEP include decreased cardiac output,^{46–51} increased intracranial pressure,^{52,53} renal dysfunction,^{54,55} and decreased splanchnic perfusion^{56–58} and oxygenation.^{43,44} Although the benefits and detriments of PEEP in patients with ARDS have been variously weighted for many years, debate as to optimal PEEP and the best method for detecting it is still ongoing.

In the presence of airflow obstruction, the lungs may fail to deflate to *FRC* at end-expiration and alveolar pressure remain positive to an extent that depends on the volume of trapped air, a phenomenon referred to as *auto* or *intrinsic PEEP (PEEPi)*.⁵⁹ In this setting, application of external PEEP may be beneficial. Benefit is evident during spontaneous breathing: CPAP reduces dyspnea^{9,60} and work of breathing.^{9,60} Benefit is also evident with any patient-triggered mode of ventilation:^{2,61–63} PEEP enhances triggering function,^{2,61–63} reduces patient's respiratory drive,⁶² reduces inspiratory muscle effort,^{2,61–63} and improves patient-ventilator interaction.^{63,64} PEEP replaces the amount of pressure that must be generated by the inspiratory muscles to offset *PEEPi* (necessary to initiate inspiratory flow or trigger the ventilator).^{2,62} Figure 10-1B summarizes the principal effects of PEEP in patients with chronic obstructive pulmonary disease (COPD).

PEEP has substantial effects on hemodynamics. When left-ventricular function is normal, an increase in intrathoracic pressure causes a fall in cardiac output, largely secondary to decrease in venous return (i.e., preload).^{47,65,66} In patients with poor left-ventricular function, when filling pressure and left-ventricular diastolic volume are elevated, cardiac output is relatively insensitive to decline in venous return.⁶⁷ In this condition, the dominant effect of an increase in intrathoracic pressure is to decrease left-ventricular transmural pressure (i.e., afterload), which improves left-ventricular performance.^{68–71} These effects (Fig. 10-2C) are also observed with relatively moderate levels of CPAP; the impact on afterload is likely to result primarily from a reduction in the inspiratory negative swings of intrathoracic pressure,⁷² consequent to improvement in respiratory mechanics.⁷² Application of CPAP in patients with severe cardiogenic pulmonary edema may accelerate the physiologic improvement and reduce the need for endotracheal intubation.^{10,73–75}

Figure 10-2

Normal subject**Cardiogenic pulmonary edema****CPAP in cardiogenic pulmonary edema**

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Heart–lung interactions during spontaneous breathing in a normal subject (**A**), and in a patient with cardiogenic pulmonary edema resulting from a low cardiac output in the absence (**B**) and presence (**C**) of CPAP. The effects of variations in intrathoracic pressure (ITP) on left-ventricular systolic function are shown at end-expiration (*left panels*) and end-inspiration (*right panels*). LVTM, left-ventricular transmural pressure (SAP – ITP); SAP, systolic arterial (intraventricular) pressure. In a normal subject (**A**), tidal excursions in ITP have little effect on left-ventricular systolic function. With pulmonary edema (**B**), more negative swings in ITP are needed to achieve the same tidal volume. As a result, LVTM (afterload) is increased independently of systemic vascular resistance. The rise in ITP produced by CPAP 10 cm H₂O (**C**) produces a slight reduction in LVTM during expiration. CPAP remarkably decreases LVTM during inspiration by decreasing negative swings in ITP, because of improved lung mechanics.

Several investigators have studied the use of CPAP in the treatment of the obstructive sleep apnea syndrome, which is characterized by reduction or cessation of breathing secondary to periodic narrowing and/or collapse of the upper airway. Detailed discussion of this topic is beyond the scope of this chapter. Likewise, we do not provide detailed discussion of positive expiratory pressure, a physiotherapeutic technique aimed at clearance of secretions.

Physiologic Rationale for Positive End-Expiratory Pressure

Effect of Positive End-Expiratory Pressure on Gas Exchange

The role of PEEP in correcting hypoxemia in patients with pulmonary edema and ARF has long been recognized^{11,12} and extensively documented.^{24,42,76–79} The chief mechanism by which PEEP improves arterial oxygenation is through an increase in the amount of alveoli reached by the air. By decreasing cardiac output, however, PEEP may negatively affect mixed venous O₂ tension, which, in turn, decreases the partial pressure of arterial oxygen (PaO₂) to an extent that depends on alveolar ventilation–perfusion ($\dot{V}_A/\dot{Q}$) mismatch.^{80–83}

Effect in Anesthetized Subjects

Arterial oxygenation is often impaired during anesthesia^{84,85} as a result of the intrapulmonary shunt^{86–90} caused by atelectasis, predominantly in the dependent zones.^{85,91–97} Areas of collapsed parenchyma occur in 90% of all intubated anesthetized subjects, both during spontaneous breathing and muscle paralysis.^{90,98–100} Other mechanisms, such as regional distribution of ventilation,¹⁰¹ airway closure,^{102–112} and pharmacologic inhibition of hypoxic vasoconstriction^{113–116} may also contribute. Application of PEEP during anesthesia can prevent or reverse closing of peripheral airways,¹⁰⁵ decrease atelectasis,^{90,100,117,118} and improve regional $\dot{V}_A/\dot{Q}$ ratios and oxygenation,^{118–120} although shunt may only be partially improved.⁹⁰ In normal anesthetized dogs, however, Dueck et al found that PEEP ≥ 10 cm H₂O produced a reduction in PaO₂ and increases in dead space ventilation and partial pressure of arterial carbon dioxide (PaCO₂). This was mainly caused by a drop in cardiac output associated with an increase in alveolar ventilation of both high $\dot{V}_A/\dot{Q}$ and unperfused lung regions, suggesting overdilation of nondependent lung regions.¹²¹ Similarly, Pelosi et al showed that PEEP values up to 10 cm H₂O decreased, although not significantly, PaO₂ during general anesthesia in normal subjects.¹²²

Alterations in diaphragmatic position and mechanics during anesthesia also contribute to impaired gas exchange by inducing ventilation abnormalities in the dependent and caudal zones.^{123,124} This phenomenon has been elegantly studied in anesthetized rabbits by Heneghan et al who compared PEEP and phrenic nerve stimulation, adjusted to produce the same increase in lung volume. Phrenic stimulation caused a greater caudal movement of the diaphragm, particularly in the dependent regions, and produced greater improvement in gas exchange than did PEEP.¹²³

Beneficial effects of PEEP on oxygenation have been reported in obese patients, who are at greater risk to develop caudal atelectasis during anesthesia, and during pneumoperitoneum induced for laparoscopic surgery causing cephalad displacement of the diaphragm.^{122,125,126} Pelosi et al found that PEEP induced an increase in PaO₂ in obese patients; this increase was correlated with recruitment of atelectatic lung regions.¹²² Other authors found that the sole application of 10 cm H₂O PEEP was able to reduce atelectasis but not to improve oxygenation in these patients.^{125,127,128} Oxygenation improved and atelectasis was further decreased, however, when PEEP application was preceded by a recruitment maneuver or associated with vertical positioning both in obese patients and healthy subjects undergoing laparoscopic surgery. In the former case, use of high airway pressures opened atelectatic lung areas subsequently kept open by PEEP, while in the latter case vertical positioning decreased the negative effect of abdominal pressure on diaphragmatic displacement. In morbidly obese patients, PEEP has also been found beneficial during induction of general anesthesia. Gander et al showed that applying 10 cm H₂O PEEP during induction of anesthesia increased the duration of apnea without hypoxia by approximately 50%.¹²⁹

Conflicting data on the effect of PEEP have been obtained in postoperative patients who had developed respiratory failure. In two studies, PEEP up to 10 cm H₂O did not significantly modify PaO₂ during the early postoperative course of patients who underwent cardiac or vascular surgery.^{130,131} Conversely, other investigators found that PEEP, combined with bilevel positive airway pressure ventilation administered via a nasal mask, improved oxygenation in the vast majority of patients who exhibited respiratory distress after various types of surgery.^{132–135} A recent randomized, clinical trial found that the use of noninvasive positive-pressure ventilation with low PEEP (approximately 4 cm H₂O) increased PaO₂, reduced the need for intubation, and improved survival of patients who developed ARF after lung resection.¹³⁶

In summary, use of PEEP to improve oxygenation during and after anesthesia in normal subjects has generated contradictory results, probably because of the deleterious effects of PEEP on hemodynamics and lung overdilation. PEEP, however, can be effective in improving gas exchange in selected patients.

Effect in Hypoxemic Respiratory Failure

Most forms of hypoxemic respiratory failure, such as cardiogenic pulmonary edema, ARDS, and unilateral pneumonia, are characterized by a decrease in lung volume caused by atelectasis, interstitial and alveolar edema, and small airway closure. The pivotal mechanism for hypoxemia in these conditions is intrapulmonary shunt as demonstrated by the small increase in PaO₂ when pure O₂ is administered,¹³⁷ and, to a lesser extent, $\dot{V}_A/\dot{Q}$ mismatch.^{138,139} Moreover, the high fractional inspired oxygen concentration (FiO₂) frequently used in these patients, may decrease lung volume by promoting alveolar denitrogenation and reabsorption atelectasis.^{137,140,141} In ARDS, the altered blood flow distribution, resulting from widespread involvement of the pulmonary vasculature¹⁴² and impaired hypoxic vasoconstriction, contributes to worsening of gas exchange.¹⁴⁰

PEEP has long been recognized as an effective means of increasing lung volume and improving gas exchange.¹ In cardiogenic pulmonary edema, CPAP (alone or in association with an inspiratory assistance) improves gas exchange^{10,73,143–154} by increasing aerated lung volume,¹⁵⁵ and improving cardiac output^{73,143,148} and $\dot{V}_A/\dot{Q}$ distribution.^{155,156} A decrease in extravascular lung water is another possible mechanism whereby PEEP reduces intrapulmonary shunt.¹⁵⁷ Yet, PEEP does not decrease, and sometimes increases, extravascular lung water when microvascular hydrostatic pressure is high.¹⁵⁸ Malo et al¹⁵⁶ clarified this issue. They showed that PEEP 13 cm H₂O decreased shunt and alveolar flooding without modifying extravascular lung

water; indeed, perivascular cuff edema increased.¹⁵⁶ These findings indicate that PEEP redistributes the excess alveolar water into the compliant perivascular space, thus reinflating previously flooded and collapsed air spaces, without decreasing the overall amount of lung edema.¹⁵⁹

Unilateral pneumonia and ARDS are characterized by severe impairment of gas exchange because of increased intrapulmonary shunt,^{138,140} resulting mainly from flooded and/or collapsed alveoli.^{160–163} This observation is corroborated by the correlation between the extent of lung densities mainly found in the dependent regions on a computed tomography (CT) scan and deterioration in arterial blood-gases.^{18,78} In fact, the edema increases the total mass of the lung (up to more than twice that of normal lungs); consequently, the dependent zones progressively collapse under the weight of the superimposed lung (compression atelectasis), and aerated lung decreases (“baby lung” concept).^{164–166}

Atelectasis in the lower lobes and in dependent lung regions may be caused by other mechanisms, such as external compression by the heart^{167,168} and the abdominal compartment.^{93,169–172} The severity of intrapulmonary shunt may also be affected by surfactant abnormalities,¹⁷³ increased airway resistance,^{174–187} hyaline membranes, and inflammatory proteins and cells accumulating within the alveoli. Interstitial edema and the deposition of cells and connective tissue in the interalveolar septa may further contribute to gas exchange impairment in ARDS. The amount of nonaerated lung volume may also be increased by the combined effects of reabsorption atelectasis^{89,138,140,188} and ventilation occurring at low lung volumes.^{141,189,190} Hypoxic pulmonary vasoconstriction appears greatly impaired in most patients with ARDS.^{140,191} In some patients with diffuse lung injury, however, hypoxic pulmonary vasoconstriction is partially preserved, explaining a lack of hypoxemia despite diffuse loss of aeration. Conversely, hypoxic pulmonary vasoconstriction is constantly impaired in some patients with localized lung injury, explaining severe hypoxemia despite well-preserved lung aeration.¹⁹¹

Use of PEEP to correct gas exchange impairment in ARDS was initially proposed by Asbaugh et al.¹ It remains the cornerstone of ventilatory management of these patients. Extensive literature supports the use of PEEP for improving oxygenation in hypoxemic respiratory failure, alone or combined with various ventilator modes, during both invasive^{5,18,42,76,78,184,192–209} and noninvasive ventilation.^{210–213} Several mechanisms may explain the effect of PEEP on gas exchange. PEEP promotes alveolar recruitment and increases aerated lung volume, therefore decreasing intrapulmonary shunt.^{214,215} PEEP-induced recruitment of aerated lung volume is strongly correlated with arterial oxygenation (Table 10-1).^{18,197,200,215,216} Prevention of atelectasis consequent to pure O₂ breathing helps explain the beneficial effect of PEEP on oxygenation.¹⁴¹ In addition, a redistribution of alveolar edema to the interstitial spaces may also explain the benefit of PEEP on gas exchange.¹⁵⁶ Following the pioneering work of Webb and Tierney,²¹⁷ PEEP has been shown to decrease pulmonary edema,^{218–220} partly because of a concomitant reduction in cardiac output.²¹⁹ By recruiting nonaerated alveoli and stabilizing airways, PEEP also affects the regional distribution of tidal ventilation.^{169,199,221,222} When the predominant effect of PEEP is recruitment, alveolar ventilation is expected to become more homogeneous, particularly in the dependent zones. Although an increase in lung volume is the main mechanism for PEEP-induced changes in oxygenation, a small decrease in cardiac output also reduces intrapulmonary shunt and improves PaO₂.⁴⁸ Finally, different pathophysiologic mechanisms of ARDS influences the response to PEEP,^{31,33,198} although understanding is far from being conclusive.²²³

Table 10-1: Studies Assessing the Effect of Positive End-Expiratory Pressure on Lung Volumes and Other Physiologic Variables

Reference	Technique	PEEP cm H ₂ O	FRC mL	Δ EELV mL	Vrec mL	Vrec/ Δ EELV	Voverdist ml	Δ PaCO ₂ mm Hg
Ranieri 1991 (200)	P-V curve	15	–	720	230	0.32	–	50
Ranieri 1994 (201)	P-V curve	15	–	690	248	0.36	–	–
Ranieri 1995 (202)	P-V curve	10	–	1274	756	0.59	–	55
Jonson 1999 (25)	P-V curve	10	–	–	205	–	–	–
Richard 2001 (204)	P-V curve	11	–	–	175	–	–	–
Maggiore 2001 (197)	P-V curve	15	–	764	304	0.40	–	79*
Mergoni 2001 (315)	P-V curve	15	–	–	379	–	–	29
Koutsoukou 2002 (184)	P-V curve	15	–	660	457	0.69	–	57
Richard 2003 (203)	P-V curve	14	–	–	384	–	–	20†
Maggiore 2003 (296)	P-V curve	15	–	650	235	0.36	–	–
Vieillard-Baron 2003 (316)	P-V curve	12	–	358	69	0.19	–	44
Thille 2007 (345)	P-V curve	11	–	–	226	–	–	–
Valta 1993 (208)	Static compliance	14	–	500	126	0.25	–	–
Gattinoni 1998 (282)	Static compliance	15	576	721	131	0.18	–	–
Chelucci 2000 (314)	Static compliance	13	–	446	102	0.23	–	–
Vieira 1998 (36)	CT scan	13	–	–	320	–	238	117
Vieira 1999 (225)	CT scan	16	–	–	332	–	13	113
Puybasset 2000 (199)	CT scan	10	1621	659	187	0.28	41	45
Malbouisson 2001 (18)	CT scan	15	1553	820	499	0.61	24	99
Nieszkowska 2004 (311)	CT scan	15	1105	1115	369	0.33	63	80
MEAN		13	1214	721	287	0.37	76	66
SD		2	483	249	161	0.16	93	32

Abbreviations: CT, computed tomography; Δ EELV, change in end-expiratory lung volume; FRC, functional residual capacity; Δ PaO₂, change in arterial oxygen tension compared to PEEP 0 cm H₂O; P-V, pressure-volume; SD, standard deviation; Voverdist, volume of lung overdistension; Vrec, recruited volume.

* Compared to PEEP 5 cm H₂O; † compared to PEEP 10 cm H₂O.

Improved efficiency of alveolar ventilation and an expected decrease in alveolar dead space should produce a reduction in PaCO₂. Several authors, however, have failed to find any significant relationship between PaCO₂ and PEEP. The lack of response probably arises because PEEP both favors lung recruitment, which theoretically reduces PaCO₂, and promotes pulmonary overdistension, which theoretically increases PaCO₂.^{33,169,224}

The benefit of PEEP on gas exchange might also be expected in unilateral pneumonia and localized lung injuries. In these conditions, however, high PEEP is more likely to produce overdistension of normally aerated regions, because of the coexistence of areas with normal, low, and very low compliances.^{169,199} The net effect of PEEP on gas exchange depends on the balance between overdistension of already aerated alveolar units and recruitment of collapsed (nonaerated) alveoli, and is also influenced by the level of PEEP.^{169,225} In patients with unilateral lung disease, PEEP may be detrimental whenever it hyperinflates the normal lung, thus directing the blood flow to the diseased lung and increasing intrapulmonary shunt.^{226–230} To limit this risk, unilateral delivery of PEEP to the injured lung has been proposed;²³¹ the technical complexity of this approach greatly limits its feasibility in the clinical practice.

In summary, PEEP improves gas exchange in hypoxemic ARF, mainly through recruitment of previously nonaerated lung areas and the homogenization of regional distribution of tidal ventilation. PEEP remains the cornerstone of ventilator treatment of most forms of hypoxemic ARF. Debate continues about the optimal level of PEEP, not about its usefulness.

Effect in Obstructive Lung Disease

Impaired gas exchange in obstructive lung diseases is caused by complex disturbances of regional $\dot{V}_A/\dot{Q}$ relationships associated with alveolar hypoventilation; diffusion impairment is not a major factor.²³² During an acute exacerbation of COPD, high $\dot{V}_A/\dot{Q}$ areas predominate in patients who mostly have the emphysematous variant with loss of blood flow consequent to alveolar wall destruction.^{232–234} Conversely, low $\dot{V}_A/\dot{Q}$ regions are mostly seen in patients with a predominant obstructive component.^{232–234} In general, application of PEEP does not affect gas exchange.^{2,235–238} Nevertheless, Rossi and coworkers reported that low PEEP (50% of PEEP_i) induced a moderate increase in PaO₂ and slight decrease in PaCO₂ secondary to improved $\dot{V}_A/\dot{Q}$ distribution without alteration in respiratory mechanics or hemodynamics.²³⁹ Further increases in PEEP (up to 100% of PEEP_i) did not produce further improvement in gas exchange.²³⁹ In other small studies, low to moderate PEEP (5 to 9 cm H₂O) produced similar improvements in PaCO₂ and PaO₂.^{240–243} Other authors, however, reported little or no change in arterial oxygenation with PEEP^{2,60,237,238} unless high levels were applied;^{237,238} PaCO₂ did not change with up to 15 cm H₂O PEEP.^{2,60,236–238}

PEEP may have some benefit on gas exchange. Unless hypoxemia is caused by a large intrapulmonary shunt, the benefit is small and of little clinical importance and PEEP may have negative hemodynamic effects that worsen oxygen delivery.^{236–238} The balance between positive and negative effects depends to some extent on the level of applied PEEP; nevertheless, the chief reason for using PEEP in COPD is not to improve gas exchange.

Effect of Positive End-Expiratory Pressure on Respiratory Mechanics

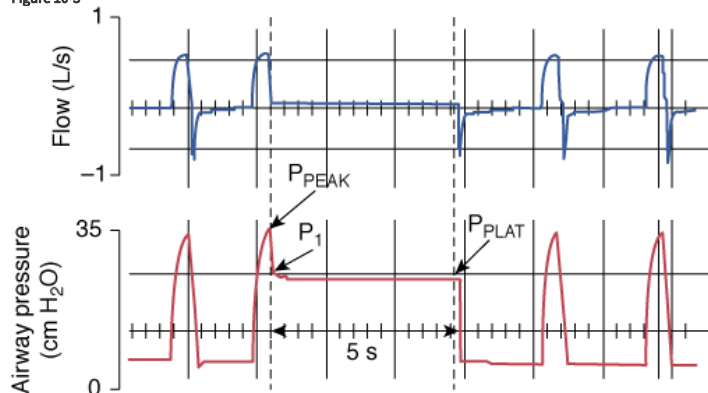
Assessment of Respiratory Mechanics

The static behavior of the respiratory system is described by the pressure–volume curve.^{244–246} In healthy subjects, the pressure–volume curve, from residual volume to total lung capacity, has a sigmoidal shape.²⁴⁶ Above FRC, however, the curve is linear. Thus, the tangential slope of the curve (i.e., the linear or chord compliance, reflecting the elasticity of the respiratory system) is constant over the range of tidal ventilation. FRC is the volume of gas in the lungs at the end of a passive expiration. It corresponds to the point of equilibrium between lung and chest-wall elastic recoil.

The shape of the pressure–volume curve reveals two important features: stiffening of the lungs at high lung volumes, and closure of peripheral airways and lung units at low lung volumes.²⁴⁷ At volumes above 75% to 80% of total lung capacity, the slope of the pressure–volume curve varies, with a sharp decrease in compliance, indicating that the lungs are close to their maximum stretch limit, although the chest wall is normally not stiffer. Further increases in pressure beyond this point, the so-called *upper inflection point* (UIP), have progressively less effect on lung volume. When lung volume falls below FRC and approaches residual volume, the smallest airways tend to collapse under the influence of surface tension of alveolar lining fluid.^{248–251} During the subsequent inflation, these structures remain closed until a much higher pressure is applied. From this point on, compliance rapidly increases during inflation, as the closed lung units progressively pop open. This action produces a knee on the pressure–volume curve, termed the *lower inflection point* (LIP).

During constant-flow controlled ventilation, the elastic and resistive properties of the respiratory system can be assessed using the occlusion technique, which consists of occluding the airway at end-inspiration for 3 to 5 seconds.^{252–258} As soon as flow is interrupted, there is a rapid drop in airway pressure from the peak (P_{PEAK}) to a lower value (P₁), followed by a gradual decay to an apparent pressure plateau (P_{PLAT}) (Fig. 10-3). Ohmic (flow-dependent) resistance (R_{max}) is calculated as the initial drop in airway pressure (P_{PEAK} – P₁) divided by inspiratory flow. The slower pressure change (P₁ – P_{PLAT}) divided by the inspiratory flow preceding the occlusion yields the additional tissue resistance (ΔR). The sum of airflow and tissue resistances provides the total respiratory resistance (R_{TOT} = (P_{PEAK} – P_{PLAT})/ $\dot{V}$). By occluding the airway at end-expiration, it is possible to determine the total PEEP (PEEP_{TOT}), which is the sum of the externally applied PEEP and PEEP_i (Fig. 10-4). The difference between P_{PLAT} and PEEP_{TOT} is the recoil pressure. It is then possible to compute the static compliance (C_{st}) of the respiratory system according to the equation C_{st} = V_T/(P_{PLAT} – PEEP_{TOT}). The failure to take into account PEEP_i may result in a substantial underestimation of C_{st}, especially when low or no PEEP is used.^{259,260}

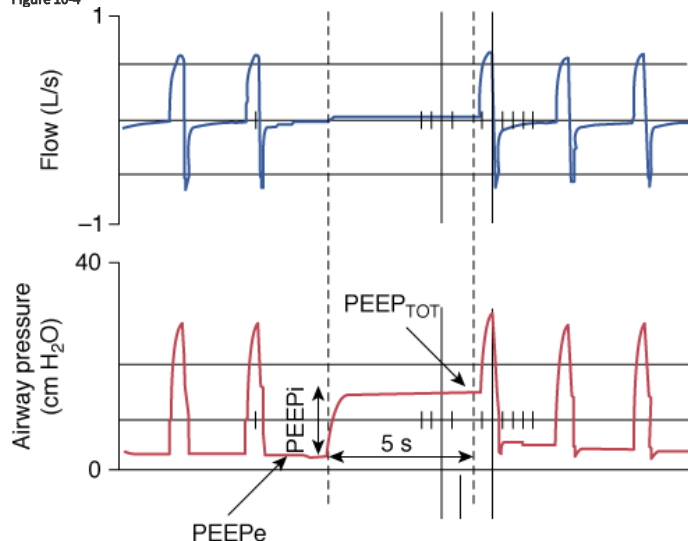
Figure 10-3



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End-inspiratory occlusion during constant flow controlled ventilation. As soon as the airway is occluded, flow suddenly falls to zero and airway pressure drops from a peak (P_{PEAK}) to a lower value (P_1), and then slowly declines to an apparent plateau (P_{PLAT}).

Figure 10-4



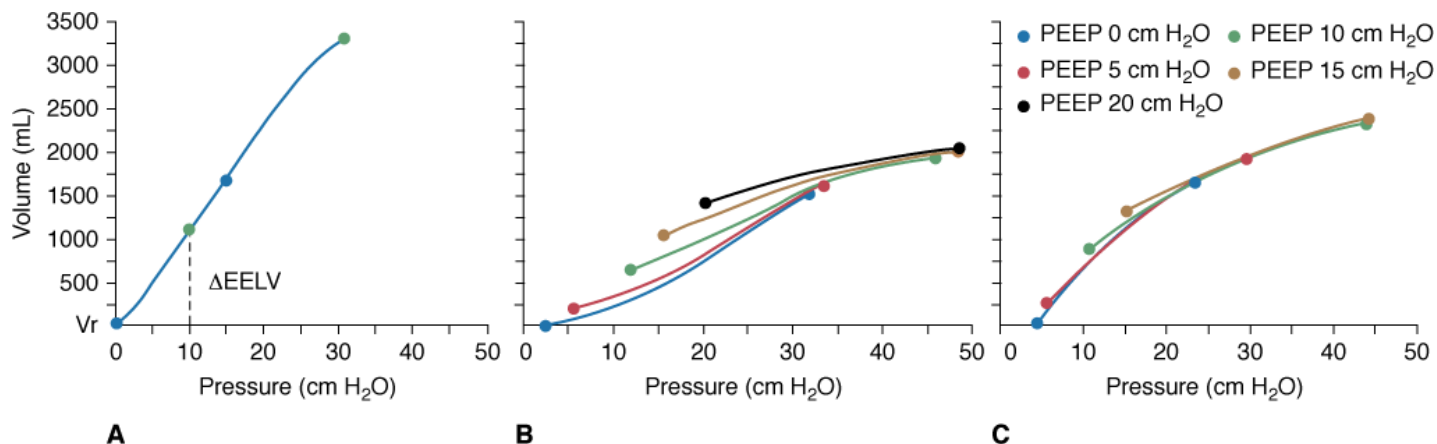
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End-expiratory occlusion. Airway pressure rises following an end-expiratory occlusion and reaches a plateau, which corresponds to total PEEP ($PEEP_{TOT}$). Auto or intrinsic PEEP ($PEEP_i$) is the difference between $PEEP_{TOT}$ and the externally applied preset PEEP ($PEEP_e$). $PEEP_i$ measured by an end-expiratory occlusion is termed static.

Effects of Positive End-Expiratory Pressure in Anesthetized Subjects

Anesthetized healthy subjects may have decreased aerated lung volume, consequent to the decline in FRC associated with the supine posture^{93,127,128,261,262} and the potential for airway closure and atelectasis.^{85,96,100,127,128,190} Both anesthesia and paralysis affect the mechanical characteristics of the respiratory system,^{85,263,264} which can be also impaired by abdominal and thoracic surgical procedures.^{125,128,265–267} It is thought that anesthesia primarily alters the elastic properties of the chest wall, causing a fall in FRC, whereas the changes in lung compliance may result from breathing at low lung volumes.^{128,263,268,269} Application of PEEP in anesthetized and paralyzed subjects produces an increase in end-expiratory lung volume^{125,128,270–272} and upward displacement of respiratory system, lung and chest wall pressure–volume relationships (Fig. 10-5A).^{272–274} Several authors, however, report little or no increase in static respiratory compliance with PEEP.^{272,275,276} In some studies, PEEP increased both chest wall and lung compliance.²⁷² In other studies, chest wall compliance increased and lung compliance decreased,²⁷⁶ or no effect was observed.²⁷⁵

Figure 10-5



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Pressure-volume (P-V) curves recorded from different PEEP levels in a healthy subject (A), a patient with ARDS (B), and a patient with an acute exacerbation of COPD (C). All curves are related to the relaxation volume of the respiratory system. For each curve, end-expiratory and end-inspiratory points are indicated by circles. In the healthy subject, PEEP 10 cm H₂O generates an increase in end-expiratory lung volume and a rightward shift of the P-V curve. The new curve is superimposed on the curve acquired without PEEP, indicating the absence of recruitment. In the patient with ARDS, increasing PEEP produces increases in end-expiratory lung volume and significant lung recruitment, as indicated by the upward shift of the P-V curves. In the patient with COPD, the application of PEEP results in end-expiratory lung volume increase with worsening of hyperinflation, as indicated by the superposition of P-V curves and the progressive decrease in the slope (compliance) of the curves. The reduction in aerated volume in ARDS produces a lower inflection point and flattening of the P-V curve at PEEP 0 cm H₂O. The presence of PEEPi during ventilation without PEEP in ARDS, and more so in COPD, induces a rightward shift of the initial P-V curve. In the healthy subject (*left panel*), the PEEP-induced increase in end-expiratory lung volume is indicated by the dotted line. *V_r*, Relaxation volume of the respiratory system.

The effect of PEEP on the elastic properties of the respiratory system varies with the amount of applied PEEP²⁷³ and preexisting derangement in respiratory mechanics. Compared with zero end-expiratory pressure (ZEEP), D'Angelo et al²⁷³ found that PEEP of 9 cm H₂O increased both respiratory system and lung static compliance; the compliances, however, decreased when PEEP was raised above 20 cm H₂O, indicating lung overdistension. Dechman et al²⁷⁷ also observed an increase in lung compliance at PEEP 10 cm H₂O in patients undergoing closed-chest surgery, but not in patients undergoing open-chest surgery, who exhibited a progressive decrease in dynamic lung compliance. The mechanical characteristics of the chest wall may influence the effect of PEEP in subjects with normal lungs during anesthesia.²⁶³ In patients undergoing abdominal surgery, Pelosi et al¹²² reported that 10 cm H₂O PEEP did not improve respiratory function in normal subjects, although it increased end-expiratory lung volume and lung and chest wall compliance in morbidly obese patients. Others reported similar improvements in end-expiratory lung volume and/or in respiratory system compliance both in obese patients^{125,127,128} and in healthy subjects with impaired chest wall compliance caused by pneumoperitoneum during laparoscopic surgery.^{125,267}

In general, PEEP decreases airway resistance.^{122,267,272,273} The decrease in airway resistance is generally small and mainly related to increase in lung volume,²⁷² although other mechanisms, such as a PEEP-induced modification of basal vagal tone,²⁷⁸ might also play a role.

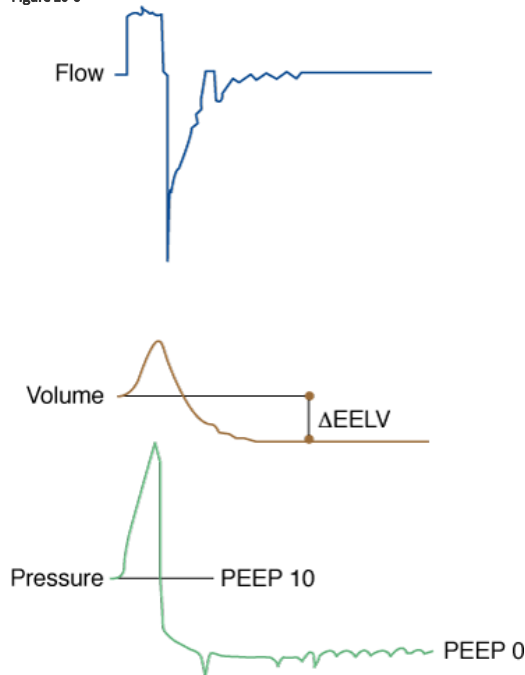
Effect of Positive End-Expiratory Pressure on Lung Volume in Acute Respiratory Distress Syndrome

Since the first description by Asbaugh et al,¹ ARDS has been recognized as a condition characterized by reduction in aerated lung and respiratory system mechanical derangements.^{76,79,279,280} In ARDS, massive lung edema, atelectasis, and tissue consolidation cause a marked decrease in FRC.^{76,79} This fall in normally ventilated lung volume is the main cause of the impaired respiratory mechanics.¹⁶⁵ Conversely, an increase in aerated lung volume is long recognized as the main cause of the benefit of PEEP on lung function.^{42,76} The increase in lung volume may result from two different mechanisms: recruitment of terminal lung units not accessible to ventilation because of collapse or flooding, and distension or overdistension of already open lung units.

Several authors reported an increase in end-expiratory lung volume and FRC following the application of PEEP in patients with ARDS (see Fig. 10-5B).^{14,18,36,42,76,199,214,281,282} In absence of PEEPi, the increase in lung volume above FRC at end-expiration, or change in end-expiratory lung volume (ΔEELV), depends on the amount of applied PEEP and the compliance of the respiratory system. Different techniques can be used to measure FRC and EELV, such as helium dilution,²⁸³⁻²⁸⁸ nitrogen^{285,288-290} and sulphur hexafluoride^{288,291-294} washout, positron emission tomography,²⁹⁵ and CT scan.^{36,78,199} These measurements are not routinely obtained in the clinical setting because of technical limitations and/or logistic complexities.

A surrogate approach has been used for assessing PEEP-induced Δ EELV; this involves measuring the volume exhaled between PEEP and elastic equilibrium volume of the respiratory system (Fig. 10-6).^{25,197,200–204,266,296,297} While the ventilator is delivering V_T , the preset PEEP is rapidly brought to zero. During the following expiration, sufficient time (5 to 10 seconds) is provided to enable complete exhalation; thus the elastic equilibrium, relaxation volume, will be reached.^{25,176,197,200,202} The difference between the volume exhaled during this prolonged expiration and the volume insufflated during the preceding inspiration represents the Δ EELV produced by PEEP_{TOT}. The excess volume caused by PEEPi should be separately assessed, and accounted for, to avoid overestimation of PEEP-induced Δ EELV. PEEP-induced variations in EELV can also be measured by inductive^{208,296,298–301} or optoelectronic plethysmography.^{19,302}

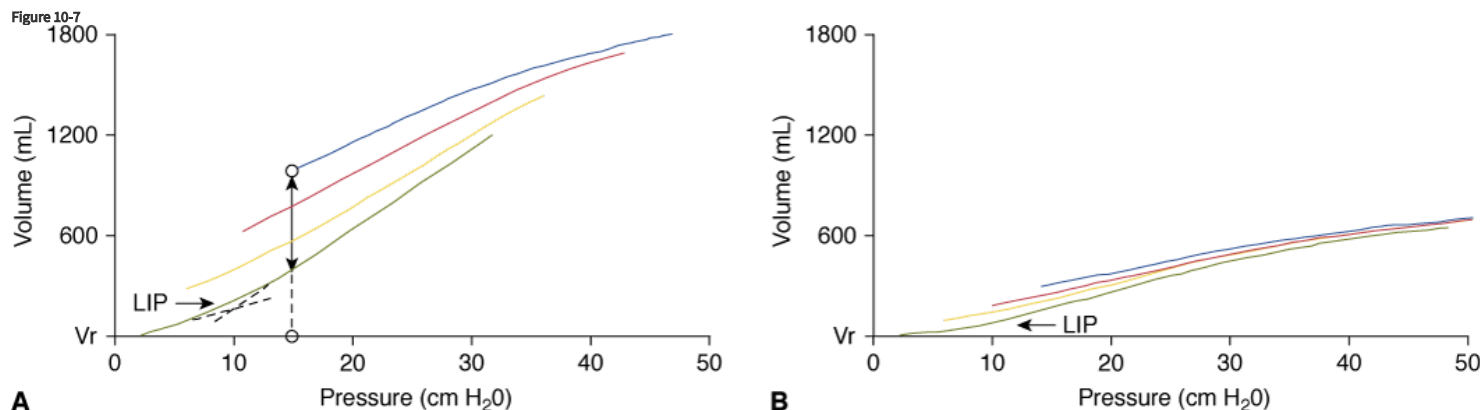
Figure 10-6



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Measurement of PEEP-induced change in end-expiratory lung volume (Δ EELV). PEEP 10 cm H₂O is abruptly brought to zero. A prolonged expiration ensues, and relaxation volume is achieved. The difference between the volume exhaled during this expiration and the tidal volume delivered by the ventilator equals the increase in EELV induced by PEEP.

An increase in EELV may also occur when PEEP distends or overdistends already aerated lung regions (Fig. 10-7 and see Table 10-1). Hence, measurement of Δ EELV, without quantification of lung recruitment, may be misleading. *Lung recruitment* refers to re-aeration of collapsed or fluid-filled terminal airways and alveoli. These areas become accessible to ventilation (anatomic recruitment) and may participate in gas exchange if local perfusion is preserved (functional recruitment). Lung recruitment can be quantified by using pressure-volume curves, traced from different levels of PEEP, or by CT scan analysis (see Table 10-1). Other techniques have been proposed, but their application in clinical practice is at present very limited.^{295,303–306}



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Pressure-volume (P-V) curves at different levels of PEEP (0, 5, 10, and 15 cm H₂O), related to relaxation volume (V_r) of the respiratory system, in two patients with ARDS. In the left panel (A), PEEP induces progressive upward shifts of P-V curves, indicating recruitment. At PEEP 15 cm H₂O, PEEP-induced increase in

end-expiratory lung volume above V_r is indicated by the *dotted line* between the two *open circles*. Alveolar recruitment induced by PEEP 15 cm H₂O, compared to PEEP 0 cm H₂O, and measured at a pressure of 15 cm H₂O is shown by the *double-arrow solid line*. In the right panel (B), P-V curves are particularly flattened, suggesting a marked decrease in aerated lung volume. Increasing PEEP results in minimal recruitment: P-V curves are almost totally superimposed and progressively shifted to right. In both panels, all P-V curves tend to converge at high lung volumes, suggesting that total lung capacity is approached. PEEP-induced recruitment is greater when the slope of the linear segment above the lower inflection point (LIP)—that is, linear compliance—of curve recorded from 0 cm H₂O (the lowest curve) is high (A). When the curve at 0 cm H₂O has a very low linear compliance, PEEP-induced recruitment is trivial (B). PEEP-induced recruitment also proceeds far above LIP (up to pressures greater than 40 cm H₂O), suggesting that lung reopening is a pan-inspiratory phenomenon (A).

Measurement of PEEP-induced lung recruitment using pressure–volume curve stems from the pioneering observations of Katz et al¹⁴ on the time course of PEEP-mediated increase in lung volume. After applying PEEP, these authors observed that end-expiratory lung volume increased to approximately 66% of its total change within the first breath; it reached 90% in approximately five breaths, and was fully complete only after several minutes, suggesting time-dependence for lung recruitment.¹⁴ For a given pressure, lung volume measured during ventilation with 13 cm H₂O PEEP was larger than the volume predicted for that pressure by the static compliance computed at 3 cm H₂O PEEP. This observation suggests that some lung units were not ventilated at the lower PEEP, but were recruited by the higher PEEP.¹⁴ Quantification of lung recruitment induced by PEEP was not attempted until Ranieri et al^{200–202} proposed a simple method using pressure–volume curve analysis. This approach consists of aligning on the same pressure–volume diagram curves obtained at ZEEP and varying PEEP levels. With this technique, which assumes that FRC (i.e., the relaxation volume at ZEEP) remains constant,^{14,200,201} the Δ EELV induced by PEEP is added to the volume insufflated during the maneuver for recording pressure–volume curve at the corresponding PEEP level.^{197,247,297} Lung recruitment is then computed as the difference in volume between pressure–volume curves obtained at different PEEP levels at a given value of elastic recoil pressure (generally 20 cm H₂O). Anatomic recruitment is reflected by an upward shift of the pressure–volume curve (see Fig. 10-7). If PEEP fails to recruit new regions, no volume gain at a given pressure will be observed; the resulting pressure–volume curve will be shifted rightward and superposed on the curve acquired at ZEEP.^{197,201}

The CT scan is used to measure recruitment.^{215,307,308} PEEP-induced lung recruitment is computed as the decrease in volume of nonaerated lung parenchyma^{78,164,216,221,307} or, as recently proposed, the increase in volume of gas penetrating nonaerated and poorly aerated lung regions.^{18,308–311} With this technique, however, recruitment is assessed by comparing scans taken at different pressures (ZEEP and PEEP), and not at the same pressure as is done with the pressure–volume curve technique. Consequently, no matter how it is computed, recruitment at a given PEEP will include also a certain volume of gas corresponding to lung units that inflate normally from end-expiration at ZEEP up to the pressure corresponding to that PEEP. This helps explain some of the differences in the amounts of recruitment with the CT and pressure–volume techniques.³¹² Conversely, the CT scan offers a unique opportunity to estimate the amount of PEEP-induced hyperinflation (see Table 10-1),^{18,36,78,199,225,311,313} computed as the increase in gas volume within normally aerated lung regions; hyperinflation, however, does not necessarily mean overstretching.³⁰⁷

Bedside lung ultrasound has been proposed as a mean to estimate PEEP-induced recruitment. In forty patients with ARDS, Bouhemad et al compared the pressure–volume curve method and lung ultrasound to assess recruitment with PEEP 15 cm H₂O.³⁰³ Significant correlations were found between an ultrasound re-aeration score and recruitment measured by pressure–volume curve, and between ultrasound score and oxygenation. This new technique did not, however, allow assessment of lung hyperinflation. Further studies are needed to evaluate the usefulness of this technique.

In patients with ARDS, PEEP increases end-expiratory lung volume and produces varying recruitment (see Table 10-1).^{18,25,76,78,169,197,199–204,208,216,221,225,275,282,296,311,313–317} Recent work¹⁹⁷ shows that PEEP achieves progressive recruitment, from 84 mL at PEEP 5 cm H₂O (range: 41 to 156 mL) to 304 mL at 15 cm H₂O (range: 114 to 545 mL). The large interindividual differences suggest that, in certain patients, PEEP increases lung volume without achieving significant recruitment, likely reflecting overdistension of previously aerated lung regions. Indeed, PEEP-induced overdistension has been shown in several studies.^{200–202 18,36,199,225,311} In most cases, recruitment and overdistension occur concomitantly, rather than sequentially, in different regions of the lung following PEEP application.^{199,225,308,309,311,313} Vieira et al³⁶ found that applying PEEP 10 and 15 cm H₂O in patients with ARDS who did not have a clear LIP on the pressure–volume curve produced progressive recruitment of nonaerated lung areas and overdistension of aerated lung zones. The interplay between these two phenomena depends on several factors, such as lung and chest wall mechanics,^{266,318} stage of disease,^{319,320} lung morphology,^{199,225,309} and amount of PEEP.²²⁵

Recent studies highlight the importance of assessing lung recruitability for an individually tailored optimization of PEEP. By using CT scan in sixty-eight patients with ARDS, Gattinoni et al. assessed the percentage of potentially recruitable lung, as defined by the proportion of lung tissue in which re-aeration was restored at airway pressure between 5 and 45 cm H₂O, and the effects of PEEP.³³ Lung recruitability was 13% $\pm$ 11% of the lung weight, varied widely across the studied population (from 0% to more than 50%), and was highly correlated with lung recruitment induced by an increase in PEEP from 5 to 15 cm H₂O. Patients with higher lung recruitability had more severe disease (lower oxygenation and respiratory system compliance) and a higher mortality rate.

Assessment of lung recruitability is important for setting PEEP: the use of high PEEP in patients with a low percentage of potentially recruitable lung provides little, if any, benefit and may actually be harmful if it produces overdistension.³²¹ Lung recruitability may influence the effects of PEEP on alveolar strain (the ratio between the amount of gas delivered during a tidal breath and the amount of aerated lung available for ventilation) and alveolar opening and closing.

Caironi et al reported that an increase in PEEP from 5 to 15 cm H₂O in patients with a lower percentage of potentially recruitable lung did not significantly affect the amount of opening and closing lung tissue, which was already negligible at lower PEEP.²⁷ Conversely, in patients with a higher recruitability, opening and closing was dramatically reduced (approximately equal to 50%) by increase in PEEP. Interestingly, increases in PEEP led to an identical increase in alveolar strain in the two groups of patients. In patients with higher recruitability, the lung volume available for ventilation conceivably increases during inspiration as recruitment of nonaerated lung regions at end-expiration progressively occurs. These findings further suggest a relationship between the amount of intratidal lung recruitment and the effective end-inspiratory alveolar strain, such that the larger the amount of potentially recruitable lung, the lower the risk of PEEP-induced overdistension. In addition, opening and closing lung tissue turned out to be an independent risk factor for death (odds ratio 1.10 for each 10 g increase), while alveolar strain did not.²⁷ This suggests that in patients with higher lung recruitability, the beneficial impact of reducing intratidal opening and closing by increasing PEEP prevails over the effects of increasing alveolar strain. In this regard, lung imaging morphology (chest radiograph or CT scan) may help in assessing lung recruitability: patients with a lower percentage of potentially recruitable lung generally show a “lobar” pattern of lung consolidation, whereas patients with greater lung recruitability often present diffuse lung collapse.^{27,199,225,321}

The integrated analysis of respiratory mechanics, lung volumes, and lung imaging enhances recognition of the effects of PEEP on recruitment and overdistension, and hence facilitates to set optimal PEEP at the bedside.

Effect of Positive End-Expiratory Pressure on Respiratory Mechanics in Acute Respiratory Distress Syndrome

Contrasted with normal subjects, the rate of change per unit of pressure is smaller, and linear compliance of the pressure–volume curve falls to very low values in patients with ARDS (because of the decrease in normally ventilated alveolar units); the whole curve is also flattened and shifted downward, and the inflection knees, normally seen at low (LIP) and high (UIP) lung volumes, may appear in the range of tidal ventilation²⁴⁷ (see Figs. 10-5B and 10-7). The presence of LIP and UIP in the V_T range suggest considerable susceptibility of the ARDS lung to detrimental mechanical shear forces, generated by cyclic end-expiratory alveolar opening and collapse^{78,165,322} and end-inspiratory overstretching.^{201,323} These forces are considered the main mechanism of ventilator-induced lung injury.^{324,325} Recent evidence suggests that airspace reopening can occur along the entire pressure–volume curve.^{197,216,307,326–329} Moreover, the shape and characteristics of the pressure–volume curve can be greatly influenced by several factors including pathophysiologic mechanisms, stage of disease,^{282,322} chest wall mechanics,^{198,266} breathing pattern, and differences in measurement technique.³³⁰

As already mentioned, one typical feature of ARDS is the fall in respiratory system compliance (see Figs. 10-5B and 10-7), described both in experimental^{331–334} and clinical settings.^{42,76,176,322,335} Decreased compliance was initially attributed to increased lung stiffness.²⁸⁰ Subsequent studies have demonstrated that the decreased compliance in ARDS has different pathophysiologic mechanisms.^{333,334} In an animal model, Grossman et al³³³ showed that the loss of aerated lung units, secondary to alveolar flooding, accounted for virtually all of the decreased static lung compliance. With CT scan analysis, Gattinoni et al¹⁶⁵ found that several parameters of the pressure–volume curve, including linear compliance, were correlated with the amount of aerated areas (mainly distributed in the nondependent, ventral lung regions) but not with nonaerated areas (preferentially distributed in the dependent, dorsal lung regions). Indeed, compliance was in the normal range when normalized to FRC (so-called specific compliance), suggesting that the aerated areas had normal intrinsic elasticity.¹⁶⁵ According to this concept, respiratory system compliance provides an indirect estimate of the amount of aerated lung (see Figs. 10-5B and 10-7).

Through recruitment, PEEP prevents the fall in static compliance,⁵ and restores the normal pressure–volume pattern, by suppressing small airway closure and collapse of unstable lung units.^{281,336–339} Data suggest that the effects of PEEP on recruitment and overdistension can be predicted by the shape of the pressure–volume curve at ZEEP.^{197,200,201,225,326,328,340} Three distinct patterns of pressure–volume curve have been described in patients with ARDS.^{200,201,225} A concave shape, with a clear LIP, indicates that compliance is progressively increasing above this point, suggesting ongoing recruitment during inflation (see Fig. 10-7).^{200,201,340} A convex shape, showing an UIP, denotes progressive decrease in compliance, and likely overstretching (see Fig. 10-7).^{200,201} A linear pressure–volume curve, with absent or very low LIP, may be observed in localized lung injury.²²⁵

Rouby et al^{191,309,310} hypothesized that a linear pressure–volume curve results from two separate regional curves: a concave curve, related to normally aerated lung regions, and a convex curve, related to nonaerated areas. PEEP may have different effects on lung volumes and respiratory mechanics according to these patterns. When a concave shape is observed at ZEEP, PEEP results in lung recruitment, as documented by an upward shift of the pressure–volume curve and disappearance of LIP (see Fig. 10-7).^{197,200,201,225} With a convex profile and low linear compliance at ZEEP, PEEP-induced recruitment is minimal or zero: the pressure–volume curve is superimposed on that obtained at ZEEP, with a further decrease in compliance (see Fig. 10-7).^{197,200,201,225}

A linear compliance on the pressure–volume curve at ZEEP may be a useful indicator of lung recruitability (see Fig. 10-7). A tight correlation between linear compliance at ZEEP and amount of PEEP-induced recruitment has been suggested.¹⁹⁷ Because compliance, as assessed on the pressure–volume curve, is a function of normally aerated lung regions,¹⁶⁵ two different situations can be imagined: (a) the lung is principally characterized by unstable lung regions, which are collapsed at the end expiration but progressively pop open during tidal inflation, and (b) the lung is predominantly consolidated, with no large areas expanding during inflation (compliance is very low). Application of PEEP results in significant recruitment in the former case; in the latter case, PEEP may overstretch already ventilated areas without producing significant recruitment.^{200,201} Independently of its effect on recruitment and overdistension, PEEP constantly and significantly decreases the linear compliance of the pressure–volume curve.^{25,197,201,326,328} This contrasts with the classical notion that compliance increases with recruitment and decreases with overdistension.⁴² Lung units that pop open at a certain pressure have an infinitely high compliance and contribute to the higher compliance observed at ZEEP, as opposed to that observed when the lung is fully recruited at PEEP. A corollary is that UIP may indicate the end of tidal recruitment and/or the beginning of alveolar overdistension, although both mechanisms may coexist at high pressure (see Fig. 10-7).^{25,326} Therefore, PEEP-related decrease in pressure–volume linear compliance may predominantly reflect recruitment below UIP; a decrease in compliance above this point may reflect overdistension.

LIP has long been considered the pressure at which collapsed lung units open during inflation, it is therefore considered the ideal pressure at which to set PEEP.³²² No relationship between LIP and either opening or closing alveolar pressures, however, was subsequently found,¹⁹⁷ indicating that LIP is not useful for optimizing PEEP setting. Other studies confirmed that recruitment and derecruitment are continuous processes unrelated to LIP (see Fig. 10-7).^{25,204,215,216,326,328,329} Nevertheless, the presence of a clear LIP followed by a high inflation compliance on the pressure–volume curve recorded from ZEEP may predict the effect of PEEP on recruitment (see Fig. 10-7).^{165,197,225,328} Vieira et al²²⁵ showed that patients with a diffuse loss of aeration had a marked LIP at ZEEP, whereas LIP was absent or blunted in patients with a focal loss of aeration. High PEEP induced progressive recruitment without overdistension in the former group, whereas it caused a significant overdistension of normally aerated regions in the latter group.²²⁵ Some think that LIP is caused by time-constant inequalities within the lung and/or reopening of compressed peripheral airway associated with expiratory flow limitation.^{183,184,341,342} Accordingly, the disappearance of LIP following application of low or moderate PEEP might indicate that PEEP can prevent small airway closure and make the distribution of tidal ventilation more homogeneous.^{184,342} This could also explain the difference between the reported values of LIP, 5 to 15 cm H₂O, and the substantially higher pressures required to reverse atelectasis.^{216,245,343}

Some studies suggest that the point of maximum curvature of the deflation limb of the pressure–volume curve may be used to set PEEP so as to optimize lung recruitment, but the feasibility of this approach is limited.^{215,344}

The mechanical properties of the chest wall may influence the effects of PEEP.^{198,266,282} Mergoni et al¹⁹⁸ found that in some patients the LIP on the pressure–volume curve of the respiratory system at ZEEP reflected the chest wall, rather than the lung. Application of PEEP abolished LIP in all instances, but oxygenation significantly improved, suggesting recruitment, only in patients exhibiting a LIP on the lung pressure–volume curve.¹⁹⁸ Gattinoni et al²⁸² found that increasing levels of PEEP decreased respiratory and lung static compliance, without any recruitment, in patients with primary pulmonary ARDS; in patients with secondary extrapulmonary ARDS, PEEP induced significant recruitment and increased respiratory, lung, and chest wall compliance.²⁸² These findings have been confuted by other studies, even by the same group.^{25,33,197,199,225,345}

The change in static compliance produced by PEEP has been proposed as a guide for optimizing PEEP⁴² and as a prognostic factor for predicting mortality in patients with ARDS.³⁴⁶ Because of nonlinearity of the pressure–volume relationship in ARDS, however, and the aforementioned complex effect of recruitment on the pressure–volume relationship,^{326–328} static compliance (calculated as the pressure-to-volume differences between two points) may not describe adequately the changes in respiratory mechanics produced by PEEP.^{247,297} Indeed, as previously shown³⁴⁷ and recently outlined,^{182,327,345} static compliance may vary for different V_Ts. A mathematical model suggests that the highest static compliance computed at a low V_T during a decremental, but not incremental, PEEP trial may help to identify the level of PEEP needed to prevent end-expiratory alveolar collapse.³²⁷ These theoretical findings are supported by data in animals,^{348,349} but have not been sufficiently tested in humans.

Some authors recommend the monitoring of dynamic compliance as a means of assessing the mechanical properties of the respiratory system under dynamic condition (ongoing ventilation) and to identify the level of PEEP that prevents alveolar derecruitment.^{350,351} Data on this issue, however, are far from conclusive.

Several studies^{179,182,185,314,352} reveal that PEEP increases additional resistance through increased viscoelastic dissipation¹⁸⁰ that is consequent to a PEEP-mediated increase in end-expiratory lung volume, as suggested by the correlation between changes in tissue resistance and end-expiratory lung volume.³⁵² Other studies,^{275,353} although confirming the rise in tissue resistance, also report a decrease in airway resistance, especially at high PEEP. Normalizing resistances for lung volume (i.e., specific resistance) caused these changes to disappear.²⁷⁵ Partitioning resistance into lung and chest wall components, Pelosi et al²⁷⁵ found that the increase in total and tissue respiratory system resistance, consequent to PEEP, could be mainly ascribed to the

lung rather than the chest wall. Gattinoni et al²⁸² found that total respiratory resistance increased with PEEP in pulmonary ARDS, mainly because of a marked increase in lung additional resistance.²⁸² In contrast, patients with extrapulmonary ARDS, characterized by higher chest wall tissue resistance and lower lung tissue resistance than pulmonary ARDS, did not show any change in total respiratory resistance with PEEP.²⁸²

Effect of Positive End-Expiratory Pressure on Respiratory Mechanics in Chronic Obstructive Pulmonary Disease

In COPD patients, lung and chest-wall static compliance are reported to be similar to those of normal subjects,³⁵⁴ unless expiratory flow limitation and dynamic hyperinflation causes lung overdistension. Depending on the level of PEEP, static compliance may decrease (if overdistension occurs),^{238,294,355,356} stay unchanged,³⁵⁷ or even slightly increase (if recruitment of new lung units occurs)²⁵² (see Fig. 10-5C).

Total end-expiratory alveolar pressure ($PEEP_{TOT}$) may markedly exceed preset PEEP when PEEPi is present (see Fig. 10-4). Rossi et al²⁶⁰ demonstrated that failure to take PEEPi into account caused static compliance to be underestimated by up to 48%. During volume-targeted controlled ventilation, an increase in inspiratory flow resulting in a prolonged expiratory time causes PEEPi to decrease. Thus, the value of static compliance calculated without correcting for PEEPi will falsely (and paradoxically) increase.^{260,252} The presence of a LIP on the pressure–volume curve at ZEEP may be partly caused by PEEPi, at least in patients with expiratory flow limitation.³⁵⁸

Patients with COPD have a high resistance to air flow.^{59,238,357,359,360} In patients with expiratory flow limitation, PEEP should have little influence on expiratory flow until PEEP exceeds PEEPi.²⁶⁰ Other studies, however, suggest that PEEP 5 to 10 cm H₂O might partly decrease expiratory resistance, particularly at the end-expiration, and result in faster and more uniform lung emptying.^{294,357,361,362}

Effects of Positive End-Expiratory Pressure on Diaphragmatic Activation

Recent studies provide novel information about the effects produced by PEEP application on diaphragmatic electrical activity, assessed by transesophageal electromyography. Allo et al³⁶³ induced ARDS in rabbits by intratracheal instillation of hydrochloric acid: eleven animals, five of which had been previously vagotomized, underwent a short-term protocol where PEEP and inspiratory assistance delivered by neurally adjusted ventilatory assist (NAVA) were varied over a few minutes. After hydrochloric acid instillation in the nonvagotomized rabbits, tonic (i.e., continuous throughout the respiratory cycle) diaphragmatic electrical activity increased without PEEP, while phasic (i.e., timed with inspiration) diaphragmatic electrical activity became smaller and irregular. A stepwise increase in PEEP progressively attenuated tonic diaphragmatic electrical activity and increased phasic diaphragmatic electrical activity; when PEEP was subsequently reduced, tonic and phasic diaphragmatic electrical activity returned back to the initial values. In the group of vagotomized rabbits, conversely, tonic diaphragmatic electrical activity did not rise after lung injury and was consequently unaffected by PEEP, while phasic diaphragmatic electrical activity and V_T were on average much higher than in the nonvagotomized animals. These data may suggest that tonic diaphragmatic electrical activity could result from vagally mediated reflexes acting to conserve end-expiratory lung volume and PEEP might decrease tonic diaphragmatic electrical activity by recruiting lung volume.

Using the same animal model, Beck et al³⁶⁴ applied the same level of inspiratory assistance by NAVA through an endotracheal tube in the presence of PEEP (8.5 ± 3.3 cm H₂O) and in the absence of PEEP. An instantaneous increase in tonic diaphragmatic electrical activity was observed upon PEEP removal. Following extubation and initiation of NAVA without PEEP via a mask, however, tonic diaphragmatic electrical activity returned to the previous values. The authors considered the increase in tonic diaphragmatic electrical activity during invasive ventilation without PEEP as a mechanism through which the diaphragm assumes the role of conserving end-expiratory lung volume, replacing the activity of the glottal muscles that is lost when the endotracheal tube is in place.

In twenty adult patients with ARF of varied etiology undergoing NAVA, Passath et al³⁶⁵ evaluated the effects of different levels of PEEP. A stepwise reduction of PEEP from 20 to 1 cm H₂O increased inspiratory diaphragmatic electrical activity by 34%, while V_T and breathing rate did not significantly change. The ratio of V_T to diaphragmatic electrical activity, an approximation of neuroventilatory efficiency, showed broad individual variability at different PEEP levels and may help in identifying the level of PEEP at which tidal breathing occurs at minimal diaphragmatic electrical activity cost.

Effect of Positive End-Expiratory Pressure on Ventilator-Induced Lung Injury

Ventilator-Induced Lung Injury

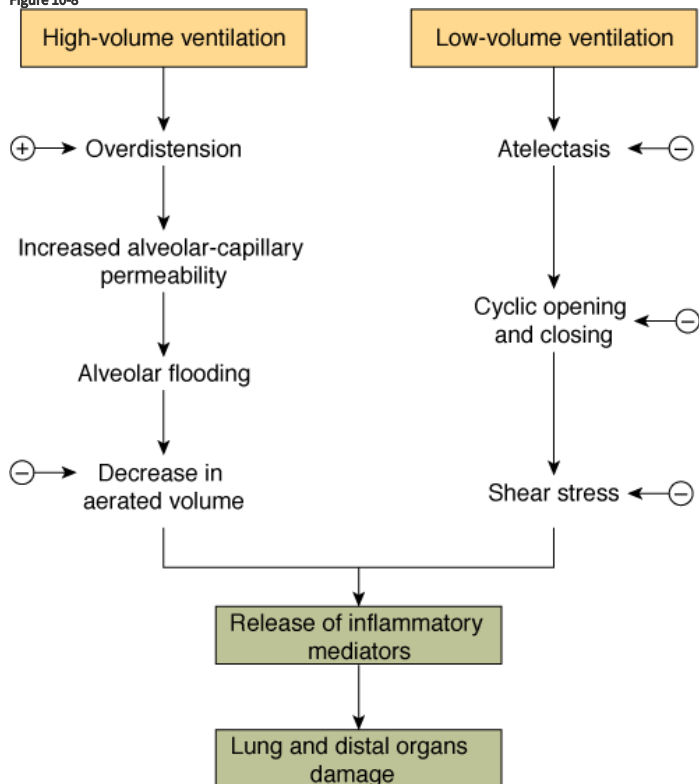
Recent decades have brought the recognition that mechanical ventilation per se, in ARDS, may damage the lung and/or aggravate preexisting lung injury.³²⁵ This so-called ventilator-induced lung injury (VILI)^{325,366,367} is now widely recognized and supported by a large number of animal experimental studies^{220,368} and recent data in patients.^{6,369,370} Attempts to prevent VILI have modified the ventilator approach to patients with ARDS.³²⁴ Two main mechanisms for VILI are: (a) alveolar^{220,368} and bronchial overdistension^{371,372} occurring at high volume and transpulmonary pressure, and (b) repeated alveolar and small airways collapse and reopening at low end-expiratory volume.^{217,371,373} The common pathway for these two mechanisms is mechanical

stress on the terminal units (including bronchiolar and alveolar walls).^{374–376} Other factors, including elevated FI_{O_2} , high blood flow, high inspiratory flow, and intensity of local inflammation, may play a role in aggravating or inducing lung injury.

Effect of Positive End-Expiratory Pressure on Ventilator-Induced Lung Injury

PEEP may have opposing consequences, depending on its effects on the two aforementioned mechanisms of VILI (alveolar overdistension and cyclic alveolar collapse and reopening) (Fig. 10-8).^{225,313} In one study, PEEP (13 cm H_2O) caused recruitment of nonaerated regions, and induced overdistension of already aerated lung areas in three patients.³⁶

Figure 10-8



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Postulated mechanisms by which mechanical ventilation may lead to ventilator-associated lung injury and organ injury. Beneficial (*minus sign*) and detrimental (*plus sign*) effects of PEEP on these mechanisms are shown.

Since the 1970s, high ventilator pressures have been known to rupture alveoli and cause air leaks.³⁷⁷ By increasing airway pressure, PEEP may promote alveolar overstretching,^{39,201,202,225,311} ultrastructural damage,³⁷² worsening of pulmonary edema,¹⁵⁷ and triggering of local and systemic inflammation.³⁷⁸ Some early studies suggest that end-inspiratory lung volume, rather than high intrathoracic pressure, may be the major determinant of ventilator-induced lung edema (“volutrauma”), at least in normal lungs.^{220,379,380} PEEP, however, may also protect against edema accumulation during ventilation at high end-inspiratory pressure and end-inspiratory lung volume,^{217–220,381} although alteration in microvascular permeability may not be prevented.^{219,220} Several factors may explain these apparent contradictions: (a) differences in experimental setup (isolated lung vs. intact animals), (b) levels of PEEP and inspiratory airway pressure, and (c) driving pressure (difference between end-inspiratory and end-expiratory alveolar pressures, which, for a given pulmonary elastance, depends on V_T size).³²⁵ In isolated lung with constant perfusion, PEEP augments edema formation probably because of increased filtration across extraalveolar vessels associated with lung overdistension.^{382,383} Conversely, in intact animals, PEEP does not affect edema formation, probably because of balancing between the PEEP-induced increase in end-inspiratory lung volume (which increases fluid filtration), and concomitant reduction in cardiac output and blood pressure (which reduce filtration pressure).^{384–386}

The role of PEEP-induced changes in hemodynamics on edema accumulation has been suggested by several authors.^{219,387} In an animal model, Dreyfuss and Saumon²¹⁹ showed that lung edema was reduced during ventilation with PEEP 10 cm H_2O , as opposed to ZEEP, despite identical end-inspiratory pressures. When the drop in arterial pressure produced by PEEP was corrected by dopamine infusion, pulmonary edema increased in direct proportion to systemic blood pressure. The increase in permeability edema was, however, less than that observed during ventilation at ZEEP, suggesting that hemodynamic modification is not the only factor explaining the effect of PEEP on edema formation.^{219,388} Indeed, reduction in driving pressure while keeping end-

inspiratory lung volume and pressure constant (as obtained by increasing PEEP while reducing V_T) may reduce edema and the severity of cellular damage; these findings suggest that a decrease in tissue stress may explain this protective effect of PEEP.^{217–220,389} The greater amount of alveoli available to clear edema, consequent to PEEP-induced alveolar recruitment, might also contribute.³⁸⁸

PEEP may also exert beneficial effects in localized lung injury. De Prost et al assessed the dispersion of localized alveolar flooding and alveolar protein permeability in rats undergoing local instillation of albumin solution followed by different ventilator strategies with varying levels of V_T and PEEP.³⁹⁰ High V_T ventilation caused alveolar liquid dispersion, which PEEP had prevented; this occurred even when V_T was kept constant, thus producing higher end-inspiratory pressures that did not exceed the safe limit of 30 cm H₂O. PEEP also reduced the increase in alveolar protein permeability caused by high V_T . In an animal model of unilateral lung injury, Schreiber et al reported that stepwise increases in PEEP, from zero up to a relatively moderate level of 13 cm H₂O, reduced the inflammatory response in the injured lung, did not result in detrimental effects on inflammation in the healthy lung, and preserved oxygenation.³⁹¹

PEEP-associated preservation of surfactant may further explain the benefit of PEEP on high-volume mediated VILI.²¹⁷ Thus, severity of overinflation is one major determinant of VILI.^{321,378,392} When ventilator-induced lung edema is produced by high V_T , the combination of PEEP and V_T reduction decreases the severity of injury at the same end-inspiratory pressure. When PEEP, however, is not accompanied by V_T reduction, excessive inspiratory pressures and additional overstretching occur, further increasing the rate of edema formation and tissue injury²¹⁹ and triggering systemic inflammation.³⁷⁸

In a mechanically heterogeneous lung, noninflating collapsed tissue is surrounded by open airspaces. Mead et al³⁷⁴ suggested that recruitment of nonaerated lung units induces a local stress to alveoli and bronchioles, which is substantially higher than the average transpulmonary or transbronchial pressure, respectively, because of alveolar and small airways interdependence.^{371,375} If this process recurs cyclically, high shear forces may damage the alveolar and airway epithelium, overstretch fragile microvessels, deplete surfactant, and initiate or worsen inflammation. By recruiting nonaerated portions, and stabilizing airways and lung units prone to repetitive opening and collapse, PEEP plays a key role in protecting the lung from the mechanical shear stress and tissue injury produced by ventilation at low end-expiratory lung volume (“atelectrauma”).^{373,374,393–398}

A large body of animal data supports the protective effect of PEEP on low-volume mediated VILI.^{217,218,338,373,389,393,395,398–402} By minimizing shear stress associated with cyclic opening and collapse, PEEP may attenuate inequalities in the regional distribution of tidal ventilation, thus avoiding or limiting overdistension in the less-injured lung zones.^{374,375} By increasing end-expiratory lung volume, PEEP can also prevent surfactant loss and preserve surfactant function.^{400,403–406} By minimizing mechanical stress, PEEP may also ease the intensity of ventilator-induced lung inflammation^{393,396,407} and remodeling,³⁹⁷ and reduce the decompartmentalization of a number of inflammatory mediators⁴⁰⁸ and bacteria from the lung into the circulation (“biotrauma”).^{409–411} These aforementioned mechanisms may play a role in initiating or propagating the systemic inflammatory response, which contributes to the multiple organ failure often observed in the terminal stage of ARDS.^{412–414} The local and systemic beneficial effects of PEEP are, however, dependent on the level of applied PEEP. Several authors, in fact, have suggested that higher-than-optimal levels of PEEP, defined as the PEEP required for maximum oxygenation and/or minimum elastance, may worsen lung mechanics, increase the amount of atelectasis and the lung-tissue expression of type III procollagen messenger RNA (an early marker of lung parenchyma remodeling), and also promote bacterial translocation.^{415,416}

Recent clinical trials show that protective ventilator strategies, reduced V_T (to limit the end-inspiratory stretch),^{6,369,370} and high PEEP (to avoid cyclic opening and closing)^{6,369} are associated with decreased pulmonary and systemic cytokine response,^{369,370} less organ dysfunction,⁴¹⁷ and reduced mortality in patients with ARDS,^{6,370} compared to conventional, injurious mechanical ventilation. Although the mechanisms whereby VILI causes organ dysfunction are not completely understood, a recent study suggests that cell apoptosis may be involved.⁴¹⁴ Rabbits ventilated with a low V_T and high PEEP showed less epithelial apoptosis in the kidney and small intestine than did rabbits ventilated with a high V_T and low PEEP.⁴¹⁴ The former rabbits had more apoptotic cells in the lung, whereas the latter rabbits had less lung apoptosis and necrosis of alveolar epithelial type III cells.⁴¹⁴ Analysis of plasma samples from a previous clinical trial³⁶⁹ revealed higher levels of soluble Fas ligand (a proapoptotic factor) in patients who received conventional ventilation than in patients who received protective ventilation; changes in soluble Fas ligand were correlated with changes in plasma creatinine.⁴¹⁷

Effects of Positive End-Expiratory Pressure on the Cardiovascular System

The lungs and heart are subject to variations in intrathoracic pressure. How alterations in intrathoracic pressure affect the heart and intrathoracic vessels varies substantially and depends on several factors, such as mechanical properties of the lung and chest wall, type of ventilation (spontaneous vs. mechanical), blood volume, and left-ventricular function. The complex interplay of these factors and the effects of PEEP on heart–lung interactions have been extensively reviewed.^{418–426}

Effect on Venous Return

Positive airway pressure causes a drop in cardiac output secondary to a decrease in cardiac filling (preload); this was initially attributed to a reduction in the pressure gradient for venous return, determined by the rise in right-atrial pressure consequent to increased intrathoracic pressure.^{50,66,427–430} The PEEP-mediated decrease in the gradient for venous return, however, is less than expected because PEEP produces a concomitant rise in mean systemic pressure⁴³¹ (the circulatory filling pressure representing the upstream pressure for venous return). In patients without lung disease undergoing implantation of defibrillator devices under general anesthesia, Jellinek et al⁴³² measured right-atrial pressure and mean systemic pressure at airway pressure zero and 15 cm H₂O during 15-second periods of apnea when ventricular fibrillation was induced to test the defibrillator. Rising airway pressure produced a drop in left-ventricular stroke volume.⁴³² Right-atrial and mean systemic pressure, however, increased equally, showing that the reduction in venous return was not determined by a decrease in pressure gradient.⁴³² The rise in mean systemic pressure may result from a reduction in vascular capacitance determined by neurovascular reflexes,⁴³³ displacement of blood from the pulmonary to the systemic circulation,⁴³⁴ and descent of the diaphragm, which increases the upstream pressure for venous return by augmenting intraabdominal pressure.⁴³⁵ These homeostatic adaptations, however, may be counteracted by a concomitant increase in venous resistance,^{431,436} suggesting that PEEP may alter venous return by affecting the peripheral venous circulation.

The effects of PEEP on hemodynamics strongly depend on intravascular volume. Cardiac output can be restored by increasing the ventricular filling through volume infusion.^{47,437–440} PEEP discontinuation produces a rise in cardiac filling, proportional to the pressure withdrawn and the circulating blood volume.^{47,441} In hypervolemic and hemodynamically stable patients who underwent cardiac surgery, van der Berg et al found that maintaining airway pressure up to 20 cm H₂O for 25 seconds produced minimal variations in right ventricular output despite a concomitant rise in right atrial pressure.⁴³⁵

In summary, PEEP reduces cardiac output through a decrease in venous return that is not a primary consequence of a decrease in its pressure gradient. Regardless of the cause, the drop in cardiac output can be counteracted by blood volume expansion.

Effect on Right-Ventricular Afterload and Ventricular Interdependence

PEEP alters both left- and right-ventricular configurations and reduces left-ventricular diastolic compliance by augmenting right-ventricular afterload.^{49,442–444} An increase in lung volume causes a rise in pulmonary vascular resistance by directly compressing the alveolar vessels.⁴⁴⁵ The interconnections between lung volume and pulmonary blood flow are not straightforward, because blood expelled from the alveolar vessels can be retained in the extraalveolar vessels.^{446,447} When airway pressure is augmented, right-ventricular outflow impedance (i.e., afterload) is also increased.^{426,448} In patients with ARDS, high levels of PEEP can cause or worsen tricuspid regurgitation.⁴⁴⁹

The fall in cardiac output with PEEP has been also attributed to the stress exerted by the right ventricle on the interventricular septum,⁴⁴³ which is displaced leftward and restricts left-ventricular filling.⁴⁹ Culver et al⁴⁵⁰ altered right-ventricular afterload, through a partial occlusion of the main pulmonary arteries, and found that this did not produce the same hemodynamic changes induced by increases in lung volume; these findings suggest that ventricular interdependence (caused by an increased right-ventricular afterload) was unlikely to be the main mechanism for PEEP-mediated fall in cardiac output. Wise et al⁶⁹ evaluated the impact of PEEP 15 cm H₂O on left-ventricular compliance in a study where they bypassed the right heart to exclude the effects of interventricular interdependence; they concluded that the elevation in left-ventricular filling pressure was mainly consequent to pericardial compression. In patients with ARDS, Dhainaut et al⁴⁴⁰ did not find any changes in ventricular diastolic compliance and ejection fraction when PEEP was increased up to 20 cm H₂O during controlled ventilation; they attributed the reduction in cardiac output mainly to the preload effect.

More recently, Gernoth et al reported that an open-lung strategy with high PEEP (20 cm H₂O), which improved oxygenation without significantly affecting PaCO₂, caused no major hemodynamic alteration, while right- and left-ventricular function, respectively, were improved and unchanged.⁴⁵¹ An impairment in right-ventricular function and hemodynamics, however, has been reported when PEEP is increased at constant plateau pressure, with significant worsening of respiratory acidosis despite improvement in oxygenation.⁴⁵²

In summary, PEEP increases pulmonary vascular resistance and right-ventricular afterload and may therefore increase the stress exerted by the right ventricle, which has to maintain an adequate output to guarantee left-ventricular filling. Right-ventricular systolic overload may result in leftward displacement of the interventricular septum; the effect on left-ventricular function seems to be adjunctive, rather than a major determinant of the PEEP-mediated reduction in cardiac output.

Impact of Respiratory Mechanics on the Hemodynamic Effect of Positive End-Expiratory Pressure

A reduction in lung compliance reduces the transmission of airway pressure to the pericardial space,^{453,454} and to the pleural space. The transmission of PEEP to the pleural space increases when chest wall compliance is reduced⁴⁵⁵ and decreases when chest wall compliance is increased.⁴⁵⁴ Venus et al⁴⁵⁶ found that the fraction of PEEP transmitted to the pleura was reduced from 62% to 34% and to the pericardium from 54% to 36%, in intact and acutely injured

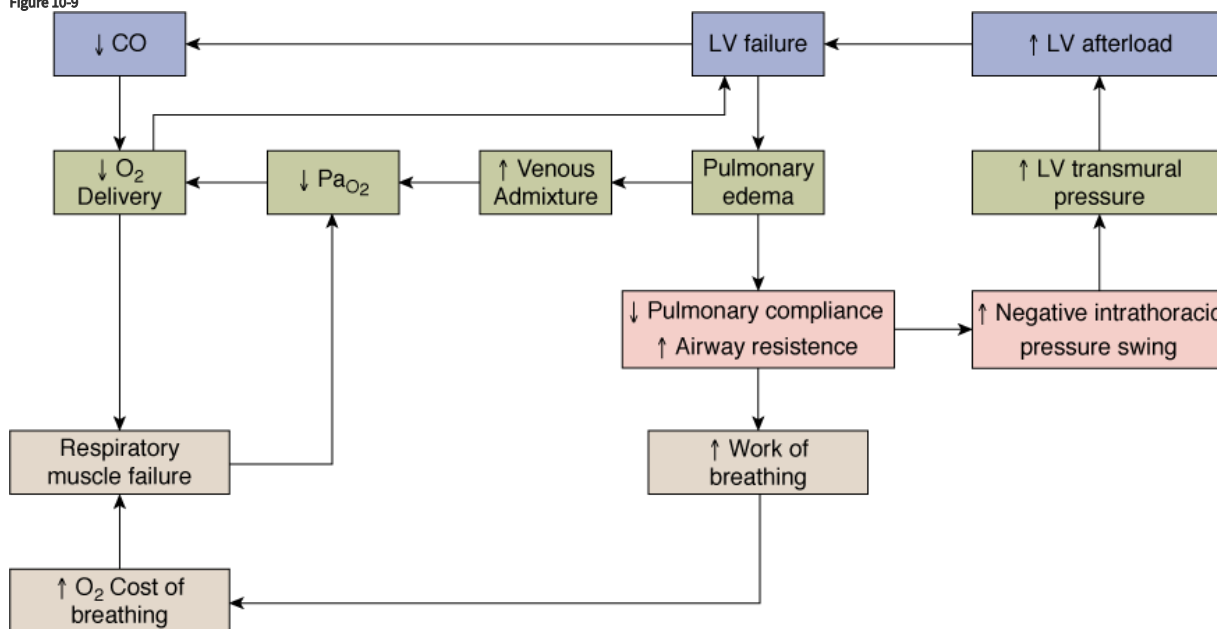
lungs, respectively. V_T was kept constant throughout the study and the reduction in cardiac output produced by PEEP did not differ before and after inducing lung injury, because absolute pericardial pressure at end-expiration did not differ.⁴⁵⁶

Effect on Afterload

Left-ventricular afterload is the force opposing contraction. It corresponds to the tension developed by the contracting cardiac muscle. It is determined by both the systemic arterial resistance and the transmural pressure exerted on the left-ventricular wall; that is, the difference between the systolic pressure and the pressure surrounding the heart (i.e., intrathoracic pressure). A reduction in left-ventricular afterload is achieved either by decreasing systemic arterial resistance (through vasodilator administration)⁴⁵⁷ or increasing intrathoracic pressure.^{69–71} In healthy subjects with normal cardiac function the dominant action of increase in intrathoracic pressure is reduction in venous return; the consequences of the lowered transmural pressure are rather small (see Fig. 10-2A). In patients with poor left-ventricular function and congestive heart failure, cardiac output is relatively insensitive to reduction in venous return because left-ventricular filling pressure and diastolic volume are elevated.⁶⁷ Thus, the net effect of a rise in intrathoracic pressure is a reduction in left-ventricular transmural pressure (see Fig. 10-2B).^{68,458,459} Conversely, a decrease in intrathoracic pressure raises afterload by augmenting left-ventricular transmural pressure (see Fig. 10-2B).^{332,460}

When afterload exceeds the capacity of the left ventricle, pulmonary edema ensues, establishing a vicious circle (Fig. 10-9). Because of worsening pulmonary compliance^{332,460,461} and resistance,^{460,462–464} the inspiratory muscles must exert a stronger effort to achieve adequate alveolar ventilation. Thus, inspiratory intrathoracic pressure becomes more negative, and afterload increases. The combination of reduced O_2 delivery⁴⁶⁵ and increased metabolic cost of breathing^{464–466} may precipitate respiratory muscle failure,⁴⁶⁶ which further worsens gas exchange.

Figure 10-9

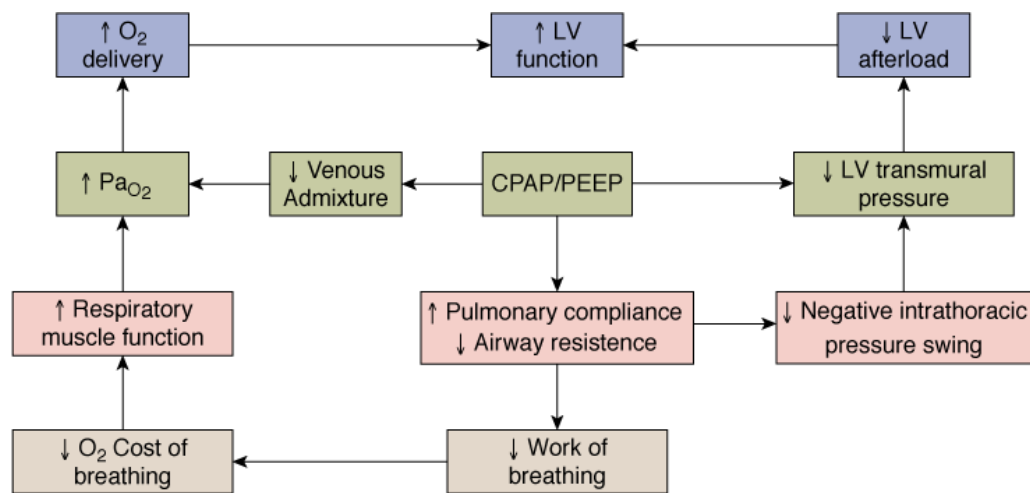


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Pathophysiologic mechanisms and pathways of cardiogenic pulmonary edema. Worsening in pulmonary mechanics increases negative swings in intrathoracic pressure and left-ventricular afterload. Worsening in gas exchange decreases oxygen delivery to the heart and the respiratory muscle. See text for further explanation. CO, Cardiac output; LV, left ventricle; O_2 , oxygen; Pa_{O_2} , partial pressure of arterial oxygen.

The impact on left-ventricular transmural pressure of the increase in intrathoracic pressure produced by CPAP 10 cm H_2O during expiration is rather small (see Fig. 10-2C). Conversely, the improvement in pulmonary mechanics⁴⁶⁶ generated by the same CPAP may minimize the swings in intrathoracic pressure^{466,467} and hence decrease afterload,^{467,468} thereby improving left-ventricular function (see Figs. 10-2C and 10-10).^{72,468} CPAP may simultaneously improve gas exchange^{72,73,146} by reducing venous admixture^{10,146} and avert failure of the respiratory muscles by improving their O_2 balance¹⁰ (Fig. 10-10).

Figure 10-10



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Cardiopulmonary effects of CPAP in cardiogenic pulmonary edema. By improving pulmonary mechanics, CPAP reduces negative swings in intrathoracic pressure and left-ventricular afterload. By improving gas exchange CPAP increases [oxygen](#) delivery to the heart and the respiratory muscle. See text for further explanation. LV, left ventricle; O₂, [oxygen](#); PaO₂, partial pressure of arterial [oxygen](#).

Effect on Oxygen Delivery

The effect of PEEP on O₂ delivery depends on its relative effects on cardiac output and arterial [oxygen](#) content: in other words, on the balance between the pulmonary and hemodynamic consequences of PEEP. PEEP-induced increase in PaO₂ and arterial [oxygen](#) content may be accompanied by a decrease in O₂ delivery, because of concomitant decrease in cardiac output.⁴⁶ The impact of PEEP on O₂ delivery depends primarily on its hemodynamic effect. By decreasing cardiac output and consequently O₂ delivery, PEEP may worsen O₂ balance and promote pathologic supply-dependency of O₂ delivery.^{430,469–472}

The effect of protective ventilation on O₂ delivery was evaluated in patients with ARDS.^{473–475} In forty-eight patients with severe ARDS, Carvalho et al⁴⁷⁴ found that PEEP up to 24 cm H₂O (average: 16 cm H₂O) produced an increase in cardiac output, heart rate, and O₂ delivery, and decreased systemic vascular resistance and plasma lactate. The authors speculated that the good tolerance of high PEEP may have been a consequence of the acute hemodynamic effects of hypercapnia and the observed very low lung compliance.⁴⁷⁴ No correlation was found between changes in PEEP and cardiac output. A strong inverse correlation existed between changes in plateau pressure and cardiac output. These findings suggest, in line with other reports,^{202,318} that the negative hemodynamic effect classically attributed to PEEP may depend more on the associated high inspiratory pressure (if V_T is not concomitantly reduced).⁴⁷⁴

Noncardiorespiratory Effects of Positive End-Expiratory Pressure

Renal and Hormonal Effects

PEEP decreases urinary output,^{47,54,55,476–500} sodium excretion,^{55,476–478,480–482,484–490,493,500} and creatinine clearance.^{476,480,481,484,490} Although there is general consensus about these effects of PEEP, studies investigating the mechanisms are conflicting. Discrepancies are related to differences in experimental design, such as diverse subjects, trial duration, type and severity of underlying disorder, volemic state, use of drugs and anesthetics (which commonly cause sympathetic depression), and level of PEEP. The decrease in urinary output caused by PEEP has been attributed to multiple factors, such as fall in cardiac output^{54,477,491} and renal blood flow,^{477,479,487,491,495} reduced intravascular volume,⁵⁴ reflex sympathetic nerve activation,⁴⁹⁷ and altered release of hormones including catecholamines,⁴⁷⁶ renin–angiotensin–aldosterone system,^{476,488,498–500} antidiuretic hormone,^{476,488} and atrial natriuretic factor.^{485,486,488–490,493}

In dogs receiving a constant fluid infusion, Berry et al⁵⁵ observed that the time needed for urinary output to approximate the rate of fluid infusion was 20, 27, and 46 hours during spontaneous breathing, positive-pressure ventilation and positive-pressure ventilation plus PEEP 10 cm H₂O, respectively. Priebe et al⁵⁴ found that renal impairment caused by PEEP 10 cm H₂O in anesthetized dogs primarily resulted from a reduction in intravascular volume. In swine, Venus et al⁴⁷⁶ found that the addition of PEEP to positive-pressure ventilation caused a fall in cardiac and urinary output, associated with increase in plasma antidiuretic hormone, renin, [epinephrine](#), and norepinephrine in normovolemic animals, but not in animals that received volume expansion through crystalloids. In six neurologic patients with intact cardiopulmonary and renal function who were receiving positive-pressure ventilation plus PEEP 15 cm H₂O, an increase in lower body pressure via military antishock trousers improved systemic and renal hemodynamics and decreased plasma norepinephrine, with

no change in total blood volume.⁴⁹⁹ Urinary output and sodium excretion did not improve. The authors concluded that increased plasma renin and sympathetic activation were the main determinants of renal impairment with PEEP.⁴⁹⁹

Annat et al⁴⁷⁷ evaluated the short-term effects of PEEP 10 cm H₂O in normovolemic patients without cardiac and renal abnormalities and little or no respiratory failure who were receiving positive-pressure ventilation. PEEP decreased urinary output by 34%, renal blood flow by 32%, and sodium excretion by 33%; the associated drop in cardiac output was 15%.⁴⁷⁷ PEEP also increased urinary antidiuretic hormone and plasma renin and aldosterone.⁴⁷⁷ In a similar setting, Payen et al⁴⁹¹ found that PEEP 15 cm H₂O reduced urinary output (55%) and fractional sodium excretion (39%); cardiac index fell by 21%. Plasma antidiuretic hormone did not vary, and norepinephrine increased.⁴⁹¹

PEEP increases sympathetic activity.⁵⁰¹ Nevertheless, the observation by Fewell et al⁴⁹⁴ that the effects of PEEP on renal function are mediated by sympathetic activity has not been confirmed.^{481,492} PEEP decreases plasma atrial natriuretic factor.^{478,485,486,488,493} This may contribute to the reduction in urinary output and sodium excretion, probably as a result of the reduced atrial transmural pressure consequent to the raised intrathoracic pressure.⁴⁸⁵

In summary, PEEP may alter renal function by reducing cardiac output and renal blood flow to an extent that depends on the volemic state and on the amount of applied pressure, although other mechanisms, including several hormones, may play a role.

Effects on Splanchnic Circulation and Oxygenation

PEEP decreases splanchnic blood flow^{43,57,58,502–504} and causes hepatic congestion.⁵⁰³ The reduction in splanchnic blood flow is consequent to a fall in cardiac output secondary to impaired central hemodynamic^{43,57,58,502–504}; it is less pronounced when cardiac output is maintained either by expanding blood volume^{57,58} or inotropic drugs.⁵⁰⁴ In hemodynamically stable patients with ARDS, Kiefer et al⁵⁰⁵ found that PEEP 13 cm H₂O (5 cm H₂O increment above the preset values), did not decrease cardiac index or splanchnic blood flow and metabolism, suggesting that PEEP does not affect splanchnic blood flow. Other factors, however, such as hepatic outflow resistance, may play a role.^{58,506} In swine, Brienza et al⁵⁰⁶ found that PEEP 15 cm H₂O reduced portal vein flow through an increase in liver venous resistance; the authors postulated a direct compressive effect caused by diaphragmatic descent on the liver.

Sha et al⁵⁰² applied increasing PEEP (up to 20 cm H₂O) and found proportional decreases in hepatic blood flow. Because the rate of decrease in hepatic O₂ delivery exceeded that of cardiac output and hepatic blood flow, the authors concluded that hepatic O₂ delivery was reduced primarily because of the drop in portal venous O₂ content.⁵⁰² Aneman et al⁵⁰⁷ evaluated the impact of PEEP 10 cm H₂O on mesenteric and hepatic blood flow in patients under general anesthesia (during elective surgery for gastric or pancreatic neoplasm). Neither arterial pressure nor arterial, portal, and hepatic venous norepinephrine were significantly affected by PEEP.⁵⁰⁷ A decrease in portal blood flow was associated with a rise in mesenteric vascular resistance; conversely, an increase in hepatic arterial flow was associated with a drop in hepatic arterial resistance.⁵⁰⁷ Mesenteric O₂ delivery was reduced and hepatic O₂ delivery remained unchanged.⁵⁰⁷ Mesenteric and hepatic O₂ did not significantly vary with PEEP.⁵⁰⁷ In similar patients, Berendes et al⁴³ found that hepatic venous lactate did not change with PEEP up to 15 cm H₂O, and concluded that a critical reduction in splanchnic oxygenation did not occur.

In spontaneously breathing normal subjects, Fournell et al⁵⁰⁸ reported that CPAP levels up to 10 cm H₂O progressively altered gastric mucosal microvascular O₂ saturation, as assessed by reflectance spectrophotometry. Lehtipalo et al⁴⁴ found that PEEP 10 cm H₂O had little effect on intestinal perfusion pressure; mesenteric blood flow and oxygenation were maintained until the intestinal perfusion pressure exceeded 33 mm Hg. In patients with septic shock, Träger et al⁵⁰⁹ increased PEEP up to 15 cm H₂O. PEEP decreased cardiac index and hepatic venous O₂ saturation.⁵⁰⁹ Hepatic metabolic performance, assessed by glucose production, fell at each PEEP level in patients who did not survive, whereas it decreased only at the highest PEEP among survivors.⁵⁰⁹ In 141 unselected critically ill patients, PEEP greater than 5 cm H₂O during mechanical ventilation was found to be a risk factor for liver dysfunction, as defined by serum bilirubin equal to or greater than 2 mg/dL for at least 48 hours.⁵¹⁰

In sixty-five patients who underwent liver transplantation, Saner et al found in the immediate postoperative period that application of PEEP up to 10 cm H₂O did not affect venous liver outflow, portal vein and hepatic artery flow, and systemic hemodynamics.⁵¹¹ In a retrospective observation study, Kocabayoglu et al reviewed the records of 50 patients who underwent mechanical ventilation for more than 7 days because of pulmonary complications following liver transplantation and found that PEEP equal to or greater than 10 cm H₂O did not harm liver-graft function.⁵¹²

In summary, PEEP reduces hepatic and splanchnic blood flow when determining a drop in cardiac output, although other factors, such as regional outflow resistance or derangements in microcirculation consequent to sepsis, are likely to play a role.

Effects on Intracranial Pressure and Cerebral Perfusion

High intracranial pressure (ICP) and reduced cerebral perfusion pressure (the difference between mean arterial pressure and ICP) are common in several acute neurologic and neurosurgical intensive care unit patients, particularly those who have focal or generalized cerebral edema. Hypoxemic respiratory failure secondary to ARDS is a fearful complication in these patients leading to use of PEEP.⁵¹³

ICP may be altered by PEEP through a rise in right-atrial pressure, which increases superior vena cava pressure and hence reduces cerebral venous return.^{53,514,515} The effects of PEEP on ICP depend on the amount of intracranial compression, and are of little importance when cerebral compliance is not altered.⁵² In a study evaluating twenty-five head-injured patients, Apuzzo et al⁵² found that PEEP caused an increase in ICP in only twelve patients who had low cerebral compliance, and reduced cerebral perfusion pressure below 60 mm Hg in only six of the twelve patients.⁵² In normal healthy volunteers, Hormann et al⁵¹⁶ found that alterations in ICP and cerebral perfusion pressure produced by CPAP 12 cm H₂O were slight and not of clinical relevance. Body position also affects the impact of PEEP on ICP. PEEP raises ICP through an increase in the downstream venous pressure; thus, elevating the head above the chest should lessen the transmission of intrathoracic pressure to the venous sinuses and mitigate the effects of PEEP on ICP. In supine anesthetized patients undergoing posterior fossa surgery, Lodrini et al⁵¹⁷ found that PEEP up to 15 cm H₂O caused proportional increases in central venous pressure and ICP; in the sitting position, PEEP also increased central venous pressure, but ICP changed little in most of the patients.

Muench et al found that increases in PEEP up to 25 cm H₂O in healthy pigs did not produce any adverse effect on ICP, cerebral blood flow, or tissue oxygenation. On the contrary, in patients with subarachnoid hemorrhage, reduction in mean arterial pressure consequent to application of PEEP up to 20 cm H₂O caused a significant decrease in cerebral blood flow attributable to the altered cerebrovascular autoregulation.⁵¹⁸ Caricato et al studied the effects of increasing PEEP up to 12 cm H₂O in twenty-one patients with severe head injury or subarachnoid hemorrhage and who were receiving mechanical ventilation; static compliance of the respiratory system was normal in thirteen patients and reduced in the other eight.⁵¹⁹ The authors found that PEEP caused a decrease in mean arterial pressure and cerebral perfusion pressure in the first subgroup of thirteen patients, whereas it did not produce any significant change in the eight patients with reduced respiratory compliance.⁵¹⁹ In twelve severely brain-injured patients with ARDS and high ICP, Mascia et al randomly applied 5 and 10 cm H₂O of PEEP; recruited volume and elastance were used to classify patients as recruiters ($n = 6$) and nonrecruiters ($n = 6$).⁵²⁰ PaCO₂, ICP, Doppler flow velocity, and jugular saturation increased significantly with higher PEEP in the nonrecruiters, but did not change in the recruiters.⁵²⁰

Several studies in neurologic, neurosurgical, or brain-injured patients found that judicious use of PEEP was not detrimental.^{521–526} As PEEP may affect cerebral circulation, adequate monitoring of cerebral and respiratory function should be used to titrate its application.⁵²⁷

Effects on Bronchial Circulation and Thoracic Lymph Drainage

By increasing intrathoracic pressure, PEEP alters bronchial blood flow^{528–530} and thoracic lymph drainage, which could affect the process of lung repair and edema removal.⁵³¹ Blood from the bronchial vessels is drained to the right side of the heart via the azygos and bronchial veins and to the left atrium through anastomoses between systemic and pulmonary circulation. In ventilated open chest dogs, Baile et al⁵²⁸ investigated the effect of increasing PEEP on systemic to pulmonary (anastomotic) flow (Qbra) and on bronchial blood flow (Qbr), which was further partitioned into tracheal, bronchial, and parenchymal fractions. Bronchial and parenchymal, but not tracheal, fractions of Qbr decreased with PEEP.⁵²⁸ Qbr and Qbra did not differ at PEEP 3 and 10 cm H₂O.⁵²⁸ At PEEP 15 cm H₂O, Qbr exceeded Qbra, indicating a decrease in the anastomotic drainage.⁵²⁸ In patients undergoing cardiopulmonary bypass, Agostoni et al⁵³⁰ observed that an increase in alveolar pressure of approximately 10 cm H₂O decreased Qbra by 40%.

Local release of mediators, rise in pulmonary vascular resistance, and reflex bronchial vasoconstriction may explain Qbra reduction.⁵²⁹ To assess the influence of vagal reflexes on PEEP-mediated reduction in Qbra, Lakshminarayan et al⁵³² isolated and perfused the left lower lobe in open chest dogs, and found that increasing PEEP from 5 to 15 cm H₂O halved Qbra. At PEEP 5 cm H₂O, bilateral cervical vagal cooling decreased Qbra.⁵³² An increase of PEEP to 15 cm H₂O did not further diminish it, suggesting that the effect of PEEP on Qbra might be partly vagally mediated.⁵³²

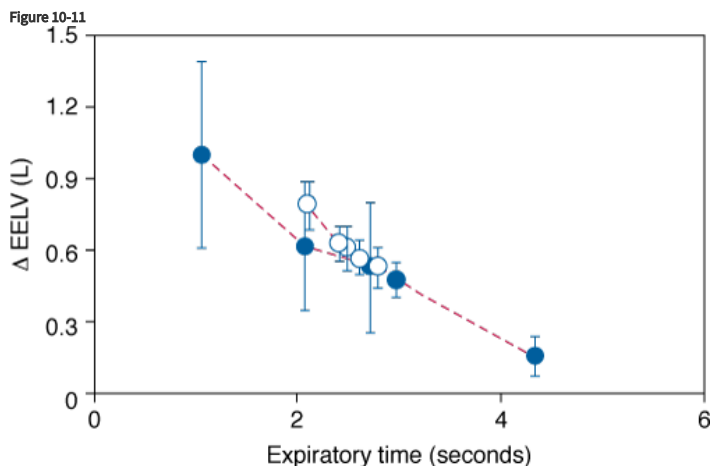
PEEP may obstruct lymph drainage from the thoracic duct into the jugular vein.^{531,533} By altering thoracic lymph return, PEEP might affect edema removal from the lungs.⁵³¹

Intrinsic Positive End-Expiratory Pressure

During passive exhalation, airflow is driven by the difference between alveolar and airway opening pressures and opposed by expiratory resistance. In absence of expiratory muscle activity and externally applied PEEP, alveolar pressure corresponds to the elastic recoil of the respiratory system for a given V_T: alveolar pressure is thus equal to respiratory system elastance/V_T or V_T/respiratory system compliance. The product of respiratory system compliance and airway expiratory resistance is the expiratory time constant (τ). During expiration, $V = V_T e^{-(t/\tau)}$ where V is the amount of V_T not yet exhaled (i.e., above FRC),

t is the time elapsed from the onset of expiration, and e is the base of natural logarithms (2.7189). When $t = \tau$, V is roughly 37% ($1/2.7189$) of V_T . It takes five τ to almost entirely (99%) exhale V_T .

In healthy adult subjects, τ is approximately 0.3 second. Therefore, the time required to exhale V_T is approximately 1.5 seconds. Assuming a respiratory rate of 12 breaths/min and an inspiratory duty cycle (T_I/T_{TOT}) of 40%, expiratory time (T_E) is 3 seconds, which considerably exceeds the time necessary for the lungs to deflate to FRC. If either breathing frequency or T_I/T_{TOT} increase, T_E diminishes. Any condition causing an increase in expiratory resistance, including an artificial airway,⁵³⁴ or a loss in elastic recoil prolongs τ and thus the time required for complete V_T exhalation. When shortened T_E and/or prolonged τ impede a complete exhalation to FRC, dynamic hyperinflation occurs. End-expiratory relaxation volume will exceed FRC to an extent that is proportional to τ and V_T and inversely related to T_E (Fig. 10-11).



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Effects of varying expiratory time on end-expiratory trapped volume in patient with airway obstruction. $\Delta EELV$ indicates end-expiratory lung volume above functional residual capacity. The shorter the expiratory time, the greater the amount of end-expiratory trapped volume. Data are expressed as mean $\pm$ standard deviation (*closed circles*) and mean $\pm$ standard error (*open circles*), and taken from references 237 and 359, respectively. Tidal volume was always 1 L in reference 237, and averaged 0.576 L (± 0.026) in reference 359. In reference 359, changes in expiratory time were obtained by manipulating the ventilator preset inspiratory time, while leaving respiratory rate (and hence minute ventilation) unmodified. In reference 237, inspiratory duty cycle remained unchanged and expiratory time was changed by modifying respiratory rate; accordingly minute ventilation also varied.

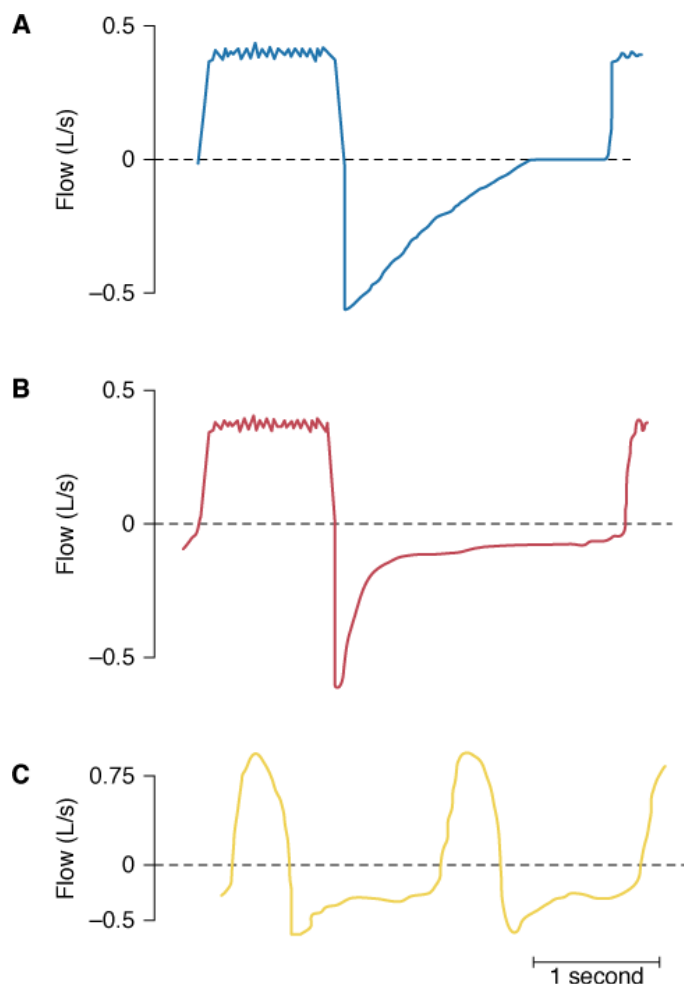
Expiratory flow may also be limited in the course of V_T exhalation because the airways collapse at a choke point where intrathoracic pressure equals intrabronchial pressure (so-called equal pressure point). The airways located downstream of the equal pressure point are then compressed because intrathoracic pressure exceeds the intrabronchial pressure. Thus, alveolar pressure–airway opening pressure no longer represents the pressure driving expiratory flow. Expiratory flow limitation is very common in patients with an acute exacerbation of COPD.^{238,535–538} It has also been reported in patients with asthma,^{539,540} obesity,^{271,541} and ARDS.^{183,184,535,542}

Regardless the underlying mechanism, after a few breaths a steady state is achieved that depends on the variations in the different components of respiratory mechanics. For a certain trapped volume, the amount of PEEPi is determined by the corresponding respiratory system compliance.

Measurement of Intrinsic Positive End-Expiratory Pressure

PEEPi can be detected qualitatively by observing the flow-time curve, available on most ventilators. In absence of PEEPi, expiratory flow is zero before the onset of the subsequent inspiration (Fig. 10-12A). The persistence of expiratory flow at end-expiration indicates the presence of PEEPi, during either mechanical ventilation (Fig. 10-12 B) or spontaneous breathing (Fig. 10-12C).

Figure 10-12



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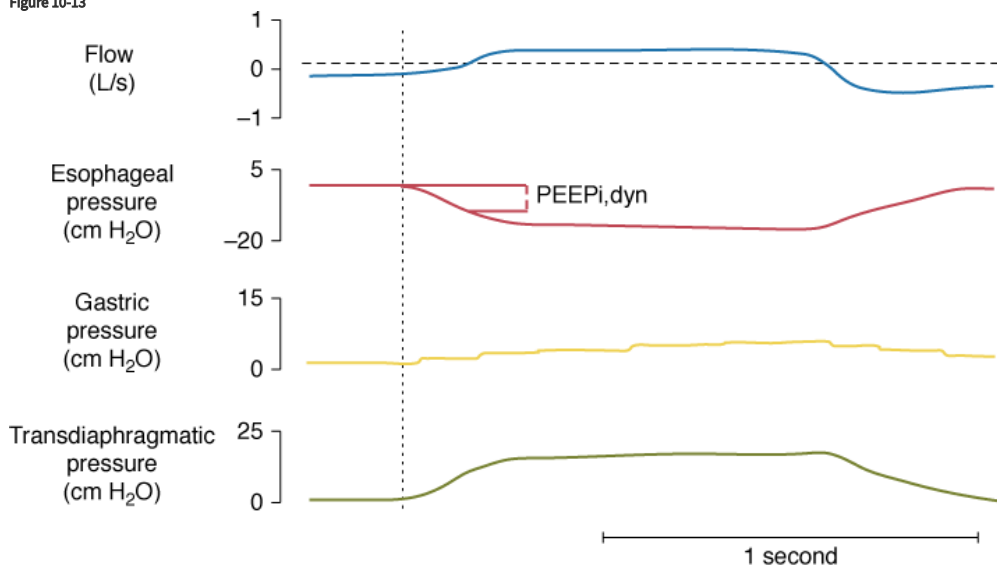
Flow-time tracings during controlled ventilation in a normal anesthetized subject (A), in a patient with severe COPD during controlled ventilation (B) and during spontaneous breathing (C). In the normal subject (A), expiratory flow reaches zero before expiration ends. In the patient with COPD, expiratory flow remains below zero value before the subsequent inspiration, denoting dynamic hyperinflation, during both controlled ventilation and spontaneous breathing.

Because the ventilator built-in pressure gauge is open to atmosphere, PEEPi cannot be detected by simply inspecting airway pressure. During controlled ventilation, PEEPi can be measured by occluding the airway opening at end-expiration for a few seconds^{59,543} (see Fig. 10-4), the end-expiratory occlusion maneuver. This value is static PEEPi (PEEPi,stat). Unless some alveolar units are not at all communicating with the central airways (because the corresponding peripheral airways are completely occluded),⁵⁴⁴ PEEPi,stat represents the average pressure in the different lung regions.^{9,238,545-547} A second approach is to measure the difference between preset PEEP and airway pressure at the onset of inspiratory flow^{260,546,547}; this value is dynamic PEEPi (PEEPi,dyn). PEEPi,dyn can be considered as the lowest regional end-expiratory alveolar pressure that has to be overcome to begin inspiratory flow.^{9,238,545-547} PEEPi,dyn and PEEPi,stat approximate each other, in animals⁵⁴⁶ and in human subjects⁵⁴⁷ in the absence of severe airway disorders, when PEEPi is consequent to T_E shortening and/or increasing minute ventilation (VE). In animals receiving high doses of aerosolized methacoline⁵⁴⁶ and patients affected by severe airway obstruction,⁵⁴⁷ PEEPi,dyn is considerably lower than PEEPi,stat. A low ratio of PEEPi,dyn to PEEPi,stat suggests time constant inequalities within the lung and high lung tissue viscoelastic pressure losses.^{183,546-549}

When the inspiratory muscles are contracting, assessing PEEPi is more complex. When inspiration is active and expiration relaxed, PEEPi,dyn can be measured with a balloon-tipped esophageal catheter.^{2,545,550,551} PEEPi,dyn equals the difference in esophageal (pleural) pressure between the onset of negative deflection (indicating the start of the inspiratory effort), and the pressure at the transition between expiratory and inspiratory flow (Fig. 10-13).^{2,545,551} When the expiratory muscles are contracting, which is common in flow-limited patients,⁵⁵²⁻⁵⁵⁵ especially during partial ventilator support,^{2,554,556,557} the pressure developed in the abdomen is transmitted through the relaxed diaphragm and raises intrathoracic pressure. Consequently, assessing PEEPi via esophageal pressure will overestimate the true end-expiratory elastic recoil pressure.^{2,552-558} With a second balloon-tipped catheter in the stomach, it is possible to measure, and correct for, the amount of intrathoracic pressure rise caused by expiratory muscles contraction^{2,554,556}; briefly, the expiratory rise in gastric pressure is subtracted from the esophageal pressure at the onset of the subsequent inspiration (Fig. 10-14).^{554,556} Another

method is to subtract the rise in gastric pressure, during an end-expiratory occlusion, from airway pressure.⁵⁵⁹ Partitioning the rise in intrathoracic pressure from recording of airway, esophageal and gastric pressures is complex, and infrequently performed in the clinical setting.

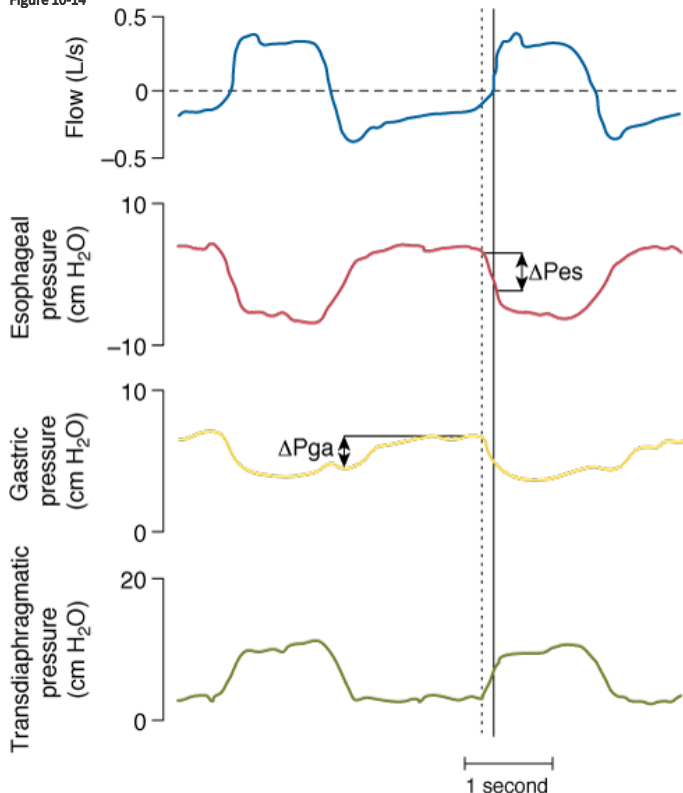
Figure 10-13



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Measurement of dynamic intrinsic PEEP (PEEPi,dyn) in a spontaneously breathing patient without expiratory muscle activity. The *dashed horizontal line* indicates zero flow. The *dotted vertical line* indicates the onset of the inspiratory effort. The *solid vertical line* indicates the start of inspiratory flow. Gastric pressure rises during inspiration and descends during expiration, indicating that expiratory muscles were not active. PEEPi,dyn is the difference in esophageal pressure between onset of inspiratory effort and start of inspiratory flow.

Figure 10-14



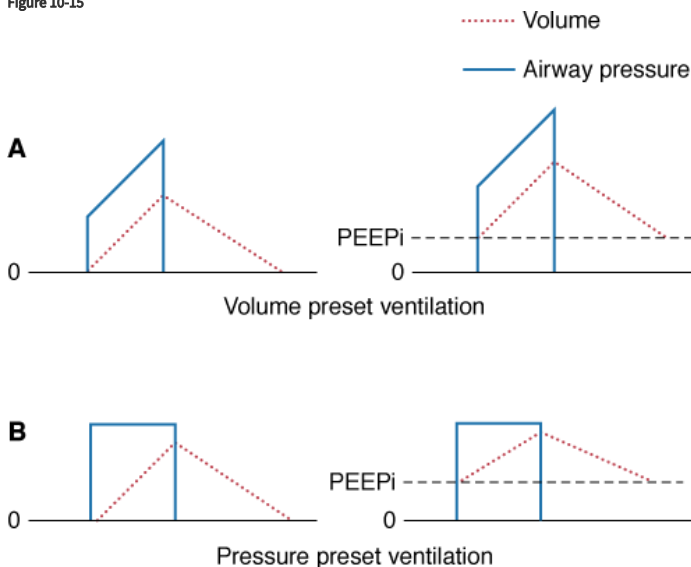
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Measurement of dynamic intrinsic PEEP (PEEPi,dyn) in a spontaneously breathing patient with expiratory muscle activity. The *dashed horizontal line* indicates zero flow. The *dotted vertical line* indicates onset of inspiratory effort. The *solid vertical line* indicates the start of inspiratory flow. Unlike in Figure 10-13, gastric pressure rises during expiration (ΔPga) and descends at the following inspiration, indicating expiratory muscle recruitment. ΔPga must be subtracted from the difference in esophageal pressure between onset of inspiratory effort and start of inspiratory flow (ΔPes) to determine PEEPi,dyn.

Physiologic Effects of Intrinsic Positive End-Expiratory Pressure

During volume-targeted controlled ventilation PEEPi does not affect V_T , whereas it elevates peak airway pressure, which is a dependent variable (i.e., it varies depending on the impedance of the respiratory system) (Fig. 10-15A). Conversely, during pressure-targeted controlled ventilation, the independent variable is the preset positive pressure applied to the airway during inspiration, the presence of PEEPi decreases the actual lung distending pressure and therefore V_T (Fig. 10-15B).

Figure 10-15



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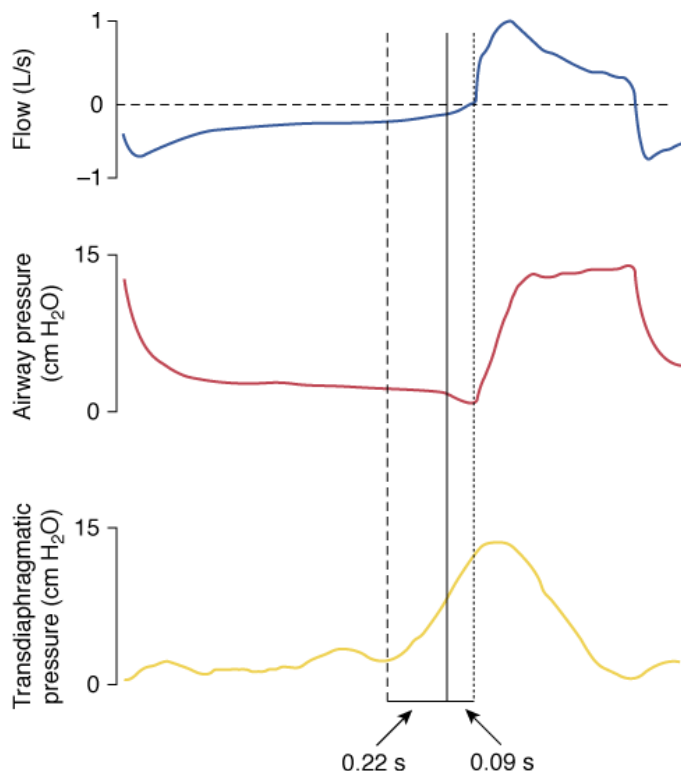
Simulation of the effects of PEEPi on volume-targeted (A) and pressure-targeted (B) controlled ventilation. Volume and airway pressure are indicated by dotted and solid lines, respectively. In the left panels, when PEEPi is absent, tidal volume is the same for both modes. In the upper (A) right panel, during volume-targeted ventilation, PEEPi cause a rise in airway pressure, while preset tidal volume is unchanged. During pressure-targeted ventilation, lower (B) right panel, preset pressure does not vary, and, consequently, tidal volume falls.

PEEPi resembles in several respects external PEEP. In patients with hypoxemic ARF consequent to ARDS or cardiogenic pulmonary edema, an increase in end-expiratory lung volume, produced by T_E shortening, secondary to tachypnea or inverse-ratio ventilation, can decrease intrapulmonary shunt as much as can external PEEP.⁵⁶⁰ In patients without COPD, Brandolese et al¹⁹² found that external PEEP produced more consistent increases in Pa_{O_2} than did an equivalent level of PEEPi. The authors attributed the less-favorable impact of PEEPi on Pa_{O_2} to less-homogeneous gas distribution between different lung units.¹⁹²

PEEPi decreases cardiac output^{59,192,238,560} and blood pressure^{59,238} to an extent similar to that produced by external PEEP for the same increase in lung volume.⁵⁶⁰ Connery et al⁵⁶¹ reported fatal cardiac arrest and life-threatening hypotension during manual ventilation with large V_T and high respiratory frequencies in two patients with severe COPD. By increasing alveolar pressure, PEEPi may also cause barotrauma.⁵⁶²

In spontaneously breathing patients, PEEPi poses a threshold load that has to be overcome before inspiratory flow can be initiated;^{9,545,550,563} this load is perceived as inspiratory difficulty.⁵⁶⁴ With any mode of partial ventilator support, this threshold must be overcome before the ventilator can be triggered (Fig. 10-16).^{2,61,63,551,565–568} Inspiratory efforts that are not intense enough to overcome PEEPi fail to trigger the ventilator,^{63,64,569–571} a phenomenon referred to as *wasted or ineffective inspiratory efforts* (Fig. 10-17). Wasted efforts occur with all triggered modes,⁵⁷⁰ and are more frequent at higher ventilator assistance^{63,64,570} because of higher PEEPi, consequent to increased lung volume,^{63,64,570} and diminished respiratory drive.⁵⁷⁰

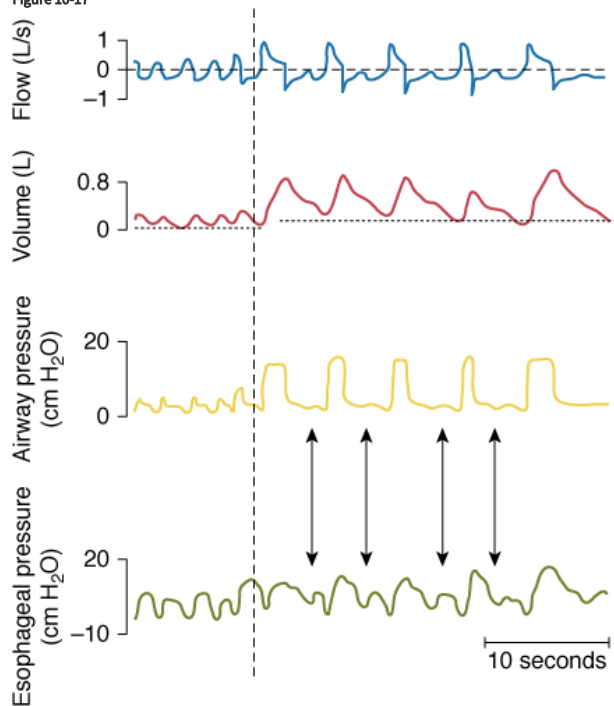
Figure 10-16



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Effect of intrinsic PEEP (PEEPi) on ventilator triggering. The *dashed horizontal line* indicates zero flow. The *dashed vertical line* on the left indicates the onset of the inspiratory effort. The *dotted vertical line* on the right corresponds to the start of inspiratory flow. The *solid vertical line* indicates the point at which the ventilator is triggered and the mechanical assistance initiated. The amount of effort spent in overcoming PEEPi corresponds to the difference in transdiaphragmatic pressure between the points crossed by the *dashed* and *solid* lines. PEEPi increases the magnitude of the inspiratory effort and remarkably lengthens the effort-to-assist delay.

Figure 10-17



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Ineffective or wasted inspiratory efforts. The *dashed horizontal line* on the flow tracing indicates zero. The *dashed vertical line* indicates the onset of the ventilator assistance. To the left of the dashed line, the patient is spontaneously breathing with the ventilator set in CPAP mode. The breathing pattern is rapid and shallow. Initiation of pressure support increases tidal volume and decreases respiratory rate, as assessed on the flow and airway pressure tracings. The respiratory rate, assessed by swings in esophageal pressure is higher. The *arrows* show that half the inspiratory efforts fails to trigger the ventilator. The

dotted horizontal lines on the volume tracing indicate that end-expiratory lung volume increased after institution of pressure support (dynamic hyperinflation).

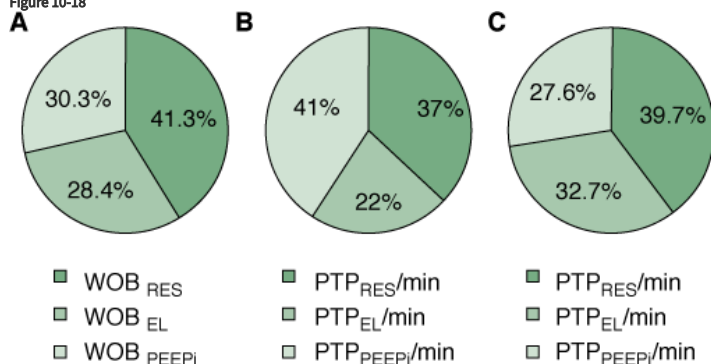
Acute hyperinflation impairs the force-generating capacity of the diaphragm by moving the diaphragm to an inefficient part of its force-length relationship^{572–574} and increasing the O₂ cost of breathing.⁵⁷⁵ In healthy volunteers, Beck et al⁵⁷⁶ showed that greater diaphragmatic activation to generate a certain amount of pressure increased with increasing lung volume, and pressure generated by maximal diaphragmatic activation decreased with increasing lung volume. In spontaneously breathing dogs, Kawagoe et al⁵⁷⁷ found that a threshold load associated with PEEP_i produced pulmonary artery hypertension and decreases in cardiac output and blood flow to the liver, pancreas, sternomastoid and parasternal muscles, while little change in blood flow to the diaphragm.⁵⁷⁷ An equivalent resistive load increased diaphragmatic blood flow, without changes in cardiac output or regional blood flow.⁵⁷⁷ Both loads produced noncompensated respiratory acidosis, but the severity was greater for the threshold load.⁵⁷⁷ An acute inspiratory threshold load causes a greater diaphragmatic sarcomere disruption in patients with COPD than in patients without lung disease⁵⁷⁸; hyperinflation explains approximately 40% of the injury.⁵⁷⁸

The complex interactions of dynamic hyperinflation and PEEP_i on preload were reviewed by Ranieri et al.⁵⁷⁹ During spontaneous breathing or triggered ventilation, venous return is decreased by the higher end-expiratory lung volume and intrathoracic pressure or increased by the larger negative inspiratory swings in intrathoracic pressure, with latter dominating.⁵⁷⁹

Clinical Consequences of Intrinsic Positive End-Expiratory Pressure

The extra load imposed on the respiratory muscles by PEEP_i may be large (Fig. 10-18) and have relevant clinical consequences. In patients with COPD in ARF, Coussa et al⁵⁸⁰ found that work of breathing was increased because of increased resistance and PEEP_i. PEEP_i accounted for more than 30% of the overall workload (Fig. 10-18A). In a ventilator-dependent patients with COPD, Appendini et al⁶⁴ found that the effort to overcome PEEP_i accounted for more than 40% of overall inspiratory effort⁶⁴ (Fig. 10-18B).

Figure 10-18



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Fractions of the respiratory load in ventilator-dependent patients with COPD. **A.** Elastic (EL), resistive (RES), and PEEP_i fractions of work of breathing (WOB) in sedated and paralyzed patients with COPD in acute respiratory failure while receiving controlled ventilation. **B.** Elastic (EL), resistive (RES), and PEEP_i components of inspiratory effort, assessed as esophageal pressure-time product (PTP), during a brief period of spontaneous breathing in patients with COPD who could not be weaned. **C.** Same measurement as in B in patients with COPD who failed a T-tube trial. Despite the different settings, PEEP_i accounted for a considerable fraction of inspiratory load.

Jubran and Tobin⁵⁸¹ studied thirty-one patients with COPD undergoing a T-tube trial, seventeen of whom failed and fourteen were successfully extubated. At the onset of the trial, PEEP_i was higher in the failure group than in the success group.⁵⁸¹ Between the start and end of the trial, the fraction of inspiratory effort caused by PEEP_i more than doubled in the failure group, while it did not change in the success group.⁵⁸¹

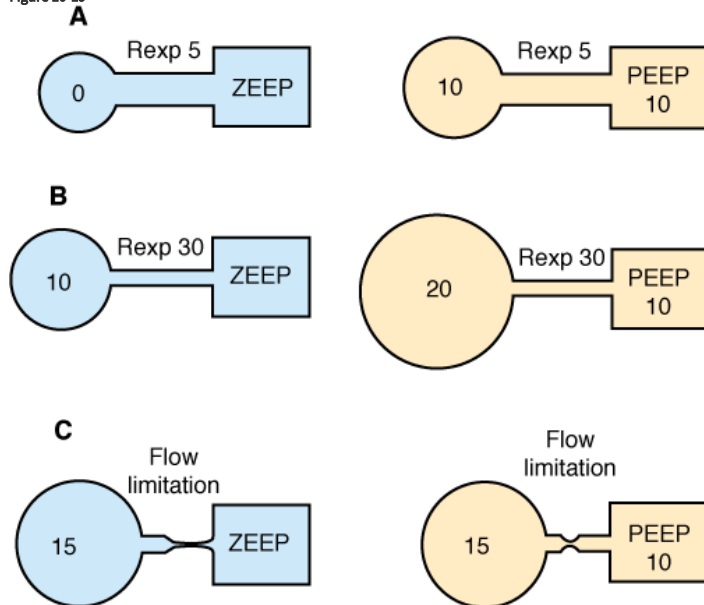
In thirty patients who failed an initial weaning trial but who later succeeded, Vassilakopoulos et al⁵⁸² found that dynamic hyperinflation and PEEP_i were higher at the time of weaning failure.

Impact of External Positive End-Expiratory Pressure on Dynamic Hyperinflation and Intrinsic Positive End-Expiratory Pressure

Depending on the mechanisms of dynamic hyperinflation and PEEP_i, external PEEP can have different effects, as elegantly pointed out by Marini (Fig. 10-19).⁵⁸³ When PEEP_i is not caused by flow limitation, external PEEP is entirely transmitted to the distal airway, producing a concomitant rise in alveolar pressure (Fig. 10-19B). In six patients with acute asthma or a COPD exacerbation, Tuxen et al²³⁷ increased dynamic hyperinflation and PEEP_i by increasing V_E

from 10 L/min to 16 L/min and 22 L/min. Increasing levels of PEEP up to 15 cm H₂O produced increases in end-expiratory lung volume and intrathoracic pressure, accompanied by a parallel hemodynamic deterioration.²³⁷ At the lowest V_E, trapped volume was small, suggesting little if any flow limitation.²³⁷

Figure 10-19



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Schematic of the effects of PEEP on alveolar pressure in normal conditions (A), high airway resistance but no expiratory flow limitation (B), and expiratory flow limitation and dynamic airway compression (C). The *circles* represent the alveoli, the *rectangles* represent the ventilator, and the *tubes* represent the bronchi. PEEP, Positive end-expiratory pressure expressed in cm H₂O; Rexp, expiratory resistance expressed in cm H₂O/L/s; ZEEP, zero positive end-expiratory pressure. **A.** At ZEEP (*left panel*), the alveolar pressure falls to zero. Applied PEEP (*right panel*) is entirely transmitted to the alveoli. **B.** At ZEEP (*left panel*), the high expiratory resistance causes alveolar pressure to remain positive at end-expiration. Applied PEEP (*right panel*) is entirely transmitted to the alveoli. **C.** At ZEEP (*left panel*), the alveolar pressure is positive. Applied PEEP does not exceed the critical pressure at the choke point and does not affect alveolar pressure.

When flow limitation and dynamic airway compression are present, PEEP is not transmitted to the distal airway and does not increase alveolar pressure as long as it does not exceed the critical pressure corresponding to the choke point (see Fig. 10-19C).⁵⁸³ The waterfall analogy helps in understanding this concept. The equal-pressure point is viewed as the crest of a waterfall, and the airway segment between the choke point and the airway opening as the downstream water.^{584,585} Increasing the level of downstream water (external PEEP) does not alter upstream water flow (air flow), as long as it does not exceed the crest (critical pressure at the choke point).⁵⁸⁵ In flow-limited patients with COPD who had PEEPi 10 cm H₂O and trapped volume of 1 L, Ranieri et al²³⁸ applied PEEP 5, 10, and 15 cm H₂O. Up to PEEP 10 cm H₂O, changes in peak and plateau airway pressure, end-expiratory lung volume, and total PEEP (i.e., PEEPi + external PEEP) were minimal; hemodynamic impairment was absent. PEEP 15 cm H₂O increased peak and plateau airway pressure, trapped volume and total PEEP, with worsening of hemodynamics.²³⁸ Changes in end-expiratory lung volume and cardiac index started to occur when external PEEP matched approximately 85% of PEEPi,stat.²³⁸ In eight patients with severe airway obstruction undergoing controlled mechanical ventilation, Carames et al evaluated the application of a PEEP level that exceeded by 50% the PEEPi value; in five patients, such PEEP levels produced a reduction in airway plateau pressure and total PEEP.⁵⁸⁶ In flow-limited patients with COPD receiving a triggered ventilation, Appendini et al found that PEEP slightly lower than PEEPi,dyn did not cause further hyperinflation.²

Flow limitation associated with small airway closure and PEEPi may be present in ventilated patients with ARDS.¹⁸³ In patients with ARDS who have flow limitation and PEEPi, Koutsoukou et al¹⁸⁴ found that PEEP 10 cm H₂O, produced a small increase in total PEEP and alveolar recruitment, but a significant increase in PaO₂; the latter was attributed to improved regional PEEPi in homogeneity. In another group of patients who did not exhibit flow limitation and had little PEEPi, PEEP 10 cm H₂O increased total PEEP and produced greater alveolar recruitment, which was the chief reason for an increase in PaO₂.¹⁸⁴ Analogous results were reported by Vieillard-Baron et al,³⁴² who found flow limitation and PEEPi in 73% of patients with ARDS receiving volume-controlled ventilation.

An increase in respiratory rate is commonly used during lung-protective ventilation in patients with ARDS to limit the hypercapnia consequent to V_T reduction. de Durante et al reported the occurrence PEEPi when patients with ARDS were ventilated with the low-V_T protocol of the ARDS Network (6 mL/kg) as opposed to a traditional V_T (12 mL/kg); respiratory rate was adjusted to achieve a V_E that maintained arterial pH between 7.30 and 7.45 (rate: 34 ± 1

breaths/min) during the low- V_T protocol, and rate was decreased to maintain the same V_E (rate: 14 ± 1 breaths/min) during the traditional- V_T protocol.⁵⁸⁷ PEEPi averaged 5.8 cm H₂O with low V_T and 1.4 cm H₂O with a traditional V_T .⁵⁸⁷ Analogous results were obtained by Richard et al⁵⁸⁸ and, though to a lesser extent, by Hough et al.⁵⁸⁹ Preliminary data indicate that PEEPi is reduced when higher PEEP levels are applied, confirming that flow limitation occurs also in ARDS patients.⁵⁹⁰

In actively breathing patients, the threshold load imposed by PEEPi may be offset by applying CPAP^{9,60,591,592} or external PEEP.^{2,61,63,566} If excessive or improperly applied,⁵⁸³ external PEEP can worsen hyperinflation^{238,593,594} placing the diaphragm at an additional disadvantage^{573,574,576} and causing further hemodynamic impairment.^{238,593}

Use of Positive End-Expiratory Pressure in the Clinical Setting

Acute Respiratory Distress Syndrome

The rationale for PEEP in patients with ARDS is to increase the amount of aerated lung volume,^{197,200,221} improve respiratory mechanics,^{197,200,338} reduce intrapulmonary shunt^{140,595} and ameliorate gas exchange,^{1,18,76,197,200} stabilize unsteady lung units,⁵⁹⁶ and reduce the risk of VILI.^{338,393} From its introduction almost fifty years ago,¹ PEEP remains a cornerstone in management of patients with ARDS.⁵⁹⁷ It is easy to see no randomized trial evaluated ventilator strategies without PEEP in patients with ARDS, except in the very early phase.^{151,598} A few studies have assessed the effect of PEEP on clinical outcome of patients with ARDS (Table 10-2).^{6,599–604} One early study reported that high PEEP levels did not improve survival and caused more adverse events, when compared to patients treated with low PEEP.⁵⁹⁹

Table 10-2: Randomized Controlled Trials Evaluating the Effect of Positive End-Expiratory Pressure on Clinical and Physiologic Outcomes

Reference	Population	Arm	No. of patients	Associated settings	PEEP setting	Mean PEEP (cm H ₂ O)	Outcomes
Carroll 1988 (599)	Surgical	Intervention	22	VT = 12 mL/kg PIP unlimited	Titrated to reduce shunt or to reach an oxygenation target	15	ICU mortality: 27%
		Control	28	VT = 12 mL/kg PIP unlimited	Minimal PEEP to reach an oxygenation target	4	ICU mortality: 4%
Amato 1998 (6)	Medical	Intervention	29	VT < 6 mL/kg PIP < 40 cm H ₂ O CPAP recruiting	2 cm H ₂ O > LIP (minimum 5 cm H ₂ O)	16	Mortality: 45% Barotrauma: 7%
		Control	24	VT = 12 mL/kg PIP unlimited	Titrated to P/F ratio (minimum 5 cm H ₂ O)	7	Mortality: 71% Barotrauma: 42%
Ranieri 1999 (369)	Medical-surgical	Intervention	18	VT for Pplat < UIP (or VT = 5 to 8 mL/kg)	2–3 cm H ₂ O > LIP (minimum 15 cm H ₂ O)	15	BAL/systemic levels of inflammatory mediators: ↓ Ventilator-free days: 12* Mortality: 38%* Organ failure score: unchanged
		Control	19	VT for PaCO ₂ 35 to 40 mm Hg Pplat < 35 cm H ₂ O	Titrated to SaO ₂ (minimum 5 cm H ₂ O)	7	BAL/systemic levels of inflammatory mediators: ↑ Ventilator-free days: 4* Mortality: 58%* Organ failure score: increased (mainly renal)
Brower (ARDS Network) 2004 (601)	Medical-surgical	Intervention	276	VT = 6 mL/kg Pplat < 30 cm H ₂ O	Titrated to oxygenation (minimum 12 cm H ₂ O)	13†	Mortality: 25%‡ Unassisted breathing: 72% Barotrauma: 11% Days without organ failure: 16
		Control	273	VT = 6 mL/kg Pplat < 30 cm H ₂ O	Titrated to oxygenation (minimum 5 cm H ₂ O)	8	Mortality: 28%‡ Unassisted breathing: 73% Barotrauma: 10% Days without organ failure: 16

Reference	Population	Arm	No. of patients	Associated settings	PEEP setting	Mean PEEP (cm H ₂ O)	Outcomes
Villar 2006 (604)	Medical-surgical	Intervention	50	VT = 5 to 8 mL/kg FiO ₂ according to oxygenation	2 cm H ₂ O > LIP	14	ICU mortality: 32% Ventilator-free days: 11 Barotrauma: 2% New organ failures: 0.3
		Control	45	VT = 9 to 11 mL/kg FiO ₂ according to oxygenation	Titrated to oxygenation (minimum 5 cm H ₂ O)	9	ICU mortality: 53% Ventilator-free days: 6 Barotrauma: 4% New organ failures: 1.2
Meade (LOVS) 2008 (602)	Medical-surgical	Intervention	475	VT = 6 mL/kg Pplat <30 cm H ₂ O	Titrated to oxygenation (minimum 5 to 10 cm H ₂ O)	14 [†]	Mortality: 36% Days of mechanical ventilation: 10 Barotrauma: 11% Refractory hypoxemia: 5%
		Control	508	VT = 6 mL/kg Pplat <30 cm H ₂ O	Titrated to oxygenation (minimum 5 cm H ₂ O)	9	Mortality: 40% Days of mechanical ventilation: 10 Barotrauma: 9% Refractory hypoxemia: 10%
Mercat (ExPress) 2008 (603)	Medical-surgical	Intervention (increased recruitment)	385	VT = 6 mL/kg Pplat = 28 to 30 cm H ₂ O	Titrated to reach the plateau pressure target	14	Mortality: 35% Ventilator-free days: 7 Pneumothorax: 7% Organ failure-free days: 6 Rescue therapies: 19%
		Control (minimal distension)	382	VT = 6 mL/kg Pplat <30 cm H ₂ O	Titrated to oxygenation (5 to 9 cm H ₂ O)	7	Mortality: 39% Ventilator-free days: 3 Pneumothorax: 6% Organ failure-free days: 2 Rescue therapies: 35%

Abbreviations: BAL, Bronchoalveolar lavage; CPAP, continuous positive airway pressure; LIP, lower inflection point on the pressure-volume curve; P/F, ratio between arterial oxygen tension and fractional inspired oxygen; PIP, peak inspiratory pressure; Pplat, plateau pressure; SaO₂, arterial oxygen saturation; VT, tidal volume.

* Post hoc analysis; [†] after protocol amendment; [‡] after adjustment for differences in baseline covariates.

In the last decade, research has focused on ventilator strategies that protect the lung from VILI. These investigations have shown that ventilator settings may per se affect outcome.⁶⁰⁵

Three randomized, controlled trials suggest that high levels of PEEP (15 to 20 cm H₂O), used in the context of a lung-protective strategy, may improve both the physiologic³⁶⁹ and clinical^{6,604} outcome in patients with ARDS. Amato et al⁶ found that low V_T combined with PEEP set 2 cm H₂O above the LIP (average: 16 cm H₂O) reduced mortality compared to a conventional approach where PEEP was titrated to optimize oxygenation (see Table 10-2). Ranieri et al³⁶⁹

compared a strategy of high PEEP and low V_T , respectively based on LIP and UIP, against a conventional approach, where lower PEEP and higher V_T were targeted to normalize gas exchange (see Table 10-2). The control group exhibited more inflammatory mediators than the lung-protective group.³⁶⁹ The control group also had a higher rate of organ failure and mortality and fewer ventilator-free days.^{369,417} The occurrence of PEEPi, consequent to high respiratory rate in the low V_T group, may have contributed^{587,588} to the positive results in the ARDS Network trial that compared low and high V_T in patients with ARDS.³⁷⁰ Villar et al compared a protective strategy, with PEEP set 2 cm H₂O above LIP (average: 14 cm H₂O) and low V_T (average: 7 mL/kg) to a conventional approach with PEEP set according to oxygenation (average: 9 cm H₂O) and higher V_T (average: 10 mL/kg).⁶⁰⁴ Intensive care unit mortality, hospital mortality, and number of organ failures after randomization were lower, while the number of ventilator-free days at day 28 was greater in the protective-strategy group than in the conventional-strategy group. Of note, patients included in this study had more severe disease ($Pa_{O_2}/F_{I_{O_2}} \leq 200$ with PEEP ≥ 5 cm H₂O) and persistent ARDS (criteria lasting 24 hours with standardized settings), than in previous studies (see Table 10-2).

Three randomized trials have compared the impact of different PEEP levels on clinical end points (see Table 10-2).⁶⁰¹⁻⁶⁰³ These trials randomized patients with ARDS to high or low PEEP levels to determine whether high PEEP, by enhancing lung recruitment, reduces VILI and improves outcome (see Table 10-2).⁶⁰¹⁻⁶⁰³ In the two arms of each trial, V_T was 6 mL/kg of predicted body weight and plateau pressure was less than 30 cm H₂O. In two studies, PEEP was titrated according to predefined combinations of PEEP and $F_{I_{O_2}}$, to achieve an identical oxygenation goal (Pa_{O_2} 55 to 88 mm Hg or arterial oxyhemoglobin saturation [Sp_{O_2}] 88% to 95% and 55 to 80 mm Hg or Sp_{O_2} 88% to 93%).^{601,602} In the third trial, PEEP (and, consequently, plateau pressure) was set as low as possible (total PEEP between 5 and 9 cm H₂O), without falling below oxygenation targets (Pa_{O_2} 55 to 80 mm Hg or Sp_{O_2} 88% to 95%), in the low-PEEP group (minimal distension strategy); in the high-PEEP group (increased recruitment strategy), PEEP was kept as high as possible without increasing plateau pressure above 28 to 30 cm H₂O, regardless of its effect on oxygenation. The ARDS Network study Assessment of Low Tidal Volume and Elevated End-Expiratory Volume to Obviate Lung Injury (ALVEOLI) was interrupted after enrolment of 549 patients because no difference was detected in any predefined clinical outcome variable between the two groups (see Table 10-2).⁶⁰¹ Several methodological biases cast doubt on the conclusions.^{606,607}

In the Lung Open Ventilation to Decrease Mortality in the Acute Respiratory Distress Syndrome (LOVS) trial, hospital mortality was not different between groups (36% and 40%, in the experimental and control groups, respectively). The higher PEEP group had better secondary end points related to hypoxemia, such as lower rates of refractory hypoxemia (5% vs. 10%), death with refractory hypoxemia (4% vs. 9%), and use of rescue therapies (5% vs. 9%).⁶⁰² The Comparison of Two Strategies for Setting Positive End-Expiratory Pressure in Acute Respiratory Distress Syndrome (ExPress Study) trial reported similar 28-day and hospital mortality in the two PEEP groups; ventilator-free days and organ failure-free days were higher, while the use of rescue therapy for severe hypoxemia was lower in the high-PEEP group as opposed to the low-PEEP group.⁶⁰³ Interestingly, improvements in these secondary end points were more pronounced in the sicker ARDS patients than in the milder ones. Patients with more severe ARDS also showed a trend toward a reduced mortality ($p = 0.08$) in the high-PEEP group. Several meta-analyses have been subsequently performed to assess the effect on clinical outcomes of low versus high PEEP,⁶⁰⁸⁻⁶¹¹ and these suggest that higher PEEP may improve survival, particularly in patients with more severe ARDS.

In summary, current evidence support the use of high PEEP, especially in the sickest patients with ARDS; in fact, high PEEP is safe, decreases the number of patients with severe hypoxemia and at high risk of death, and is associated with greater numbers of ventilator-free days and organ failure-free days. Nevertheless, it is uncertain how best to select the right patients and the optimal level of PEEP. The first issue is directly related to the current definitions of ARDS.⁶¹² Villar et al report that severity of lung injury and outcomes vary widely in patients meeting the current definition of ARDS, and proposed the adoption of standardized ventilator settings so as to identify patients with similar severity of lung injury and mortality.⁶¹³ Several approaches have been proposed to optimize PEEP in ARDS, but none has proven effective in improving clinical outcome (Table 10-3).^{6,22,42,164,171,201,215,303,309,322,327,344,351,599,601,614-623}

Table 10-3: Methods for Selecting Optimal Positive End-Expiratory Pressure

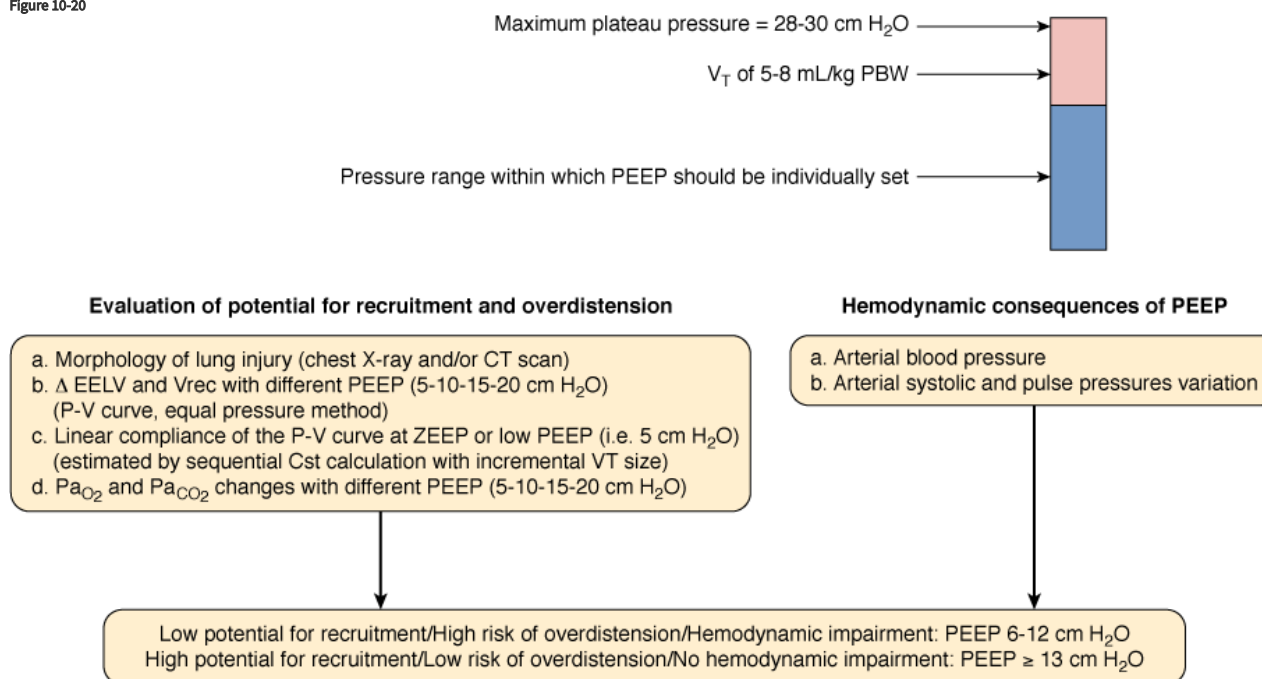
Reference	Year	PEEP target	Proposed method
Suter (42)	1975	Maximal oxygen delivery	Maximal static compliance during stepwise incremental PEEP
Kirby (22)	1975	Intrapulmonary shunt < 15%	Stepwise incremental PEEP to reach the target, while supporting cardiac output with fluids and vasopressors
Hurewitz (617)	1981	Drop of mixed venous oxygen tension	Stepwise incremental PEEP to reach the target
Weisman (618)	1982	Adequate PaO ₂ with minimal FiO ₂	Stepwise incremental PEEP to reach the target
Murray (619)	1984	Minimal arterial end-tidal CO ₂ gradient	Stepwise incremental PEEP to reach the target
Matamis (322)	1984	Best improvement in gas exchange, intrapulmonary shunt, and respiratory mechanics	PEEP set at the lower inflection point on inspiratory P-V curve
Albert (620)	1985	Adequate PaO ₂ with FiO ₂ < 0.6	Lowest PEEP to reach the target
Hartman (616)	1992	Maximal peripheral tissue perfusion	Maximal subcutaneous oxygen tension during stepwise incremental PEEP
Gattinoni (164)	1993	Prevention of compression atelectasis	Maximal PEEP (equal to superimposed hydrostatic pressure) corresponds to 70% of the dorsal-ventral height of thorax (in cm)
Ranieri (201)	1994	Linear airway pressure-time profile	PEEP variations to reach the target
Amato (6)	1998	Maximal recruitment	PEEP set 2 cm H ₂ O above the lower inflection point on the inspiratory P-V curve
Hickling (327)	2001	Maximal static compliance	Stepwise decremental PEEP trial
Rouby (309)	2002	Best oxygenation with the lowest FiO ₂ (>0.6)	Lung morphology, slope and inflection points of the P-V curve, PEEP trial
Brower (601)	2004	Oxygenation (PaO ₂ 55 to 80 mm Hg, SpO ₂ 88% to 95%)	PEEP-FiO ₂ combination table
Albaiceta (215)	2005	Expiratory point of maximum curvature	Deflation limb of the P-V curve
Suarez-Sipmann (351)	2007	Maximum dynamic compliance	Stepwise decremental PEEP trial
Talmor (626)	2008	Transpulmonary pressure between 0 and 10 cm H ₂ O	Esophageal pressure measurement

Abbreviations: CO₂, carbon dioxide; FiO₂, fractional inspired oxygen; PaO₂, arterial oxygen tension; P-V, pressure-volume; SpO₂, percutaneous oxygen saturation.

Optimal PEEP in ARDS should recruit as much nonaerated lung as possible, while avoiding lung overdistension, hemodynamic impairment, and global and regional disturbances of O₂ balance (Fig. 10-20). In the ALVEOLI and LOVS studies, PEEP was set according to preset PEEP/FiO₂ tables and PEEP-induced recruitment was estimated by improvement in oxygenation.^{601,602} It is well known that changes in oxygenation depend not only on recruitment, but also on cardiac output.^{48,138,140,191} Thus, oxygenation may have improved without any lung recruitment. Grasso et al^{318,321} analyzed the physiologic effects of the ALVEOLI protocol. They demonstrated that the protocol has unpredictable effects on lung recruitment. Indeed, the protocol may increase the risk of overdistension when high PEEP is applied in patients with a low potential for recruitment, such as those with a focal distribution of lung

atelectasis.^{27,225,321} As recently suggested,^{27,33,624} assessment of lung recruitability can be important when attempting to optimize PEEP on an individual basis. Results from the ExPress study support this reasoning and suggest that setting PEEP at the highest level compatible with a plateau pressure of 28 to 30 cm H₂O and a VT of approximately 6 mL/kg of predicted body weight is an acceptable approach (Fig. 10-20).

Figure 10-20



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Proposed algorithm for setting PEEP in ARDS. First, a safe plateau pressure limit of 28 to 30 cm H₂O is recommended; it should not be exceeded except when chest wall compliance is very low (30 to 35 cm H₂O). Second, V_T is preset, which corresponds to a certain pressure oscillation. Third, PEEP should not exceed the pressure difference between plateau pressure and the tidal pressure oscillation. After hemodynamic stabilization, the potential for recruitment, the risk of overdistension, and the hemodynamic consequences are individually assessed for different levels of PEEP. Potential for recruitment is greater when: (a) lung injury is diffuse; (b) alveolar recruitment is large (i.e., >100 mL); (c) the ratio between alveolar recruitment and change in end-expiratory lung volume is high (greater than 15% to 20%); (d) linear compliance of the pressure-volume curve at 0 cm H₂O is high (>40 mL/cm H₂O); (e) static compliance measured at low or no PEEP progressively increases with increasing V_T; and (f) PaO₂ increases and concomitantly decreases. By contrast, the risk for overdistension is greater when changes in the above parameters go in the opposite direction. PEEP-induced hemodynamic impairment may be considered clinically relevant when arterial systolic pressure falls more than 20 mm Hg and/or variations in both arterial systolic pressure or pulse pressure appear or increase. The use of standardized ventilator settings (particularly PEEP) to assess oxygenation and severity of lung injury may be useful for identifying the more severe patients who are likely to have a better response to PEEP. C_{st}, static compliance; CT, computed tomography; Δ EELV, change in end-expiratory lung volume; P-V, pressure-volume; PBW, predicted body weight; PEEP, positive end-expiratory pressure; V_{rec}, recruited volume; V_T, tidal volume; ZEEP, zero end-expiratory pressure.

In ARDS, lung units are unstable and have a strong tendency to collapse. Thus, any ventilator circuit disconnection or leakage, leading to a complete or partial drop in pressure, can cause harmful falls in lung volume and oxygenation.²⁹⁶

Another approach to setting PEEP based on real-time, continuous analysis of the dynamic airway pressure-time profile ("stress index") has been proposed for optimizing ventilator settings in ARDS.^{201,340,625} In animal studies and in a single study in patients, this approach was able to predict a ventilator strategy that minimizes VILI.^{321,340,625} The implementation of automatic tools that facilitate standardized assessment and continuous monitoring of respiratory mechanics might allow this approach to be incorporated into clinical practice. Another approach for choosing PEEP is based on the estimation of transpulmonary pressure with the placement of an esophageal-balloon catheter. Talmor et al compared a level of PEEP, where transpulmonary pressure at end-expiration was maintained between 0 and 10 cm H₂O, against standard strategy, where PEEP was set according to PEEP/FiO₂ tables.⁶²⁶ Compared to the standard approach, the transpulmonary pressure-guided strategy resulted in a higher level of PEEP (17 vs. 10 cm H₂O), and better oxygenation and respiratory system compliance. This interesting approach may prove useful in the future.

Postoperative State

Postoperative hypoxemia is mainly caused by atelectasis.^{85,96,627} After various surgical procedures, atelectasis develops in 30% to 50% of cases,⁶²⁸ and leads to ARF requiring intubation and mechanical ventilation in 8% to 10% of patients.^{628–631} The application of CPAP (5 to 10 cm H₂O) via a face mask, alone or combined with pressure support, reduces atelectasis and improves oxygenation after both cardiac^{134,632} and noncardiac surgery,^{633–637} without increasing surgical complications, such as anastomotic leaks.⁶³⁸ By reducing atelectasis, CPAP may decrease bacterial growth in the lung, mitigating bacterial translocation into the bloodstream, and normalizing alveolocapillary permeability.^{411,639}

Onset and duration of CPAP seem to be critical for avoiding or reversing atelectasis. Studies that failed to demonstrate benefit used CPAP several hours after surgery⁶⁴⁰ or for a relatively short time.^{641–643} Despite well-recognized physiologic benefits, only recently have randomized trials assessed the effects of CPAP on outcome variables.^{644,645} In seventy patients, Fagevik Olsen et al⁶⁴⁴ reported that, compared to breathing exercises, use of CPAP after thoracoabdominal resection of esophageal cancer resulted in a lower intubation rate. Squadrone et al⁶⁴⁵ randomized patients who developed hypoxemia following major abdominal surgery to either O₂ alone (FiO₂ 50%) or O₂ plus CPAP 7.5 cm H₂O. The study was interrupted after 209 patients were enrolled, because the intubation rate was lower in the CPAP group than in the O₂ group (1% vs. 10%).⁶⁴⁵ The patients who received CPAP at the end of treatment had higher oxygenation and lower rates of pneumonia, infection, and sepsis than the O₂ group.⁶⁴⁵ Although limited to selected patients undergoing elective abdominal surgery, these data show that improvement in physiologic parameters achieved with CPAP can translate into a better clinical outcome. Of note, CPAP was applied almost continuously for a prolonged time (19 ± 22 hours).⁶⁴⁵ One recent meta-analysis, based on more than nine randomized controlled trials that included a total of 654 patients undergoing abdominal surgery, suggested that CPAP decreases the risk of postoperative pulmonary complications, including atelectasis, pneumonia, and endotracheal intubation.⁶⁴⁶

In summary, CPAP is a simple, practical and safe technique for managing postoperative hypoxemia. It may improve outcome in selected patients, provided it is applied for a sufficient time.

Chronic Obstructive Pulmonary Disease

ARF in patients with COPD is characterized by high airway resistance and PEEPi,^{176,545,550,580,647} which increase respiratory muscle load.^{2,61,64,550,569} Hyperinflation simultaneously reduces the force-generating capacity of the diaphragm.^{574,576} When the force-load balance is altered to a level that severe respiratory acidosis ensues, mechanical ventilation becomes necessary. The ventilator acts as an accessory pump that helps reversing the unfavorable force-load balance. By reducing O₂ cost of breathing and correcting arterial blood gases, mechanical ventilation can improve respiratory muscle function (force). Inspiratory assistance by the ventilator, however, does not reduce the inspiratory resistive load consequent to airway resistance or the threshold load consequent to PEEPi.

Pharmacotherapy can have favorable effects on both load reduction and force restoration.⁶⁴⁸ Agents act on bronchial smooth muscle (β₂-agonists, anticholinergic)^{649–652} or inflammatory process (glucocorticoids and antibiotics).^{649,650,652,653} These agents decrease airway resistance, dynamic hyperinflation and PEEPi in ventilated patients.^{654–659} Glucocorticoids and antibiotics are commonly administered parenterally. In ventilated patients, bronchodilators are more effectively and safely given via a small volume nebulizer or a metered-dose inhaler connected to an appropriate spacer.⁶⁶⁰

External PEEP reduces the threshold load imposed on the inspiratory muscles. PEEP, however, does not reduce hyperinflation and its effects on diaphragmatic force-generating capacity. Application of PEEP in patients with COPD may cause a decrease in expiratory resistance and promote a faster and more uniform lung emptying,³⁵⁷ although the clinical relevance of these findings remains to be determined. CPAP is effective in reducing dyspnea,^{9,60,592} decreasing work of breathing,^{9,60,591,592} and improving cardiac function.⁵⁷⁹ During triggered ventilation, PEEP reduces inspiratory effort^{2,61,63,566} and ventilatory drive,⁵⁶⁶ and facilitates triggering of the ventilator^{2,61,63,566} by reducing both inspiratory effort and the delay between onset of inspiratory effort and initiation of assistance, improving patient-ventilator interaction, and reducing ineffective inspiratory efforts.^{63,64}

Despite a large body of physiologic data, no randomized trial has evaluated the effectiveness of PEEP with CPAP in improving outcome in ventilated patients with COPD. Most trials that evaluated the effectiveness of noninvasive ventilation in patients with acute hypercapnia secondary to COPD exacerbation used PEEP 4 to 5 cm H₂O in addition to pressure support.^{661–663}

NAVA is a novel mode of ventilation that utilizes diaphragmatic electrical activity, measured with an electrode array in the lower esophagus, to trigger and drive the ventilator.⁶⁶⁴ The ventilator is triggered, irrespective of PEEPi, as soon as the diaphragm contracts.⁶⁶⁴ By reducing the delay between onset of inspiratory effort and initiation of machine support,⁶⁶⁵ NAVA markedly enhances patient-ventilator coordination without the need for external PEEP, thereby eliminating the difficulties in selecting the appropriate PEEP level and avoiding the risk of further hyperinflation.⁶⁶⁴

Acute Severe Asthma

Patients with acute severe asthma have high airway resistance, dynamic hyperinflation, and PEEPi, which increase respiratory muscles load and decrease force-generation capacity.⁶⁶⁶ These patients are generally less responsive to external PEEP.²³⁷ Although some flow limitation may be present,⁵³⁹ the main cause of dynamic hyperinflation and PEEPi is the prolonged time constant, secondary to the high expiratory resistance. In the absence of flow limitation, PEEP increases hyperinflation and alveolar pressure.^{583,667} Some alveolar units may be entirely occluded by mucus plugs and thus completely disconnected from the central airway.⁵⁴⁴

Low levels of CPAP may decrease breathlessness.^{668–670} Lin et al⁶⁷¹ reported that CPAP reduced the bronchial reactivity. In patients with asthma exposed to aerosolized histamine, Martin et al⁶⁷² found that CPAP 12 cm H₂O decreased both esophageal and transdiaphragmatic pressure swings. V_E increased and thus inspiratory work did not fall.⁶⁷² Meduri et al,⁶⁷³ in a series that included seventeen patients with acute asthma, reported CPAP 3 to 5 cm H₂O combined with pressure support via a face mask improved gas exchange. Only two patients failed and required endotracheal intubation.⁶⁷³ In a randomized trial of thirty patients treated in an emergency department for acute asthma, Sorosky et al⁶⁷⁴ found that noninvasive ventilation with low CPAP produced greater and faster improvements in lung function and lower rate of hospitalization than did standard therapy.⁶⁷⁴

In summary, use of PEEP in patients receiving mechanical ventilation for acute severe asthma is often detrimental and is thus not advisable. Very few data suggest that low levels of noninvasive CPAP combined with pressure support in less severe and selected patients might be beneficial, although further evaluation is required.

Cardiogenic Pulmonary Edema

The benefit of CPAP on the failing left ventricle has already been discussed (see Fig. 10-10). Although most patients will positively respond to standard medical treatment, some do not and require endotracheal intubation and ventilator assistance.

Over the last three decades there has been growing interest in the delivery of CPAP via a noninvasive interface, as a means of avoiding endotracheal intubation and invasive mechanical ventilation. In patients with cardiogenic pulmonary edema, several randomized controlled trials have compared CPAP versus standard medical therapy,^{10,73,146,153,675} bilevel positive airway pressure (BiPAP) (i.e., PEEP plus inspiratory pressure support),^{154,676–681} or both.^{679,682,683} That is eight trials have compared CPAP versus standard treatment^{10,73,146,153,675,679,682,683} and nine trials have compared CPAP versus BiPAP.^{154,676–683} Nine studies^{153,154,676–681,683} had been powered a priori, according to their primary end points; three were prematurely interrupted after interim analysis.

The eight studies comparing CPAP versus standard treatment^{10,73,146,153,675,679,682,683} included a total of 537 patients receiving CPAP and 564 control patients; of note, one single study included 346 and 367 patients in the CPAP and standard treatment groups, respectively.⁶⁷⁹ CPAP was 10 cm H₂O in four studies,^{10,73,675,682,683} 7.5 cm H₂O in one study,¹⁵³ varied between 2.5 and 12.5 cm H₂O in one study,¹⁴⁶ and between 5 and 15 cm H₂O in another one.⁶⁷⁹ CPAP produced a greater and more rapid physiologic improvement in all studies but one.⁶⁸² Five studies^{10,73,146,153,675} reported a prompter and greater improvement in oxygenation with CPAP, although in only three studies^{10,153,675} was PaO₂/FiO₂ assessed. Bersten et al¹⁰ and Gray et al⁶⁷⁹ observed improvement in pH and PaCO₂ with CPAP. Dyspnea was reduced by CPAP in three studies,^{153,679,683} but not in a fourth study.⁶⁸² CPAP produced a lower rate of therapeutic failure and intubation in four studies^{10,146,153,683} and decreased hospital mortality in only one study.⁶⁸² None of the eight studies revealed differences in the length of hospital stay.

The nine studies comparing CPAP with BiPAP^{154,676–683} included a total of 652 CPAP and 657 BiPAP patients. Six studies used CPAP 10 cm H₂O. The setting of BiPAP varied. Park et al⁶⁸³ added a small level of inspiratory positive airway pressure to expiratory positive airway pressure 10 cm H₂O. Mehta et al¹⁵⁴ and Crane et al⁶⁸² decreased expiratory positive airway pressure from 10 to 5 cm H₂O and raised inspiratory positive airway pressure to 15 cm H₂O, in order to achieve a mean pressure of 10 cm H₂O throughout the entire breath in both groups. Bellone et al⁶⁷⁶ also used PEEP 5 cm H₂O and an initial pressure support of 15 cm H₂O (i.e., inspiratory positive airway pressure 20 cm H₂O). In the remaining studies PEEP varied from 5 to 10 cm H₂O, while the inspiratory assistance was regulated to achieve a given V_T.^{677–681}

Mehta et al¹⁵⁴ found a more rapid improvement in PaCO₂ and other physiologic variables with BiPAP than with CPAP. The other studies^{676–683} did not find differences in early physiologic variables. Eight studies^{154,676–681,683} did not reveal differences in any clinical outcome variable. Crane et al⁶⁸² found hospital survival higher in the CPAP than BiPAP group. Metha et al¹⁵⁴ interrupted their study because myocardial infarction was increased in their BiPAP group; eight following randomized trials, however, did not confirm this finding.^{676–683}

Three meta-analyses, not including the most recent studies,^{677–681} conclude that CPAP and BiPAP are equally effective in decreasing rate of intubation and mortality as compared to the sole standard medical treatment.^{684–686}

In summary, CPAP improves and expedites physiologic improvement, whereas the effects on rate of endotracheal intubation and mortality remain controversial. Several factors such as equipment, timing of intervention, underlying disease and comorbidities may explain the differences in outcome among studies. Early use of CPAP (10 cm H₂O) in addition to the medical therapy is advisable in patients with ARF secondary to cardiogenic pulmonary edema.

Prehospital Setting

Very recently, some investigators proposed early out-of-hospital application of noninvasive CPAP for treatment of hypoxemic ARF, predominantly, although not exclusively, secondary to cardiogenic pulmonary edema.

Hubble et al first reported, in a nonrandomized control trial, on the application of CPAP 10 cm H₂O via face mask in addition to standard treatment in 120 patients with presumed cardiogenic pulmonary edema presenting to an emergency service. This group of patients was compared with a group of ninety-five patients with equivalent characteristics transported by a second emergency medical service who received standard treatment only. The diagnosis was confirmed in ninety and sixty-five patients in the treatment and control groups, respectively. When considering patients with confirmed diagnosis, the treatment group had a lower rate of endotracheal intubation and mortality compared to the controls; in addition, they showed a better improvement in respiratory rate, dyspnea, and pulse heart rate.⁶⁸⁷ Subsequently, the same research group undertook a theoretical analysis in which they modeled the cost-effectiveness of prehospital CPAP.⁶⁸⁸

In an observational study, Foti et al applied prehospital noninvasive CPAP via helmet to 121 patients with suspected cardiogenic pulmonary edema; sixty-two patients were treated by physicians with CPAP (approximately equal to 10 cm H₂O) in addition to medical treatment, while the remaining fifty-nine patients were treated by nurses with CPAP (approximately equal to 7.5 cm H₂O) only. The diagnosis was confirmed in fifty-four and fifty patients in the former and latter group, respectively. In both groups, oxygen saturation, breathing frequency, and hemodynamics equally improved; rate of intubation and mortality were not different between the groups.⁶⁸⁹ These positive results were drawn into question by Frontin et al who carried out a randomized controlled trial where 124 out-of-hospital patients with presumed cardiogenic pulmonary edema received either standard treatment plus CPAP or standard treatment only. Success of treatment, as defined by a respiratory rate less than 25 breaths/min and oxygen saturation greater than 90%, was no different between the groups; rate of intubation and mortality at 5 and 30 days were also unaffected by CPAP. CPAP 10 cm H₂O was applied through a face mask fitted with a valve and controlled with a portable-flow generator. The amount of generated flow, however, was not defined; failure to control for this variable opens the possibility of overestimating the positive pressure actually applied.⁶⁹⁰

Thompson et al⁶⁹¹ performed a randomized controlled trial on patients with ARF of varied etiologies where seventy-one patients received either medical therapy plus CPAP 10 cm H₂O (intervention group) or medical therapy plus oxygen or bag-valve-mask ventilation or intubation, according to local usual care (control group). The incidence of tracheal intubation was lower in the intervention group, a result that could be strongly biased by the fact that intubation was allowed as initial treatment in the control group. Mortality was reduced in the intervention group, as opposed to controls. As in the previous study, CPAP was applied through a face mask fitted with a valve and controlled with a portable flow generator, but the generated flow was maybe not controlled.

In summary, despite a strong rationale,⁶⁹² the available data do not permit one to make definite recommendations about the appropriateness of using noninvasive CPAP in the prehospital setting.

Prophylactic Positive End-Expiratory Pressure

The term *prophylactic PEEP* was used in early studies of using PEEP in high-risk patients with the aim of reducing the incidence of ARDS.^{598,693–698} Both experimental^{386,404,699–701} and clinical studies^{693–695,702} suggest that PEEP might favorably influence the course of lung injury in patients at risk for ARDS. The hypothesis, however, was refuted by Pepe et al⁵⁹⁸ who randomized ninety-two ventilated patients at risk for ARDS to receive no PEEP or PEEP 8 cm H₂O. Early use of PEEP improved oxygenation, but it did not avert the development of ARDS⁵⁹⁸ (Table 10-4). Of note, the V_T in the study of Pepe et al was high (12 mL/kg).⁵⁹⁸ Such a high V_T might have recruited^{202,204} and overdistended the lung, masking any benefit of prophylactic PEEP. This concern is supported by barotrauma rate of around 50% in that study.⁵⁹⁸

Table 10-4: Randomized Controlled Trials Evaluating the Efficacy of Prophylactic Positive End-Expiratory Pressure

Reference	Population	Arm	No. of patients	Associated settings	PEEP setting	Mean PEEP (cm H ₂ O)	Outcomes
Pepe 1984 (598)	Medical-Surgical At risk for ARDS	Intervention	44	$V_T = 12$ mL/kg	Predetermined	8	Incidence of ARDS: 25% Mortality: 30% Barotrauma: 43%
		Control	48	$V_T = 12$ mL/kg	-	0	Incidence of ARDS: 27% Mortality: 38% Barotrauma: 50%
Delclaux 2000 (151)	Medical ARDS	Intervention	40*	O ₂ with CPAP for SaO ₂ >90%	Clinical response Tolerance	5-10	Endotracheal intubation: 38% ICU length of stay (days): 9 ICU mortality: 23% Hospital mortality: 30%
		Control	41*	O ₂ alone for SaO ₂ > 90%	-	0	Endotracheal intubation: 44% ICU length of stay (days): 9 ICU mortality: 22% Hospital mortality: 27%
Maitre 2000 (708)	Medical ARDS Undergoing FOB	Intervention	15	O ₂ with CPAP for SaO ₂ > 90%	Predetermined	7.5	Oxygenation: unchanged Ventilatory assistance after FOB: 0%
		Control	15	O ₂ alone for SaO ₂ > 90%	-	0	Oxygenation: ↓ Ventilatory assistance after FOB: 33%
Manzano 2008 (703)	Patients without ARDS	Intervention	64	$V_T = 8$ to 9 mL/kg	Predetermined According to the degree of abdominal distension	5-8	Hospital mortality: 30% Ventilator-associated pneumonia: 9% Incidence of ARDS: 5% Barotrauma: 2%
		Control	63	$V_T = 8$ to 9 mL/kg	-	0	Hospital mortality: 25% Ventilator-associated pneumonia: 25% Incidence of ARDS: 14% Barotrauma: 8%

Abbreviations: ARDS, acute respiratory distress syndrome; CPAP, continuous positive airway pressure; FOB, fiberoptic bronchoscopy; ICU, intensive care unit; O₂, oxygen; SaO₂, arterial oxygen saturation; V_T, tidal volume.

* subgroup of patients with early ARDS.

A more recent randomized controlled trial compared PEEP (8 cm H₂O) with no PEEP in 131 nonhypoxemic patients who had received mechanical ventilation for less than 24 hours.⁷⁰³ In this study, interrupted because of low patient recruitment, PEEP did not affect hospital mortality (30% vs. 25%), but decreased the rate of ventilator-associated pneumonia (9% vs. 25%) and the number of patients who developed hypoxemia (19% vs. 54%). There was a nonsignificant trend toward a reduction in the rate of ARDS (5% vs. 14%) (see [Table 10-4](#)). Three studies clarified the mechanisms through which PEEP may protect against ventilator-associated pneumonia, suggesting that fluid leakage across the cuff of the endotracheal tube is affected by PEEP.^{704–706} Two in vitro studies showed that fluid leakage, constantly present when PEEP was not applied, progressively decreased with increasing PEEP and was abolished at 15 cm H₂O.^{705,706} In intensive care unit patients, Lucangelo et al found that the minimum PEEP able to avoid fluid leakage around the cuff was 5 cm H₂O.⁷⁰⁴

PEEP equal to or less than 5 cm H₂O is sometimes used in intubated patients to counteract the fall in lung volume secondary to intubation, supine positioning, and/or muscle paralysis. A retrospective analysis of a large multinational cohort study conducted in 13,322 patients to evaluate current practice and outcomes of mechanical ventilation in the intensive care unit found that patients invasively ventilated without PEEP had significantly higher hospital mortality than those ventilated with PEEP.⁷⁰⁷

Intraoperative PEEP may have beneficial effects on the occurrence of pulmonary complications after surgery. A recent meta-analysis, including eight randomized controlled trials for a total of 330 patients, assessed the effects of intraoperative application of PEEP to mechanical ventilation on postoperative mortality and pulmonary outcomes.¹¹⁸ Mechanical ventilation with PEEP had no significant effect on mortality, compared to mechanical ventilation without PEEP, although PEEP was associated with improved oxygenation and decreased rate of postoperative atelectasis.

Use of CPAP in the very early stage of ARDS was evaluated in spontaneously breathing patients and during fiberoptic bronchoscopy. The investigators tested whether CPAP prevents the severe hypoxemia and respiratory failure requiring intubation.^{151,708} CPAP by face mask improved oxygenation, but did not improve the need for endotracheal intubation, length of hospital stay, or mortality (see [Table 10-4](#)).¹⁵¹ Early use of noninvasive CPAP, with or without pressure support, may benefit specific high-risk patients, such as immunocompromised patients.^{211,709–711} In obese patients undergoing gastroplasty, Joris et al⁷¹² found that prophylactic nasal CPAP, with pressure support, immediately after surgery reduced pulmonary dysfunction and accelerated recovery, compared with O₂ alone. CPAP is also useful during fiberoptic bronchoscopy,⁷⁰⁸ a procedure that can worsen oxygenation and respiratory mechanics.^{713–715} In a double-blind trial, Maitre et al⁷⁰⁸ compared mask CPAP to O₂ therapy alone during bronchoscopy in severe hypoxemic spontaneously breathing patients (see [Table 10-4](#)). During and immediately after the procedure oxygenation was well preserved in the CPAP group, whereas it fell in the O₂ group. CPAP prevented subsequent respiratory failure necessitating ventilator support.⁷⁰⁸

It has been suggested that PEEP may prevent VILI.⁷¹⁶ Despite robust physiologic rationale, no clinical evidence shows that PEEP protects against VILI. In a retrospective cohort study, Gajic et al⁷¹⁷ reported that approximately 25% of patients ventilated for more than 48 hours developed ARDS. V_T size was the only ventilator setting that independently identified risk for ARDS.⁷¹⁷ No relationship was found between PEEP and the risk of ARDS, possibly because the scatter in the data was insufficient to detect such an effect.⁷¹⁷

In anesthetized patients with healthy lung, Wolthius et al compared the effects on pulmonary and plasma inflammatory mediators of two ventilatory strategies, high V_T (12 mL/kg) without PEEP and low V_T (6 mL/kg) with PEEP (10 cm H₂O).⁷¹⁸ The latter strategy resulted in lower interleukin-8, myeloperoxidase, and elastase in the lung, as compared to the former approach, suggesting that ventilation with PEEP and low VT may limit VILI even in patients without preexisting lung injury. Another study, performed in animals with healthy lungs, however, showed contrasting results.⁷¹⁹

Complications and Contraindications

[Table 10-5](#) summarizes the complications and side effects of PEEP. Complications are directly related to level of PEEP. [Table 10-6](#) lists contraindications to the use of PEEP. In our opinion, there are only two absolute contraindications: life-threatening hypovolemic shock and undrained high-pressure pneumothorax. For all other conditions, there is little risk in using a low level of PEEP (up to 5 cm H₂O), but benefits and drawbacks of higher levels of PEEP need to be carefully weighed in each patient.

Table 10-5: Complications of Positive End-Expiratory Pressure

Pulmonary overdistension
Barotrauma
VILI
Increased dead space
Impaired CO ₂ elimination
Reduced diaphragm force-generation capacity
Reduced cardiac output and oxygen delivery
Impaired renal function
Reduced splanchnic blood flow
Hepatic congestion
Decreased lymph drainage

Table 10-6: Contraindications to Positive End-Expiratory Pressure

Absolute contraindications
Life-threatening hypovolemic shock
Undrained high-pressure pneumothorax
Relative contraindications
Unresolved bronchopleural fistula
Intracranial hypertension and low cerebral compliance
Chronic chest wall restrictive disorders
Dynamic hyperinflation without expiratory flow limitation

Conclusions

PEEP is widely used in the clinical setting.¹³ It is an essential component of ventilatory management of patients with both hypoxemic ARF, resulting from lung edema caused by increased microvascular permeability or hydrostatic pressure, and hypercapnic ARF, secondary to respiratory muscle failure.

Seventy-five years after the pioneering work of Poulton and Oxon¹¹ and Barach et al,¹² a large body of data has produced a general consensus on the usefulness of PEEP in the treatment of patients with acute respiratory failure. Although a few recent trials have assessed the effect of varied levels of PEEP on clinical outcomes, such as survival, complications, length of stay, and intensive care unit costs, doubts continue as to the optimal level of PEEP in the different clinical situations and on the best manner to determine it. Knowledge of pathophysiologic mechanisms and physiologic consequences of PEEP remains indispensable prerequisites for any decisions made at the bedside and must be the foundation for designing meaningful randomized, controlled trials.

References

1. Ashbaugh DG, Bigelow DB, Petty TL, Levine BE. Acute respiratory distress in adults. *Lancet*. 1967;2(7511):319–323. [PubMed: 4143721]

2. Appendini L, Patessio A, Zanaboni S, et al. Physiologic effects of positive end-expiratory pressure and mask pressure support during exacerbations of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 1994;149(5):1069–1076. [PubMed: 8173743]

3. Burchardi H. New strategies in mechanical ventilation for acute lung injury. *Eur Respir J*. 1996;9(5):1063–1072. [PubMed: 8793470]

4. Navalesi P, Hernandez P, Wongs A, et al. Proportional assist ventilation in acute respiratory failure: effects on breathing pattern and inspiratory effort. *Am J Respir Crit Care Med*. 1996;154(5):1330–1338. [PubMed: 8912744]

5. Cereda M, Foti G, Musch G, et al. Positive end-expiratory pressure prevents the loss of respiratory compliance during low tidal volume ventilation in acute lung injury patients. *Chest*. 1996;109(2):480–485. [PubMed: 8620726]

6. Amato MB, Barbas CS, Medeiros DM, et al. Effect of a protective-ventilation strategy on mortality in the acute respiratory distress syndrome. *N Engl J Med*. 1998;338(6):347–354. [PubMed: 9449727]

7. Antonelli M, Conti G, Rocco M, et al. A comparison of noninvasive positive-pressure ventilation and conventional mechanical ventilation in patients with acute respiratory failure. *N Engl J Med*. 1998;339(7):429–435. [[PubMed: 9700176](#)]

8. Gregory GA, Kitterman JA, Phibbs RH, et al. Treatment of the idiopathic respiratory-distress syndrome with continuous positive airway pressure. *N Engl J Med*. 1971;284(24):1333–1340. [[PubMed: 4930602](#)]

9. Petrof BJ, Legare M, Goldberg P, et al. Continuous positive airway pressure reduces work of breathing and dyspnea during weaning from mechanical ventilation in severe chronic obstructive pulmonary disease. *Am Rev Respir Dis*. 1990;141(2):281–289. [[PubMed: 2405757](#)]

10. Bersten AD, Holt AW, Vedig AE, et al. Treatment of severe cardiogenic pulmonary edema with continuous positive airway pressure delivered by face mask. *N Engl J Med*. 1991;325(26):1825–1830. [[PubMed: 1961221](#)]

11. Poulton EP, Oxon DM. Left-sided heart failure with pulmonary edema. Its treatment with the “pulmonary plus pressure machine.” *Lancet*. 1936;2:981–983.

12. Barach AL, Martin J, Eckman M. Positive pressure respiration and its application to the treatment of acute pulmonary edema. *Ann Intern Med*. 1938;12:754–795.

13. Esteban A, Anzueto A, Frutos F, et al. Characteristics and outcomes in adult patients receiving mechanical ventilation: a 28-day international study. *JAMA*. 2002;287(3):345–355. [[PubMed: 11790214](#)]

14. Katz JA, Ozanne GM, Zinn SE, Fairley HB. Time course and mechanisms of lung-volume increase with PEEP in acute pulmonary failure. *Anesthesiology*. 1981;54(1):9–16. [[PubMed: 7006465](#)]

15. Rose DM, Downs JB, Heenan TJ. Temporal responses of functional residual capacity and oxygen tension to changes in positive end-expiratory pressure. *Crit Care Med*. 1981;9(2):79–82. [[PubMed: 7006917](#)]

16. Gattinoni L, Mascheroni D, Torresin A, et al. Morphological response to positive end expiratory pressure in acute respiratory failure. Computerized tomography study. *Intensive Care Med*. 1986;12(3):137–142. [[PubMed: 3525633](#)]

17. East TD, in't Veen JC, Pace NL, McJames S. Functional residual capacity as a noninvasive indicator of optimal positive end-expiratory pressure. *J Clin Monit*. 1988;4(2):91–98. [[PubMed: 3131493](#)]

18. Malbouisson LM, Muller JC, Constantin JM, et al. Computed tomography assessment of positive end-expiratory pressure-induced alveolar recruitment in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2001;163(6):1444–1450. [[PubMed: 11371416](#)]

19. Dellaca RL, Aliverti A, Pelosi P, et al. Estimation of end-expiratory lung volume variations by optoelectronic plethysmography. *Crit Care Med*. 2001;29(9):1807–1811. [[PubMed: 11546992](#)]

20. Gilston A. The effects of PEEP on arterial oxygenation. An examination of some possible mechanisms. *Intensive Care Med*. 1977;3(4):267–271. [[PubMed: 338659](#)]

21. McMahon SM, Halprin GM. Modification of intrapulmonary blood shunt by end-expiratory pressure application in patients with acute respiratory failure. *Chest*. 1971;59:Suppl:27S+.

22. Kirby RR, Downs JB, Civetta JM, et al. High level positive end expiratory pressure (PEEP) in acute respiratory insufficiency. *Chest*. 1975;67(2):156–163. [[PubMed: 1090420](#)]

23. Pesenti A, Riboni A, Marcolin R, Gattinoni L. Venous admixture (Qva/Q) and true shunt (Qs/Qt) in ARF patients: effects of PEEP at constant. *Intensive Care Med*. 1983;9(6):307–311. [[PubMed: 6361086](#)]

24. Matamis D, Lemaire F, Harf A, et al. Redistribution of pulmonary blood flow induced by positive end-expiratory pressure and dopamine infusion in acute respiratory failure. *Am Rev Respir Dis*. 1984;129(1):39–44. [[PubMed: 6367569](#)]

25. Jonson B, Richard JC, Straus C, et al. Pressure-volume curves and compliance in acute lung injury: evidence of recruitment above the lower inflection point. *Am J Respir Crit Care Med*. 1999;159(4 Pt 1):1172–1178.

26. Suter PM, Fairley HB, Isenberg MD. Effect of tidal volume and positive end-expiratory pressure on compliance during mechanical ventilation. *Chest*. 1978;73(2):158–162. [[PubMed: 340159](#)]

27. Caironi P, Cressoni M, Chiumello D, et al. Lung opening and closing during ventilation of acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2010;181(6):578–586. [[PubMed: 19910610](#)]

-
28. Katz JA, Marks JD. Inspiratory work with and without continuous positive airway pressure in patients with acute respiratory failure. *Anesthesiology*. 1985;63(6):598–607. [[PubMed: 3904528](#)]
-
29. Wayne KS. Positive end-expiratory pressure (PEEP) ventilation. A review of mechanisms and actions. *JAMA*. 1976;236(12):1394–1396. [[PubMed: 785044](#)]
-
30. Perel A, Olschvang D, Eimerl D, et al. The variable effect of PEEP in acute respiratory failure associated with multiple trauma. *J Trauma*. 1978;18(3):218–220. [[PubMed: 347100](#)]
-
31. Lamy M, Fallat RJ, Koeniger E, et al. Pathologic features and mechanisms of hypoxemia in adult respiratory distress syndrome. *Am Rev Respir Dis*. 1976;114(2):267–284. [[PubMed: 788563](#)]
-
32. Matthay MA. Function of the alveolar epithelial barrier under pathologic conditions. *Chest*. 1994;105(3 Suppl):67S–74S.
-
33. Gattinoni L, Caironi P, Cressoni M, et al. Lung recruitment in patients with the acute respiratory distress syndrome. *N Engl J Med*. 2006;354(17):1775–1786. [[PubMed: 16641394](#)]
-
34. Pingleton SK. Complications of acute respiratory failure. *Am Rev Respir Dis*. 1988;137(6):1463–1493. [[PubMed: 3059862](#)]
-
35. Hasan FM, Beller TA, Sobonya RE, et al. Effect of positive end-expiratory pressure and body position in unilateral lung injury. *J Appl Physiol*. 1982;52(1):147–154. [[PubMed: 7037709](#)]
-
36. Vieira SR, Puybasset L, Richecoeur J, et al. A lung computed tomographic assessment of positive end-expiratory pressure-induced lung overdistension. *Am J Respir Crit Care Med*. 1998;158(5 Pt 1):1571–1577.
-
37. Petersen GW, Baier H. Incidence of pulmonary barotrauma in a medical ICU. *Crit Care Med*. 1983;11(2):67–69. [[PubMed: 6337021](#)]
-
38. Schnapp LM, Chin DP, Szaflarski N, Matthay MA. Frequency and importance of barotrauma in 100 patients with acute lung injury. *Crit Care Med*. 1995;23(2):272–278. [[PubMed: 7867352](#)]
-
39. Eisner MD, Thompson BT, Schoenfeld D, et al. Airway pressures and early barotrauma in patients with acute lung injury and acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2002;165(7):978–982. [[PubMed: 11934725](#)]
-
40. DiRusso SM, Nelson LD, Safcsak K, Miller RS. Survival in patients with severe adult respiratory distress syndrome treated with high-level positive end-expiratory pressure. *Crit Care Med*. 1995;23(9):1485–1496. [[PubMed: 7664550](#)]
-
41. Coffey RL, Albert RK, Robertson HT. Mechanisms of physiological dead space response to PEEP after acute oleic acid lung injury. *J Appl Physiol*. 1983;55(5):1550–1557. [[PubMed: 6358162](#)]
-
42. Suter PM, Fairley B, Isenberg MD. Optimum end-expiratory airway pressure in patients with acute pulmonary failure. *N Engl J Med*. 1975;292(6):284–289. [[PubMed: 234174](#)]
-
43. Berendes E, Lippert G, Loick HM, Brussel T. Effects of positive end-expiratory pressure ventilation on splanchnic oxygenation in humans. *J Cardiothorac Vasc Anesth*. 1996;10(5):598–602. [[PubMed: 8841866](#)]
-
44. Lehtipalo S, Biber B, Frojse R, et al. Effects of dopexamine and positive end-expiratory pressure on intestinal blood flow and oxygenation: the perfusion pressure perspective. *Chest*. 2003;124(2):688–698. [[PubMed: 12907561](#)]
-
45. Jedlinska B, Mellstrom A, Jonsson K, Hartmann M. Influence of positive end-expiratory pressure ventilation on peripheral tissue perfusion evaluated by measurements of tissue gases and pH. An experimental study in pigs with oleic acid lung injury. *Eur Surg Res*. 2000;32(4):228–235. [[PubMed: 11014924](#)]
-
46. Lutch JS, Murray JF. Continuous positive-pressure ventilation: effects on systemic oxygen transport and tissue oxygenation. *Ann Intern Med*. 1972;76(2):193–202. [[PubMed: 4550588](#)]
-
47. Qvist J, Pontoppidan H, Wilson RS, et al. Hemodynamic responses to mechanical ventilation with PEEP: the effect of hypervolemia. *Anesthesiology*. 1975;42(1):45–55. [[PubMed: 234210](#)]
-
48. Dantzker DR, Lynch JP, Weg JG. Depression of cardiac output is a mechanism of shunt reduction in the therapy of acute respiratory failure. *Chest*. 1980;77(5):636–642. [[PubMed: 6988180](#)]
-

49. Jardin F, Farcot JC, Boissante L, et al. Influence of positive end-expiratory pressure on left ventricular performance. *N Engl J Med*. 1981;304(7):387–392. [\[PubMed: 7005679\]](#)

50. Marini JJ, Culver BH, Butler J. Mechanical effect of lung distention with positive pressure on cardiac function. *Am Rev Respir Dis*. 1981;124(4):382–386. [\[PubMed: 7027851\]](#)

51. Dorinsky PM, Whitcomb ME. The effect of PEEP on cardiac output. *Chest*. 1983;84(2):210–216. [\[PubMed: 6347545\]](#)

52. Apuzzo JL, Wiess MH, Petersons V, et al. Effect of positive end expiratory pressure ventilation on intracranial pressure in man. *J Neurosurg*. 1977;46(2):227–232. [\[PubMed: 319205\]](#)

53. Huseby JS, Pavlin EG, Butler J. Effect of positive end-expiratory pressure on intracranial pressure in dogs. *J Appl Physiol*. 1978;44(1):25–27. [\[PubMed: 342464\]](#)

54. Priebe HJ, Heimann JC, Hedley-Whyte J. Mechanisms of renal dysfunction during positive end-expiratory pressure ventilation. *J Appl Physiol*. 1981;50(3):643–649. [\[PubMed: 7019173\]](#)

55. Berry AJ, Geer RT, Marshall C, Wu WH, Zbuzek VM, Marshall BE. The effect of long-term controlled mechanical ventilation with positive end-expiratory pressure on renal function in dogs. *Anesthesiology*. 1984;61(4):406–415. [\[PubMed: 6385778\]](#)

56. Arvidsson D, Almquist P, Haglund U. Effects of positive end-expiratory pressure on splanchnic circulation and function in experimental peritonitis. *Arch Surg*. 1991;126(5):631–636. [\[PubMed: 2021348\]](#)

57. Bredenberg CE, Paskanik AM. Relation of portal hemodynamics to cardiac output during mechanical ventilation with PEEP. *Ann Surg*. 1983;198(2):218–222. [\[PubMed: 6347103\]](#)

58. Matuschak GM, Pinsky MR, Rogers RM. Effects of positive end-expiratory pressure on hepatic blood flow and performance. *J Appl Physiol*. 1987;62(4):1377–1383. [\[PubMed: 2954939\]](#)

59. Pepe PE, Marini JJ. Occult positive end-expiratory pressure in mechanically ventilated patients with airflow obstruction: the auto-PEEP effect. *Am Rev Respir Dis*. 1982;126(1):166–170. [\[PubMed: 7046541\]](#)

60. Goldberg P, Reissmann H, Maltais F, et al. Efficacy of noninvasive CPAP in COPD with acute respiratory failure. *Eur Respir J*. 1995;8(11):1894–1900. [\[PubMed: 8620959\]](#)

61. Smith TC, Marini JJ. Impact of PEEP on lung mechanics and work of breathing in severe airflow obstruction. *J Appl Physiol*. 1988;65(4):1488–1499. [\[PubMed: 3053583\]](#)

62. Appendini L, Purro A, Gudjonsdottir M, et al. Physiologic response of ventilator-dependent patients with chronic obstructive pulmonary disease to proportional assist ventilation and continuous positive airway pressure. *Am J Respir Crit Care Med*. 1999;159(5):1510–1517. [\[PubMed: 10228119\]](#)

63. Nava S, Bruschi C, Rubini F, et al. Respiratory response and inspiratory effort during pressure support ventilation in COPD patients. *Intensive Care Med*. 1995;21(11):871–879. [\[PubMed: 8636518\]](#)

64. Appendini L, Purro A, Patessio A, et al. Partitioning of inspiratory muscle workload and pressure assistance in ventilator-dependent COPD patients. *Am J Respir Crit Care Med*. 1996;154(5):1301–1309. [\[PubMed: 8912740\]](#)

65. Cournand A, Motley H, Werko L, Richards D. Physiologic studies of the effects of intermittent positive pressure breathing on cardiac output in man. *Am J Physiol*. 1948;125:261–273.

66. Braunwald E, Binion JT, Morgan WL Jr, Sarnoff SJ. Alterations in central blood volume and cardiac output induced by positive pressure breathing and counteracted by metaraminol (Aramine). *Circ Res*. 1957;5(6):670–675. [\[PubMed: 13473068\]](#)

67. Fessler HE, Brower RG, Wise RA, Permutt S. Effects of systolic and diastolic positive pleural pressure pulses with altered cardiac contractility. *J Appl Physiol*. 1992;73(2):498–505. [\[PubMed: 1399972\]](#)

68. Buda AJ, Pinsky MR, Ingels NB Jr, et al. Effect of intrathoracic pressure on left ventricular performance. *N Engl J Med*. 1979;301(9):453–459. [\[PubMed: 460363\]](#)

69. Wise RA, Robotham JL, Bromberger-Barnea B, Permutt S. Effect of PEEP on left ventricular function in right-heart-bypassed dogs. *J Appl Physiol.* 1981;51(3):541–546. [\[PubMed: 7035417\]](#)

70. Pinsky MR, Summer WR, Wise RA, et al. Augmentation of cardiac function by elevation of intrathoracic pressure. *J Appl Physiol.* 1983;54(4):950–955. [\[PubMed: 6853301\]](#)

71. Pinsky MR, Matuschak GM, Klain M. Determinants of cardiac augmentation by elevations in intrathoracic pressure. *J Appl Physiol.* 1985;58(4):1189–1198. [\[PubMed: 3988674\]](#)

72. Lenique F, Habis M, Lofaso F, et al. Ventilatory and hemodynamic effects of continuous positive airway pressure in left heart failure. *Am J Respir Crit Care Med.* 1997;155(2):500–505. [\[PubMed: 9032185\]](#)

73. Rasanen J, Heikkila J, Downs J, et al. Continuous positive airway pressure by face mask in acute cardiogenic pulmonary edema. *Am J Cardiol.* 1985;55(4):296–300. [\[PubMed: 3881920\]](#)

74. Lin M, Yang YF, Chiang HT, et al. Reappraisal of continuous positive airway pressure therapy in acute cardiogenic pulmonary edema. Short-term results and long-term follow-up. *Chest.* 1995;107(5):1379–1386. [\[PubMed: 7750335\]](#)

75. Pang D, Keenan SP, Cook DJ, Sibbald WJ. The effect of positive pressure airway support on mortality and the need for intubation in cardiogenic pulmonary edema: a systematic review. *Chest.* 1998;114(4):1185–1192. [\[PubMed: 9792593\]](#)

76. Falke KJ, Pontoppidan H, Kumar A, et al. Ventilation with end-expiratory pressure in acute lung disease. *J Clin Invest.* 1972;51(9):2315–2323. [\[PubMed: 4565164\]](#)

77. Pontoppidan H. Mechanical aids to lung expansion in non-intubated surgical patients. *Am Rev Respir Dis.* 1980;122(5 Pt 2):109–119.

78. Gattinoni L, Pesenti A, Bombino M, et al. Relationships between lung computed tomographic density, gas exchange, and PEEP in acute respiratory failure. *Anesthesiology.* 1988;69(6):824–832. [\[PubMed: 3057937\]](#)

79. Kumar A, Falke KJ, Geffin B, et al. Continuous positive-pressure ventilation in acute respiratory failure. *N Engl J Med.* 1970;283(26):1430–1436. [\[PubMed: 4921310\]](#)

80. Dantzker D. The influence of cardiovascular function on gas exchange. *Clin Chest Med.* 1983;4(2):149–159. [\[PubMed: 6851459\]](#)

81. Hansen JE, Clausen JL, Levy SE, Mohler JG, Van Kessel AL. Proficiency testing materials for pH and blood gases. The California Thoracic Society experience. *Chest.* 1986;89(2):214–217. [\[PubMed: 3943381\]](#)

82. West JB. Effect of slope and shape of dissociation curve on pulmonary gas exchange. *Respir Physiol.* 1969;8(1):66–85. [\[PubMed: 5366417\]](#)

83. West JB. State of the art: ventilation-perfusion relationships. *Am Rev Respir Dis.* 1977;116(5):919–943. [\[PubMed: 921067\]](#)

84. Moller JT, Johannessen NW, Berg H, et al. Hypoxaemia during anaesthesia—an observer study. *Br J Anaesth.* 1991;66(4):437–444. [\[PubMed: 2025470\]](#)

85. Bendixen HH, Hedley-Whyte J, Laver MB. Impaired oxygenation in surgical patients during general anesthesia with controlled ventilation. A concept of atelectasis. *N Engl J Med.* 1963;269:991–996. [\[PubMed: 14059732\]](#)

86. Hedenstierna G SJ. Breathing mechanics, dead space and gas exchange in the extremely obese, breathing spontaneously and during anaesthesia with intermittent positive pressure ventilation. *Acta Anaesthesiol Scand.* 1976;20(3):248–254. [\[PubMed: 785930\]](#)

87. Rehder K, Knopp TJ, Sessler AD, Didier EP. Ventilation-perfusion relationship in young healthy awake and anesthetized-paralyzed man. *J Appl Physiol.* 1979;47(4):745–753. [\[PubMed: 511681\]](#)

88. Gunnarsson L, Tokics L, Lundquist H, et al. Chronic obstructive pulmonary disease and anaesthesia: formation of atelectasis and gas exchange impairment. *Eur Respir J.* 1991;4(9):1106–1116. [\[PubMed: 1756845\]](#)

89. Rothen HU, Sporre B, Engberg G, et al. Prevention of atelectasis during general anaesthesia. *Lancet.* 1995;345(8962):1387–1391. [\[PubMed: 7760608\]](#)

90. Tokics L, Hedenstierna G, Strandberg A, et al. Lung collapse and gas exchange during general anesthesia: effects of spontaneous breathing, muscle paralysis, and positive end-expiratory pressure. *Anesthesiology.* 1987;66(2):157–167. [\[PubMed: 3813078\]](#)

91. Nunn JF, Bergman NA, Coleman AJ. Factors influencing the arterial [oxygen](#) tension during anaesthesia with artificial ventilation. *Br J Anaesth.* 1965;37(12):898–914. [[PubMed: 5856651](#)]
92. Norlander O, Herzog P, Norden I, et al. Compliance and airway resistance during anaesthesia with controlled ventilation. *Acta Anaesthesiol Scand.* 1968;12(3):135–152. [[PubMed: 5250017](#)]
93. Froese AB, Bryan AC. Effects of anesthesia and paralysis on diaphragmatic mechanics in man. *Anesthesiology.* 1974;41(3):242–255. [[PubMed: 4604401](#)]
94. Bergman NA. Reduction in resting end-expiratory position of the respiratory system with induction of anesthesia and neuromuscular paralysis. *Anesthesiology.* 1982;57(1):14–17. [[PubMed: 7091715](#)]
95. Brismar B, Hedenstierna G, Lundquist H, et al. Pulmonary densities during anesthesia with muscular relaxation—a proposal of atelectasis. *Anesthesiology.* 1985;62(4):422–428. [[PubMed: 3885791](#)]
96. Hedenstierna G, Lundquist H, Lundh B, et al. Pulmonary densities during anaesthesia. An experimental study on lung morphology and gas exchange. *Eur Respir J.* 1989;2(6):528–535. [[PubMed: 2744136](#)]
97. Sykes MK, Young WE, Robinson BE. Oxygenation during anaesthesia with controlled ventilation. *Br J Anaesth.* 1965;37:314–325. [[PubMed: 14314887](#)]
98. Strandberg A, Tokics L, Brismar B, et al. Atelectasis during anaesthesia and in the postoperative period. *Acta Anaesthesiol Scand.* 1986;30(2):154–158. [[PubMed: 3705902](#)]
99. Gunnarsson L, Tokics L, Gustavsson H, Hedenstierna G. Influence of age on atelectasis formation and gas exchange impairment during general anaesthesia. *Br J Anaesth.* 1991;66(4):423–432. [[PubMed: 2025468](#)]
100. Hedenstierna G, Rothen HU. Atelectasis formation during anesthesia: causes and measures to prevent it. *J Clin Monit Comput.* 2000;16(5-6):329–335.
101. Heneghan CP, Bergman NA, Jones JG. Changes in lung volume and (PAO₂-PaO₂) during anaesthesia. *Br J Anaesth.* 1984;56(5):437–445. [[PubMed: 6721956](#)]
102. Milic-Emili J, Henderson JA, Dolovich MB, et al. Regional distribution of inspired gas in the lung. *J Appl Physiol.* 1966;21(3):749–759. [[PubMed: 5912744](#)]
103. Don H. The mechanical properties of the respiratory system during anesthesia. *Int Anesthesiol Clin.* 1977;15(2):113–136. [[PubMed: 856730](#)]
104. Don HF, Wahba WM, Craig DB. Airway closure, gas trapping, and the functional residual capacity during anesthesia. *Anesthesiology.* 1972;36(6):533–539. [[PubMed: 5033776](#)]
105. Hedenstierna G, McCarthy G, Bergstrom M. Airway closure during mechanical ventilation. *Anesthesiology.* 1976;44(2):114–123. [[PubMed: 1252019](#)]
106. Gilmour I, Burnham M, Craig DB. Closing capacity measurement during general anesthesia. *Anesthesiology.* 1976;45(5):477–482. [[PubMed: 973703](#)]
107. Juno J, Marsh HM, Knopp TJ, Rehder K. Closing capacity in awake and anesthetized-paralyzed man. *J Appl Physiol.* 1978;44(2):238–244. [[PubMed: 632164](#)]
108. Bergman NA, Tien YK. Contribution of the closure of pulmonary units to impaired oxygenation during anesthesia. *Anesthesiology.* 1983;59(5):395–401. [[PubMed: 6356992](#)]
109. Dueck R, Prutow RJ, Davies NJ, et al. The lung volume at which shunting occurs with inhalation anesthesia. *Anesthesiology.* 1988;69(6):854–861. [[PubMed: 3195757](#)]
110. Hirshman CA, Bergman NA. Factors influencing intrapulmonary airway calibre during anaesthesia. *Br J Anaesth.* 1990;65(1):30–42. [[PubMed: 2200482](#)]
111. Rothen HU, Sporre B, Engberg G, et al. Airway closure, atelectasis and gas exchange during general anaesthesia. *Br J Anaesth.* 1998;81(5):681–686. [[PubMed: 10193276](#)]
112. Milic-Emili J, Robatto FM, Bates JH. Respiratory mechanics in anaesthesia. *Br J Anaesth.* 1990;65(1):4–12. [[PubMed: 2200483](#)]
113. von Euler U, Liljestrand G. Observations on the pulmonary arterial blood pressure in cat. *Acta Physiol Scand.* 1946;12:310–320.
114. Thilenius OG. Effect of anesthesia on response of pulmonary circulation of dogs to acute hypoxia. *J Appl Physiol.* 1966;21(3):901–904. [[PubMed: 5912760](#)]

115. Sykes MK, Loh L, Seed RF, et al. The effect of inhalational anaesthetics on hypoxic pulmonary vasoconstriction and pulmonary vascular resistance in the perfused lungs of the dog and cat. *Br J Anaesth*. 1972;44(8):776–788. [[PubMed: 5083009](#)]
116. Bjertnaes LJ. Hypoxia-induced vasoconstriction in isolated perfused lungs exposed to injectable or inhalation anesthetics. *Acta Anaesthesiol Scand*. 1977;21(2):133–147. [[PubMed: 848256](#)]
117. Neumann P, Rothen HU, Berglund JE, et al. Positive end-expiratory pressure prevents atelectasis during general anaesthesia even in the presence of a high inspired oxygen concentration. *Acta Anaesthesiol Scand*. 1999;43(3):295–301. [[PubMed: 10081535](#)]
118. Imberger G, McIlroy D, Pace NL, et al. Positive end-expiratory pressure (PEEP) during anaesthesia for the prevention of mortality and postoperative pulmonary complications. *Cochrane Database Syst Rev*. 2010(9):CD007922.
119. Tusman G, Bohm SH, Vazquez de Anda GF, et al. “Alveolar recruitment strategy” improves arterial oxygenation during general anaesthesia. *Br J Anaesth*. 1999;82(1):8–13. [[PubMed: 10325828](#)]
120. Rusca M, Proietti S, Schnyder P, et al. Prevention of atelectasis formation during induction of general anesthesia. *Anesth Analg*. 2003;97(6):1835–1839. [[PubMed: 14633570](#)]
121. Dueck R, Wagner PD, West JB. Effects of positive end-expiratory pressure on gas exchange in dogs with normal and edematous lungs. *Anesthesiology*. 1977;47(4):359–366. [[PubMed: 332016](#)]
122. Pelosi P, Ravagnan I, Giurati G, et al. Positive end-expiratory pressure improves respiratory function in obese but not in normal subjects during anesthesia and paralysis. *Anesthesiology*. 1999;91(5):1221–1231. [[PubMed: 10551570](#)]
123. Heneghan C, Jones J. Pulmonary gas exchange and diaphragmatic position. Effect of tonic phrenic stimulation compared with that of increased airway pressure. *Br J Anaesth*. 1985;57(12):1161–1166. [[PubMed: 4084427](#)]
124. Kleinman B, Frey K, VanDrunen M, et al. Motion of the diaphragm in patients with chronic obstructive pulmonary disease while spontaneously breathing versus during positive pressure breathing after anesthesia and neuromuscular blockade. *Anesthesiology*. 2002 97(2):298–305. [[PubMed: 12151916](#)]
125. Futier E, Constantin JM, Pelosi P, et al. Intraoperative recruitment maneuver reverses detrimental pneumoperitoneum-induced respiratory effects in healthy weight and obese patients undergoing laparoscopy. *Anesthesiology*. 2010;113(6):1310–1319. [[PubMed: 21068660](#)]
126. Meininger D, Byhahn C, Mierdl S, et al. Positive end-expiratory pressure improves arterial oxygenation during prolonged pneumoperitoneum. *Acta Anaesthesiol Scand*. 2005;49(6):778–783. [[PubMed: 15954959](#)]
127. Reinius H, Jonsson L, Gustafsson S, et al. Prevention of atelectasis in morbidly obese patients during general anesthesia and paralysis: a computerized tomography study. *Anesthesiology*. 2009;111(5):979–987. [[PubMed: 19809292](#)]
128. Valenza F, Vagginelli F, Tiby A, et al. Effects of the beach chair position, positive end-expiratory pressure, and pneumoperitoneum on respiratory function in morbidly obese patients during anesthesia and paralysis. *Anesthesiology*. 2007;107(5):725–732. [[PubMed: 18073547](#)]
129. Gander S, Frascarolo P, Suter M, et al. Positive end-expiratory pressure during induction of general anesthesia increases duration of nonhypoxic apnea in morbidly obese patients. *Anesth Analg*. 2005;100(2):580–584. [[PubMed: 15673897](#)]
130. Benhamou D, Bourgain JL, Rouby JJ, Viars P. High-frequency jet ventilation vs continuous positive airway pressure for postoperative respiratory support. *Chest*. 1984;85(6):733–738. [[PubMed: 6373169](#)]
131. Michalopoulos A, Anthi A, Rellos K, Geroulanos S. Effects of positive end-expiratory pressure (PEEP) in cardiac surgery patients. *Respir Med*. 1998;92(6):858–862. [[PubMed: 9850370](#)]
132. Pennock BE, Kaplan PD, Carlin BW, et al. Pressure support ventilation with a simplified ventilatory support system administered with a nasal mask in patients with respiratory failure. *Chest*. 1991;100(5):1371–1376. [[PubMed: 1935296](#)]
133. Pennock BE, Crawshaw L, Kaplan PD. Noninvasive nasal mask ventilation for acute respiratory failure. Institution of a new therapeutic technology for routine use. *Chest*. 1994;105(2):441–444. [[PubMed: 8306743](#)]
134. Matte P, Jacquet L, Van Dyck M, Goenen M. Effects of conventional physiotherapy, continuous positive airway pressure and non-invasive ventilatory support with bilevel positive airway pressure after coronary artery bypass grafting. *Acta Anaesthesiol Scand*. 2000;44(1):75–81. [[PubMed: 10669276](#)]

135. Aguilo R, Togores B, Pons S, et al. Noninvasive ventilatory support after lung resectional surgery. *Chest*. 1997;112(1):117–121. [PubMed: 9228366]
136. Auriant I, Jallot A, Herve P, et al. Noninvasive ventilation reduces mortality in acute respiratory failure following lung resection. *Am J Respir Crit Care Med*. 2001;164(7):1231–1235. [PubMed: 11673215]
137. Dantzker DR, Wagner PD, West JB. Instability of lung units with low VA/Q ratios during O2 breathing. *J Appl Physiol*. 1975;38:886–895.
138. Dantzker DR, Brook CJ, Dehart P, et al. Ventilation-perfusion distributions in the adult respiratory distress syndrome. *Am Rev Respir Dis*. 1979;120(5):1039–1052. [PubMed: 389116]
139. Rodriguez-Roisin R. Ventilation-perfusion relationships. In: Pinsky MR, Dhainaut JF, eds. *Pathophysiologic Foundations of Critical Care*. Baltimore, MD: Williams & Williams; 1993:389–413.
140. Santos C, Ferrer M, Roca J, et al. Pulmonary gas exchange response to oxygen breathing in acute lung injury. *Am J Respir Crit Care Med*. 2000;161(1):26–31. [PubMed: 10619793]
141. Aboab J, Jonson B, Kouatchet A, et al. Effect of inspired oxygen fraction on alveolar derecruitment in acute respiratory distress syndrome. *Intensive Care Med*. 2006;32(12):1979–1986. [PubMed: 17019545]
142. Jones R, Zapol WM, Tomashefski JF, et al. Pulmonary vascular pathology: human and experimental studies. In: Zapol ZM, Falke KJ, eds. *Acute Respiratory Failure*. New York, NY: Marcel Dekker; 1985:23–160.
143. Rasanen J, Vaisanen IT, Heikkila J, Nikki P. Acute myocardial infarction complicated by left ventricular dysfunction and respiratory failure. The effects of continuous positive airway pressure. *Chest*. 1985;87(2):158–162. [PubMed: 3881227]
144. Vaisanen IT, Rasanen J. Continuous positive airway pressure and supplemental oxygen in the treatment of cardiogenic pulmonary edema. *Chest*. 1987;92(3):481–485. [PubMed: 3304850]
145. Lapinsky SE, Mount DB, Mackey D, Grossman RF. Management of acute respiratory failure due to pulmonary edema with nasal positive pressure support. *Chest*. 1994;105(1):229–231. [PubMed: 8275736]
146. Lin M, Yang YF, Chiang HT, et al. Reappraisal of continuous positive airway pressure therapy in acute cardiogenic pulmonary edema. Short-term results and long-term follow-up. *Chest*. 1995;107(5):1379–1386. [PubMed: 7750335]
147. Meduri GU, Turner RE, Abou-Shala N, et al. Noninvasive positive pressure ventilation via face mask. First-line intervention in patients with acute hypercapnic and hypoxemic respiratory failure. *Chest*. 1996;109(1):179–193. [PubMed: 8549183]
148. Takeda S, Takano T, Ogawa R. The effect of nasal continuous positive airway pressure on plasma endothelin-1 concentrations in patients with severe cardiogenic pulmonary edema. *Anesth Analg*. 1997;84(5):1091–1096. [PubMed: 9141937]
149. Hoffmann B, Welte T. The use of noninvasive pressure support ventilation for severe respiratory insufficiency due to pulmonary oedema. *Intensive Care Med*. 1999;25(1):15–20. [PubMed: 10051073]
150. Rusterholtz T, Kempf J, Berton C, et al. Noninvasive pressure support ventilation (NIPSV) with face mask in patients with acute cardiogenic pulmonary edema (ACPE). *Intensive Care Med*. 1999;25(1):21–28. [PubMed: 10051074]
151. Delclaux C, L'Her E, Alberti C, et al. Treatment of acute hypoxemic nonhypercapnic respiratory insufficiency with continuous positive airway pressure delivered by a face mask: a randomized controlled trial. *JAMA*. 2000;284(18):2352–2360. [PubMed: 11066186]
152. Guntupalli KK. Acute pulmonary edema. *Cardiol Clin*. 1984;2(2):183–200. [PubMed: 6443344]
153. L'Her E, Duquesne F, Girou E, et al. Noninvasive continuous positive airway pressure in elderly cardiogenic pulmonary edema patients. *Intensive Care Med*. 2004;30(5):882–888. [PubMed: 14991092]
154. Mehta S, Jay GD, Woolard RH, et al. Randomized, prospective trial of bilevel versus continuous positive airway pressure in acute pulmonary edema. *Crit Care Med*. 1997;25(4):620–628. [PubMed: 9142026]
155. Bredenberg CE, Kazui T, Webb WR. Experimental pulmonary edema: the effect of positive end-expiratory pressure on lung water. *Ann Thorac Surg*. 1978;26(1):62–67. [PubMed: 352287]

156. Malo J, Ali J, Wood LD. How does positive end-expiratory pressure reduce intrapulmonary shunt in canine pulmonary edema? *J Appl Physiol.* 1984;57(4):1002–1010. [PubMed: 6389451]

157. Rizk NW, Murray JF. PEEP and pulmonary edema. *Am J Med.* 1982;72(3):381–383. [PubMed: 7036733]

158. Demling RH, Staub NC, Edmunds LH Jr. Effect of end-expiratory airway pressure on accumulation of extravascular lung water. *J Appl Physiol.* 1975;38(5):907–912. [PubMed: 1126901]

159. Pare PD, Warriner B, Baile EM, Hogg JC. Redistribution of pulmonary extravascular water with positive end-expiratory pressure in canine pulmonary edema. *Am Rev Respir Dis.* 1983;127(5):590–593. [PubMed: 6342480]

160. Ware LB, Matthay MA. The acute respiratory distress syndrome. *N Engl J Med.* 2000;342(18):1334–1349. [PubMed: 10793167]

161. Lewis JF, Jobe AH. Surfactant and the adult respiratory distress syndrome. *Am Rev Respir Dis.* 1993;147(1):218–233. [PubMed: 8420422]

162. Hubmayr RD. Perspective on lung injury and recruitment: a skeptical look at the opening and collapse story. *Am J Respir Crit Care Med.* 2002;165(12):1647–1653. [PubMed: 12070067]

163. Wilson TA, Anafi RC, Hubmayr RD. Mechanics of edematous lungs. *J Appl Physiol.* 2001;90(6):2088–2093. [PubMed: 11356770]

164. Gattinoni L, D'Andrea L, Pelosi P, et al. Regional effects and mechanism of positive end-expiratory pressure in early adult respiratory distress syndrome. *JAMA.* 1993;269(16):2122–2127. [PubMed: 8468768]

165. Gattinoni L, Pesenti A, Avalli L, et al. Pressure-volume curve of total respiratory system in acute respiratory failure. Computed tomographic scan study. *Am Rev Respir Dis.* 1987;136(3):730–736. [PubMed: 3307572]

166. Pelosi P, D'Andrea L, Vitale G, et al. Vertical gradient of regional lung inflation in adult respiratory distress syndrome. *Am J Respir Crit Care Med.* 1994;149(1):8–13. [PubMed: 8111603]

167. Albert RK, Hubmayr RD. The prone position eliminates compression of the lungs by the heart. *Am J Respir Crit Care Med.* 2000;161(5):1660–1665. [PubMed: 10806172]

168. Malbouisson LM, Busch CJ, Puybasset L, et al. Role of the heart in the loss of aeration characterizing lower lobes in acute respiratory distress syndrome. CT Scan ARDS Study Group. *Am J Respir Crit Care Med.* 2000;161(6):2005–2012. [PubMed: 10852781]

169. Puybasset L, Cluzel P, Chao N, et al. A computed tomography scan assessment of regional lung volume in acute lung injury. The CT Scan ARDS Study Group. *Am J Respir Crit Care Med.* 1998;158(5 Pt 1):1644–1655.

170. Quintel M, Pelosi P, Caironi P, et al. An increase of abdominal pressure increases pulmonary edema in oleic acid-induced lung injury. *Am J Respir Crit Care Med.* 2004;169(4):534–541. [PubMed: 14670801]

171. da Silva Almeida JR, Machado FS, Schettino GP, et al. Cardiopulmonary effects of matching positive end-expiratory pressure to abdominal pressure in concomitant abdominal hypertension and acute lung injury. *J Trauma.* 2010;69(2):375–383.

172. Richard JC, Maggiore SM, Mancebo J, et al. Effects of vertical positioning on gas exchange and lung volumes in acute respiratory distress syndrome. *Intensive Care Med.* 2006;32(10):1623–1626. [PubMed: 16896856]

173. Spragg RG, Lewis JF. Pathology of the surfactant system of the mature lung: second San Diego conference. *Am J Respir Crit Care Med.* 2001;163(1):280–282. [PubMed: 11208658]

174. Bernasconi M, Ploysongsang Y, Gottfried SB, et al. Respiratory compliance and resistance in mechanically ventilated patients with acute respiratory failure. *Intensive Care Med.* 1988;14(5):547–553. [PubMed: 3065390]

175. Broseghini C, Brandolese R, Poggi R, et al. Respiratory resistance and intrinsic positive end-expiratory pressure (PEEPi) in patients with the adult respiratory distress syndrome (ARDS). *Eur Respir J.* 1988;1(8):726–731. [PubMed: 3069488]

176. Broseghini C, Brandolese R, Poggi R, et al. Respiratory mechanics during the first day of mechanical ventilation in patients with pulmonary edema and chronic airway obstruction. *Am Rev Respir Dis.* 1988;138(2):355–361. [PubMed: 3195835]

177. Chung KF, Keyes SJ, Morgan BM, et al. Mechanisms of airway narrowing in acute pulmonary oedema in dogs: influence of the vagus and lung volume. *Clin Sci (Lond)*. 1983;65(3):289–296. [PubMed: 6872463]
178. Delaunois L, Sergysels R, Martin RR. Acute effects on airways mechanics of pulmonary edema induced by intravenous oleic acid in dogs. *Bull Eur Physiopathol Respir*. 1980;16(1):47–55. [PubMed: 7357126]
179. Eissa NT, Ranieri VM, Corbeil C, et al. Effects of positive end-expiratory pressure, lung volume, and inspiratory flow on interrupter resistance in patients with adult respiratory distress syndrome. *Am Rev Respir Dis*. 1991;144(3 Pt 1):538–543.
180. Eissa NT, Ranieri VM, Corbeil C, et al. Analysis of behavior of the respiratory system in ARDS patients: effects of flow, volume, and time. *J Appl Physiol*. 1991;70(6):2719–2729. [PubMed: 1885468]
181. Esbenshade AM, Newman JH, Lams PM, et al. Respiratory failure after endotoxin infusion in sheep: lung mechanics and lung fluid balance. *J Appl Physiol*. 1982;53(4):967–976. [PubMed: 6759492]
182. Kondili E, Prinianakis G, Athanasis H, Georgopoulos D. Lung emptying in patients with acute respiratory distress syndrome: effects of positive end-expiratory pressure. *Eur Respir J*. 2002;19(5):811–819. [PubMed: 12030718]
183. Koutsoukou A, Armaganidis A, Stavrakaki-Kallergi C, et al. Expiratory flow limitation and intrinsic positive end-expiratory pressure at zero positive end-expiratory pressure in patients with adult respiratory distress syndrome. *Am J Respir Crit Care Med*. 2000;161(5):1590–1596. [PubMed: 10806160]
184. Koutsoukou A, Bekos B, Sotiropoulou C, et al. Effects of positive end-expiratory pressure on gas exchange and expiratory flow limitation in adult respiratory distress syndrome. *Crit Care Med*. 2002;30(9):1941–1949. [PubMed: 12352025]
185. Pesenti A, Pelosi P, Rossi N, et al. The effects of positive end-expiratory pressure on respiratory resistance in patients with the adult respiratory distress syndrome and in normal anesthetized subjects. *Am Rev Respir Dis*. 1991;144(1):101–107. [PubMed: 2064114]
186. Tantucci C, Corbeil C, Chasse M, et al. Flow and volume dependence of respiratory system flow resistance in patients with adult respiratory distress syndrome. *Am Rev Respir Dis*. 1992;145(2 Pt 1):355–360.
187. Wright PE, Bernard GR. The role of airflow resistance in patients with the adult respiratory distress syndrome. *Am Rev Respir Dis*. 1989;139(5):1169–1174. [PubMed: 2653150]
188. Nunn JF, Payne JP. Hypoxaemia after general anaesthesia. *Lancet*. 1962;2:631–632. [PubMed: 13939226]
189. Burger EJ Jr, Macklem P. Airway closure: demonstration by breathing 100 percent O₂ at low lung volumes and by N₂ washout. *J Appl Physiol*. 1968;25(2):139–148. [PubMed: 5666088]
190. Caro CG, Butler J, Dubois AB. Some effects of restriction of chest cage expansion on pulmonary function in man: an experimental study. *J Clin Invest*. 1960;39:573–583. [PubMed: 13807830]
191. Rouby JJ, Puybasset L, Cluzel P, et al. Regional distribution of gas and tissue in acute respiratory distress syndrome. II. Physiological correlations and definition of an ARDS Severity Score. CT Scan ARDS Study Group. *Intensive Care Med*. 2000;26(8):1046–1056. [PubMed: 11030160]
192. Brandolese R, Broseghini C, Polese G, et al. Effects of intrinsic PEEP on pulmonary gas exchange in mechanically-ventilated patients. *Eur Respir J*. 1993;6(3):358–363. [PubMed: 8472826]
193. Burns D, West TA, Hawkins K, O’Keefe GE. Immediate effects of positive end-expiratory pressure and low and high tidal volume ventilation upon gas exchange and compliance in patients with acute lung injury. *J Trauma*. 2001;51(6):1177–1181. [PubMed: 11740272]
194. Foti G, Cereda M, Sparacino ME, et al. Effects of periodic lung recruitment maneuvers on gas exchange and respiratory mechanics in mechanically ventilated acute respiratory distress syndrome (ARDS) patients. *Intensive Care Med*. 2000;26(5):501–507. [PubMed: 10923722]
195. Lim CM, Jung H, Koh Y, et al. Effect of alveolar recruitment maneuver in early acute respiratory distress syndrome according to antiderecruitment strategy, etiological category of diffuse lung injury, and body position of the patient. *Crit Care Med*. 2003;31(2):411–418. [PubMed: 12576945]
196. Luecke T, Roth H, Joachim A, et al. Effects of end-inspiratory and end-expiratory pressures on alveolar recruitment and derecruitment in saline-washout-induced lung injury—a computed tomography study. *Acta Anaesthesiol Scand*. 2004;48(1):82–92. [PubMed: 14674978]

197. Maggiore SM, Jonson B, Richard JC, et al. Alveolar derecruitment at decremental positive end-expiratory pressure levels in acute lung injury: comparison with the lower inflection point, oxygenation, and compliance. *Am J Respir Crit Care Med*. 2001;164(5):795–801. [[PubMed: 11549535](#)]

198. Mergoni M, Martelli A, Volpi A, et al. Impact of positive end-expiratory pressure on chest wall and lung pressure-volume curve in acute respiratory failure. *Am J Respir Crit Care Med*. 1997;156(3 Pt 1):846–854.

199. Puybasset L, Gusman P, Muller JC, et al. Regional distribution of gas and tissue in acute respiratory distress syndrome. III. Consequences for the effects of positive end-expiratory pressure. CT Scan ARDS Study Group. Adult Respiratory Distress Syndrome. *Intensive Care Med*. 2000;26(9):1215–1227. [[PubMed: 11089745](#)]

200. Ranieri VM, Eissa NT, Corbeil C, et al. Effects of positive end-expiratory pressure on alveolar recruitment and gas exchange in patients with the adult respiratory distress syndrome. *Am Rev Respir Dis*. 1991;144(3 Pt 1):544–551.

201. Ranieri VM, Giuliani R, Fiore T, et al. Volume-pressure curve of the respiratory system predicts effects of PEEP in ARDS: “occlusion” versus “constant flow” technique. *Am J Respir Crit Care Med*. 1994;149(1):19–27. [[PubMed: 8111581](#)]

202. Ranieri VM, Mascia L, Fiore T, et al. Cardiorespiratory effects of positive end-expiratory pressure during progressive tidal volume reduction (permissive hypercapnia) in patients with acute respiratory distress syndrome. *Anesthesiology*. 1995;83(4):710–720. [[PubMed: 7574050](#)]

203. Richard JC, Brochard L, Vandelet P, et al. Respective effects of end-expiratory and end-inspiratory pressures on alveolar recruitment in acute lung injury. *Crit Care Med*. 2003;31(1):89–92. [[PubMed: 12544999](#)]

204. Richard JC, Maggiore SM, Jonson B, et al. Influence of tidal volume on alveolar recruitment. Respective role of PEEP and a recruitment maneuver. *Am J Respir Crit Care Med*. 2001;163(7):1609–1613. [[PubMed: 11401882](#)]

205. Rimensberger PC, Cox PN, Frndova H, Bryan AC. The open lung during small tidal volume ventilation: concepts of recruitment and “optimal” positive end-expiratory pressure. *Crit Care Med*. 1999;27(9):1946–1952. [[PubMed: 10507623](#)]

206. Schreiter D, Reske A, Stichert B, et al. Alveolar recruitment in combination with sufficient positive end-expiratory pressure increases oxygenation and lung aeration in patients with severe chest trauma. *Crit Care Med*. 2004;32(4):968–975. [[PubMed: 15071387](#)]

207. Shah DM, Newell JC, Dutton RE, Powers SR Jr. Continuous positive airway pressure versus positive end-expiratory pressure in respiratory distress syndrome. *J Thorac Cardiovasc Surg*. 1977;74(4):557–562. [[PubMed: 333189](#)]

208. Valta P, Takala J, Eissa NT, Milic-Emili J. Does alveolar recruitment occur with positive end-expiratory pressure in adult respiratory distress syndrome patients? *J Crit Care*. 1993;8(1):34–42. [[PubMed: 8343857](#)]

209. Petty TL, Ashbaugh DG. The adult respiratory distress syndrome. Clinical features, factors influencing prognosis and principles of management. *Chest*. 1971;60(3):233–239. [[PubMed: 4937358](#)]

210. Ferrer M, Esquinas A, Leon M, et al. Noninvasive ventilation in severe hypoxemic respiratory failure: a randomized clinical trial. *Am J Respir Crit Care Med*. 2003;168(12):1438–1444. [[PubMed: 14500259](#)]

211. Hilbert G, Gruson D, Vargas F, et al. Noninvasive ventilation in immunosuppressed patients with pulmonary infiltrates, fever, and acute respiratory failure. *N Engl J Med*. 2001;344(7):481–487. [[PubMed: 11172189](#)]

212. Cosentini R, Brambilla AM, Aliberti S, et al. Helmet continuous positive airway pressure vs oxygen therapy to improve oxygenation in community-acquired pneumonia: a randomized, controlled trial. *Chest*. 2010;138(1):114–120. [[PubMed: 20154071](#)]

213. Isgro S, Zanella A, Sala C, et al. Continuous flow biphasic positive airway pressure by helmet in patients with acute hypoxic respiratory failure: effect on oxygenation. *Intensive Care Med*. 2010;36(10):1688–1694. [[PubMed: 20521025](#)]

214. Ralph DD, Robertson HT, Weaver LJ, et al. Distribution of ventilation and perfusion during positive end-expiratory pressure in the adult respiratory distress syndrome. *Am Rev Respir Dis*. 1985;131(1):54–60. [[PubMed: 3881062](#)]

215. Albaiceta GM, Luyando LH, Parra D, et al. Inspiratory vs. expiratory pressure-volume curves to set end-expiratory pressure in acute lung injury. *Intensive Care Med*. 2005;31(10):1370–1378. [[PubMed: 16091965](#)]

216. Crotti S, Mascheroni D, Caironi P, et al. Recruitment and derecruitment during acute respiratory failure: a clinical study. *Am J Respir Crit Care Med*. 2001;164(1):131–140. [[PubMed: 11435251](#)]

217. Webb HH, Tierney DF. Experimental pulmonary edema due to intermittent positive pressure ventilation with high inflation pressures. Protection by positive end-expiratory pressure. *Am Rev Respir Dis.* 1974;110(5):556–565. [PubMed: 4611290]

218. Corbridge TC, Wood LD, Crawford GP, et al. Adverse effects of large tidal volume and low PEEP in canine acid aspiration. *Am Rev Respir Dis.* 1990;142(2):311–315. [PubMed: 2200314]

219. Dreyfuss D, Saumon G. Role of tidal volume, FRC, and end-inspiratory volume in the development of pulmonary edema following mechanical ventilation. *Am Rev Respir Dis.* 1993;148(5):1194–1203. [PubMed: 8239153]

220. Dreyfuss D, Soler P, Basset G, Saumon G. High inflation pressure pulmonary edema. Respective effects of high airway pressure, high tidal volume, and positive end-expiratory pressure. *Am Rev Respir Dis.* 1988;137(5):1159–1164. [PubMed: 3057957]

221. Gattinoni L, Pelosi P, Crotti S, Valenza F. Effects of positive end-expiratory pressure on regional distribution of tidal volume and recruitment in adult respiratory distress syndrome. *Am J Respir Crit Care Med.* 1995;151(6):1807–1814. [PubMed: 7767524]

222. Hickling KG, Walsh J, Henderson S, Jackson R. Low mortality rate in adult respiratory distress syndrome using low-volume, pressure-limited ventilation with permissive hypercapnia: a prospective study. *Crit Care Med.* 1994;22(10):1568–1578. [PubMed: 7924367]

223. Tugrul S, Akinci O, Ozcan PE, et al. Effects of sustained inflation and postinflation positive end-expiratory pressure in acute respiratory distress syndrome: focusing on pulmonary and extrapulmonary forms. *Crit Care Med.* 2003;31(3):738–744. [PubMed: 12626977]

224. Beydon L, Uttman L, Rawal R, Jonson B. Effects of positive end-expiratory pressure on dead space and its partitions in acute lung injury. *Intensive Care Med.* 2002;28(9):1239–1245. [PubMed: 12209271]

225. Vieira SR, Puybasset L, Lu Q, et al. A scanographic assessment of pulmonary morphology in acute lung injury. Significance of the lower inflection point detected on the lung pressure-volume curve. *Am J Respir Crit Care Med.* 1999;159(5 Pt 1):1612–1623.

226. Blanch L, Roussos C, Brotherton S, et al. Effect of tidal volume and PEEP in ethchlorvynol-induced asymmetric lung injury. *J Appl Physiol.* 1992;73(1):108–116. [PubMed: 1506357]

227. Craven KD, Oppenheimer L, Wood LD. Effects of contusion and flail chest on pulmonary perfusion and oxygen exchange. *J Appl Physiol.* 1979;47(4):729–737. [PubMed: 511679]

228. Kanarek DJ, Shannon DC. Adverse effect of positive end-expiratory pressure on pulmonary perfusion and arterial oxygenation. *Am Rev Respir Dis.* 1975;112(3):457–459. [PubMed: 1099951]

229. Mink SN, Light RB, Cooligan T, Wood LD. Effect of PEEP on gas exchange and pulmonary perfusion in canine lobar pneumonia. *J Appl Physiol.* 1981;50(3):517–523. [PubMed: 7019171]

230. Remolina C, Khan AU, Santiago TV, Edelman NH. Positional hypoxemia in unilateral lung disease. *N Engl J Med.* 1981;304(9):523–525. [PubMed: 6779161]

231. Cinnella G, Dambrosio M, Brienza N, et al. Independent lung ventilation in patients with unilateral pulmonary contusion. Monitoring with compliance and EtCO₂. *Intensive Care Med.* 2001;27(12):1860–1867. [PubMed: 11797020]

232. Wagner PD, Dantzker DR, Dueck R, et al. Ventilation-perfusion inequality in chronic obstructive pulmonary disease. *J Clin Invest.* 1977;59(2):203–216. [PubMed: 833271]

233. Barbera JA, Reyes A, Roca J, et al. Effect of intravenously administered aminophylline on ventilation/perfusion inequality during recovery from exacerbations of chronic obstructive pulmonary disease. *Am Rev Respir Dis.* 1992;145(6):1328–1333. [PubMed: 1595998]

234. Barbera JA, Roca J, Ramirez J, et al. Gas exchange during exercise in mild chronic obstructive pulmonary disease. Correlation with lung structure. *Am Rev Respir Dis.* 1991;144(3 Pt 1):520–525.

235. Esteban A, De Elio FJ, Cerdá E, et al. Blood-gas changes with different end-expiratory pressures in patients with chronic bronchitis. *Br J Anaesth.* 1974;46(2):159–161. [PubMed: 4601116]

236. Barat G, Asuero MS. Positive end-expiratory pressure. Effect on arterial oxygenation during respiratory failure in chronic obstructive airway disease. *Anaesthesia.* 1975;30(2):183–189. [PubMed: 1093429]

237. Tuxen DV. Detrimental effects of positive end-expiratory pressure during controlled mechanical ventilation of patients with severe airflow obstruction. *Am Rev Respir Dis.* 1989;140(1):5–9. [[PubMed: 2665589](#)]

238. Ranieri VM, Giuliani R, Cinnella G, et al. Physiologic effects of positive end-expiratory pressure in patients with chronic obstructive pulmonary disease during acute ventilatory failure and controlled mechanical ventilation. *Am Rev Respir Dis.* 1993;147(1):5–13. [[PubMed: 8420430](#)]

239. Rossi A, Santos C, Roca J, et al. Effects of PEEP on VA/Q mismatching in ventilated patients with chronic airflow obstruction. *Am J Respir Crit Care Med.* 1994;149(5):1077–1084. [[PubMed: 8173744](#)]

240. Miro AM, Shivaram U, Hertig I. Continuous positive airway pressure in COPD patients in acute hypercapnic respiratory failure. *Chest.* 1993;103(1):266–268. [[PubMed: 8417894](#)]

241. de Lucas P, Tarancon C, Puente L, et al. Nasal continuous positive airway pressure in patients with COPD in acute respiratory failure. A study of the immediate effects. *Chest.* 1993;104(6):1694–1697.

242. Shivaram U, Cash ME, Beal A. Nasal continuous positive airway pressure in decompensated hypercapnic respiratory failure as a complication of sleep apnea. *Chest.* 1993;104(3):770–774. [[PubMed: 8365287](#)]

243. Tzoufi M, Mentzelopoulos SD, Roussos C, Armaganidis A. The effects of nebulized salbutamol, external positive end-expiratory pressure, and their combination on respiratory mechanics, hemodynamics, and gas exchange in mechanically ventilated chronic obstructive pulmonary disease patients. *Anesth Analg.* 2005;101(3):843–850, table of contents. [[PubMed: 16116002](#)]

244. Fenn WO. Mechanics of respiration. *Am J Med.* 1951;10:77–91. [[PubMed: 14799502](#)]

245. Radford Jr EP. Static mechanical properties of mammalian lungs. In: Fenn WO, Rahn H, eds. *Handbook of Physiology.* Vol. 1. Washington, DC: American Physiological Society; 1964:429–449.

246. Rahn H, Fenn WO, Otis AB. The pressure-volume diagram of the thorax and the lung. *Am J Physiol.* 1946;146:161–178. [[PubMed: 20982947](#)]

247. Maggiore SM, Brochard L. Pressure-volume curve in the critically ill. *Curr Opin Crit Care.* 2000;6:1–10.

248. Anthonisen NR. Changes in shunt flow, compliance, and volume of lungs during apneic oxygenation. *Am J Physiol.* 1964;207:235–238. [[PubMed: 14193596](#)]

249. Clements JA, Hustead RF, Johnson RP, Gribetz I. Pulmonary surface tension and alveolar stability. *J Appl Physiol.* 1961;16:444–450. [[PubMed: 13694048](#)]

250. Mead J, Collier C. Relation of volume history of lungs to respiratory mechanics in anesthetized dogs. *J Appl Physiol.* 1959;14(5):669–678.

251. Williams JV, Tierney DF, Parker HR. Surface forces in the lung, atelectasis, and transpulmonary pressure. *J Appl Physiol.* 1966;21(3):819–827. [[PubMed: 5912752](#)]

252. Rossi A, Gottfried SB, Higgs BD, Zocchi L, Grassino A, Milic-Emili J. Respiratory mechanics in mechanically ventilated patients with respiratory failure. *J Appl Physiol.* 1985;58(6):1849–1858. [[PubMed: 4008405](#)]

253. Bates JH, Baconnier P, Milic-Emili J. A theoretical analysis of interrupter technique for measuring respiratory mechanics. *J Appl Physiol.* 1988;64(5):2204–2214. [[PubMed: 3391919](#)]

254. Bates JH, Rossi A, Milic-Emili J. Analysis of the behavior of the respiratory system with constant inspiratory flow. *J Appl Physiol.* 1985;58(6):1840–1848. [[PubMed: 4008404](#)]

255. Kochi T, Okubo S, Zin WA, Milic-Emili J. Chest wall and respiratory system mechanics in cats: effects of flow and volume. *J Appl Physiol.* 1988;64(6):2636–2646. [[PubMed: 3403448](#)]

256. Kochi T, Okubo S, Zin WA, Milic-Emili J. Flow and volume dependence of pulmonary mechanics in anesthetized cats. *J Appl Physiol.* 1988;64(1):441–450. [[PubMed: 3356662](#)]

257. D'Angelo E, Calderini E, Torri G, et al. Respiratory mechanics in anesthetized paralyzed humans: effects of flow, volume, and time. *J Appl Physiol.* 1989;67(6):2556–2564. [[PubMed: 2606863](#)]

258. D'Angelo E, Robatto FM, Calderini E, et al. Pulmonary and chest wall mechanics in anesthetized paralyzed humans. *J Appl Physiol*. 1991;70(6):2602–2610. [PubMed: 1885455]

259. Jonson B, Nordstrom L, Olsson SG, Akerback D. Monitoring of ventilation and lung mechanics during automatic ventilation. A new device. *Bull Physiopathol Respir (Nancy)*. 1975;11(5):729–743. [PubMed: 1218293]

260. Rossi A, Gottfried SB, Zocchi L, et al. Measurement of static compliance of the total respiratory system in patients with acute respiratory failure during mechanical ventilation. The effect of intrinsic positive end-expiratory pressure. *Am Rev Respir Dis*. 1985;131(5):672–677. [PubMed: 4003913]

261. Leblanc P, Ruff F, Milic-Emili J. Effects of age and body position on “airway closure” in man. *J Appl Physiol*. 1970;28(4):448–451. [PubMed: 5437433]

262. Craig DB, Wahba WM, Don HF, et al. “Closing volume” and its relationship to gas exchange in seated and supine positions. *J Appl Physiol*. 1971;31(5):717–721. [PubMed: 5117187]

263. Westbrook PR, Stubbs SE, Sessler AD, et al. Effects of anesthesia and muscle paralysis on respiratory mechanics in normal man. *J Appl Physiol*. 1973;34(1):81–86. [PubMed: 4697382]

264. Svantesson C, Sigurdsson S, Larsson A, Jonson B. Effects of recruitment of collapsed lung units on the elastic pressure-volume relationship in anaesthetised healthy adults. *Acta Anaesthesiol Scand*. 1998;42(10):1149–1156. [PubMed: 9834796]

265. Locke TJ, Griffiths TL, Mould H, Gibson GJ. Rib cage mechanics after median sternotomy. *Thorax*. 1990;45(6):465–468. [PubMed: 2392792]

266. Ranieri VM, Brienza N, Santostasi S, et al. Impairment of lung and chest wall mechanics in patients with acute respiratory distress syndrome: role of abdominal distension. *Am J Respir Crit Care Med*. 1997;156(4 Pt 1):1082–1091.

267. Maracaja-Neto LF, Vercosa N, Roncally AC, et al. Beneficial effects of high positive end-expiratory pressure in lung respiratory mechanics during laparoscopic surgery. *Acta Anaesthesiol Scand*. 2009;53(2):210–217. [PubMed: 19175578]

268. Rehder K, Sittipong R, Sessler AD. The effects of thiopental-meperidine anesthesia with succinylcholine paralysis on functional residual capacity and dynamic lung compliance in normal sitting man. *Anesthesiology*. 1972;37(4):395–398. [PubMed: 5074719]

269. Rehder K, Mallow JE, Fibuch EE, et al. Effects of isoflurane anesthesia and muscle paralysis on respiratory mechanics in normal man. *Anesthesiology*. 1974;41(5):477–485. [PubMed: 4429217]

270. Dyhr T, Laursen N, Larsson A. Effects of lung recruitment maneuver and positive end-expiratory pressure on lung volume, respiratory mechanics and alveolar gas mixing in patients ventilated after cardiac surgery. *Acta Anaesthesiol Scand*. 2002;46(6):717–725. [PubMed: 12059898]

271. Koutsoukou A, Koulouris N, Bekos B, et al. Expiratory flow limitation in morbidly obese postoperative mechanically ventilated patients. *Acta Anaesthesiol Scand*. 2004;48(9):1080–1088. [PubMed: 15352952]

272. D'Angelo E, Calderini E, Tavola M, et al. Effect of PEEP on respiratory mechanics in anesthetized paralyzed humans. *J Appl Physiol*. 1992;73(5):1736–1742. [PubMed: 1474045]

273. D'Angelo E, Tavola M, Milic-Emili J. Volume and time dependence of respiratory system mechanics in normal anaesthetized paralysed humans. *Eur Respir J*. 2000;16(4):665–672. [PubMed: 11106210]

274. Sigurdsson S, Svantesson C, Larsson A, Jonson B. Elastic pressure-volume curves indicate derecruitment after a single deep expiration in anaesthetised and muscle-relaxed healthy man. *Acta Anaesthesiol Scand*. 2000;44(8):980–984. [PubMed: 10981576]

275. Pelosi P, Cereda M, Foti G, et al. Alterations of lung and chest wall mechanics in patients with acute lung injury: effects of positive end-expiratory pressure. *Am J Respir Crit Care Med*. 1995;152(2):531–537. [PubMed: 7633703]

276. Musch G, Foti G, Cereda M, et al. Lung and chest wall mechanics in normal anaesthetized subjects and in patients with COPD at different PEEP levels. *Eur Respir J*. 1997;10(11):2545–2552. [PubMed: 9426093]

277. Dechman GS, Chartrand DA, Ruiz-Neto PP, Bates JH. The effect of changing end-expiratory pressure on respiratory system mechanics in open- and closed-chest anesthetized, paralyzed patients. *Anesth Analg*. 1995;81(2):279–286. [PubMed: 7618715]

278. Cohendy R, Ripart J, Eledjam JJ. The effect of positive end-expiratory pressure on respiratory resistive properties in anaesthetized paralysed humans. *Eur Respir J*. 1994;7(2):286–291. [PubMed: 7909298]

279. Pontoppidan H, Geffin B, Lowenstein E. Acute respiratory failure in the adult. 1. *N Engl J Med.* 1972;287(14):690–698. [[PubMed: 4561570](#)]

280. Petty TL, Silvers GW, Paul GW, Stanford RE. Abnormalities in lung elastic properties and surfactant function in adult respiratory distress syndrome. *Chest.* 1979;75(5):571–574. [[PubMed: 582029](#)]

281. Dall'ava-Santucci J, Armaganidis A, Brunet F, et al. Mechanical effects of PEEP in patients with adult respiratory distress syndrome. *J Appl Physiol.* 1990;68(3):843–848. [[PubMed: 2187853](#)]

282. Gattinoni L, Pelosi P, Suter PM, et al. Acute respiratory distress syndrome caused by pulmonary and extrapulmonary disease. Different syndromes? *Am J Respir Crit Care Med.* 1998;158(1):3–11. [[PubMed: 9655699](#)]

283. Heldt GP, Peters RM. A simplified method to determine functional residual capacity during mechanical ventilation. *Chest.* 1978;74(5):492–496. [[PubMed: 738086](#)]

284. Weaver LJ, Pierson DJ, Kellie R, et al. A practical procedure for measuring functional residual capacity during mechanical ventilation with or without PEEP. *Crit Care Med.* 1981;9(12):873–877. [[PubMed: 7032846](#)]

285. Ibanez J, Raurich JM, Moris SG. Measurement of functional residual capacity during mechanical ventilation. Comparison of a computerized open nitrogen washout method with a closed helium dilution method. *Intensive Care Med.* 1983;9(2):91–93. [[PubMed: 6343447](#)]

286. Macnaughton PD, Morgan CJ, Denison DM, Evans TW. Measurement of carbon monoxide transfer and lung volume in ventilated subjects. *Eur Respir J.* 1993;6(2):231–236. [[PubMed: 8444295](#)]

287. Macnaughton PD, Evans TW. Measurement of lung volume and DLCO in acute respiratory failure. *Am J Respir Crit Care Med.* 1994;150(3):770–775. [[PubMed: 8087351](#)]

288. Kendrick AH. Comparison of methods of measuring static lung volumes. *Monaldi Arch Chest Dis.* 1996;51(5):431–439. [[PubMed: 9009635](#)]

289. Fretschner R, Deusch H, Weitnauer A, Brunner JX. A simple method to estimate functional residual capacity in mechanically ventilated patients. *Intensive Care Med.* 1993;19(7):372–376. [[PubMed: 8270715](#)]

290. Wrigge H, Sydow M, Zinserling J, et al. Determination of functional residual capacity (FRC) by multibreath nitrogen washout in a lung model and in mechanically ventilated patients. Accuracy depends on continuous dynamic compensation for changes of gas sampling delay time. *Intensive Care Med.* 1998;24(5):487–493. [[PubMed: 9660266](#)]

291. Jonmarker C, Castor R, Drefeldt B, Werner O. An analyzer for in-line measurement of expiratory sulfur hexafluoride concentration. *Anesthesiology.* 1985;63(1):84–88. [[PubMed: 4014774](#)]

292. Jonmarker C, Jansson L, Jonson B, et al. Measurement of functional residual capacity by sulfur hexafluoride washout. *Anesthesiology.* 1985;63(1):89–95. [[PubMed: 3925821](#)]

293. East TD, Wortelboer PJ, van Ark E, et al. Automated sulfur hexafluoride washout functional residual capacity measurement system for any mode of mechanical ventilation as well as spontaneous respiration. *Crit Care Med.* 1990;18(1):84–91. [[PubMed: 2293972](#)]

294. Beydon L, Svantesson C, Brauer K, et al. Respiratory mechanics in patients ventilated for critical lung disease. *Eur Respir J.* 1996;9(2):262–273. [[PubMed: 8777962](#)]

295. Richard JC, Le Bars D, Costes N, et al. Alveolar recruitment assessed by positron emission tomography during experimental acute lung injury. *Intensive Care Med.* 2006;32(11):1889–1894. [[PubMed: 17047926](#)]

296. Maggiore SM, Lellouche F, Pigeot J, et al. Prevention of endotracheal suctioning-induced alveolar derecruitment in acute lung injury. *Am J Respir Crit Care Med.* 2003;167(9):1215–1224. [[PubMed: 12615633](#)]

297. Maggiore SM, Richard JC, Brochard L. What has been learnt from P/V curves in patients with acute lung injury/acute respiratory distress syndrome. *Eur Respir J Suppl.* 2003;42:22s–26s.

298. Tobin MJ, Jenouri G, Birch S, et al. Effect of positive end-expiratory pressure on breathing patterns of normal subjects and intubated patients with respiratory failure. *Crit Care Med.* 1983;11(11):859–867. [[PubMed: 6354586](#)]

299. Hoffman RA, Ershowsky P, Krieger BP. Determination of auto-PEEP during spontaneous and controlled ventilation by monitoring changes in end-expiratory thoracic gas volume. *Chest*. 1989;96(3):613–616. [[PubMed: 2670467](#)]
300. Cereda M, Villa F, Colombo E, et al. Closed system endotracheal suctioning maintains lung volume during volume-controlled mechanical ventilation. *Intensive Care Med*. 2001;27(4):648–654. [[PubMed: 11398690](#)]
301. Nunes S, Uusaro A, Takala J. Pressure-volume relationships in acute lung injury: methodological and clinical implications. *Acta Anaesthesiol Scand*. 2004;48(3):278–286. [[PubMed: 14982559](#)]
302. Aliverti A, Dellaca R, Pelosi P, et al. Optoelectronic plethysmography in intensive care patients. *Am J Respir Crit Care Med*. 2000;161(5):1546–1552. [[PubMed: 10806152](#)]
303. Bouhemad B, Brisson H, Le-Guen M, et al. Bedside ultrasound assessment of positive end-expiratory pressure-induced lung recruitment. *Am J Respir Crit Care Med*. 2011;183(3):341–347. [[PubMed: 20851923](#)]
304. Lowhagen K, Lindgren S, Odenstedt H, et al. A new non-radiological method to assess potential lung recruitability: a pilot study in ALI patients. *Acta Anaesthesiol Scand*. 2011;55(2):165–174. [[PubMed: 21039359](#)]
305. Cereda M, Emami K, Kadlecsek S, et al. Quantitative imaging of alveolar recruitment with hyperpolarized gas MRI during mechanical ventilation. *J Appl Physiol*. 2011;110(2):499–511. [[PubMed: 21127207](#)]
306. Meier T, Luepschen H, Karsten J, et al. Assessment of regional lung recruitment and derecruitment during a PEEP trial based on electrical impedance tomography. *Intensive Care Med*. 2008;34(3):543–550. [[PubMed: 17653529](#)]
307. Gattinoni L, Caironi P, Pelosi P, Goodman LR. What has computed tomography taught us about the acute respiratory distress syndrome? *Am J Respir Crit Care Med*. 2001;164(9):1701–1711. [[PubMed: 11719313](#)]
308. Rouby JJ, Puybasset L, Nieszkowska A, Lu Q. Acute respiratory distress syndrome: lessons from computed tomography of the whole lung. *Crit Care Med*. 2003;31(4 Suppl):S285–S295.
309. Rouby JJ, Lu Q, Goldstein I. Selecting the right level of positive end-expiratory pressure in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2002;165(8):1182–1186. [[PubMed: 11956065](#)]
310. Rouby JJ, Lu Q, Vieira S. Pressure/volume curves and lung computed tomography in acute respiratory distress syndrome. *Eur Respir J Suppl*. 2003;42:27s–36s.
311. Nieszkowska A, Lu Q, Vieira S, et al. Incidence and regional distribution of lung overinflation during mechanical ventilation with positive end-expiratory pressure. *Crit Care Med*. 2004;32(7):1496–1503. [[PubMed: 15241094](#)]
312. Lu Q, Constantin JM, Nieszkowska A, et al. Measurement of alveolar derecruitment in patients with acute lung injury: computerized tomography versus pressure-volume curve. *Crit Care*. 2006;10(3):R95.
313. Dambrosio M, Roupie E, Mollet JJ, et al. Effects of positive end-expiratory pressure and different tidal volumes on alveolar recruitment and hyperinflation. *Anesthesiology*. 1997;87(3):495–503. [[PubMed: 9316952](#)]
314. Chelucci GL, Dall'Ava-Santucci J, Dhainaut JF, et al. Association of PEEP with two different inflation volumes in ARDS patients: effects on passive lung deflation and alveolar recruitment. *Intensive Care Med*. 2000;26(7):870–877. [[PubMed: 10990100](#)]
315. Mergoni M, Volpi A, Bricchi C, Rossi A. Lower inflection point and recruitment with PEEP in ventilated patients with acute respiratory failure. *J Appl Physiol*. 2001;91(1):441–450. [[PubMed: 11408462](#)]
316. Vieillard-Baron A, Prin S, Chergui K, et al. Early patterns of static pressure-volume loops in ARDS and their relations with PEEP-induced recruitment. *Intensive Care Med*. 2003;29(11):1929–1935. [[PubMed: 12923622](#)]
317. Albaiceta GM, Taboada F, Parra D, et al. Tomographic study of the inflection points of the pressure-volume curve in acute lung injury. *Am J Respir Crit Care Med*. 2004;170(10):1066–1072. [[PubMed: 15317670](#)]
318. Grasso S, Fanelli V, Cafarelli A, et al. Effects of high versus low positive end-expiratory pressures in acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2005;171(9):1002–1008. [[PubMed: 15665322](#)]

319. Gattinoni L, Bombino M, Pelosi P, et al. Lung structure and function in different stages of severe adult respiratory distress syndrome. *JAMA*. 1994;271(22):1772–1779. [PubMed: 8196122]

320. Grasso S, Mascia L, Del Turco M, et al. Effects of recruiting maneuvers in patients with acute respiratory distress syndrome ventilated with protective ventilatory strategy. *Anesthesiology*. 2002;96(4):795–802. [PubMed: 11964585]

321. Grasso S, Stripoli T, De Michele M, et al. ARDSnet ventilatory protocol and alveolar hyperinflation: role of positive end-expiratory pressure. *Am J Respir Crit Care Med*. 2007;176(8):761–767. [PubMed: 17656676]

322. Matamis D, Lemaire F, Harf A, et al. Total respiratory pressure-volume curves in the adult respiratory distress syndrome. *Chest*. 1984;86(1):58–66. [PubMed: 6734293]

323. Roupie E, Dambrosio M, Servillo G, et al. Titration of tidal volume and induced hypercapnia in acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 1995;152(1):121–128. [PubMed: 7599810]

324. International consensus conferences in intensive care medicine: ventilator-associated lung injury in ARDS. *Am J Respir Crit Care Med*. 1999;160(6):2118–2124.

325. Dreyfuss D, Saumon G. Ventilator-induced lung injury: lessons from experimental studies. *Am J Respir Crit Care Med*. 1998;157(1):294–323. [PubMed: 9445314]

326. Hickling KG. The pressure-volume curve is greatly modified by recruitment. A mathematical model of ARDS lungs. *Am J Respir Crit Care Med*. 1998;158(1):194–202. [PubMed: 9655729]

327. Hickling KG. Best compliance during a decremental, but not incremental, positive end-expiratory pressure trial is related to open-lung positive end-expiratory pressure: a mathematical model of acute respiratory distress syndrome lungs. *Am J Respir Crit Care Med*. 2001;163(1):69–78. [PubMed: 11208628]

328. Jonson B, Svantesson C. Elastic pressure-volume curves: what information do they convey? *Thorax*. 1999;54(1):82–87. [PubMed: 10343639]

329. Pelosi P, Goldner M, McKibben A, et al. Recruitment and derecruitment during acute respiratory failure: an experimental study. *Am J Respir Crit Care Med*. 2001;164(1):122–130. [PubMed: 11435250]

330. Servillo G, Svantesson C, Beydon L, et al. Pressure-volume curves in acute respiratory failure: automated low flow inflation versus occlusion. *Am J Respir Crit Care Med*. 1997;155(5):1629–1636. [PubMed: 9154868]

331. Cheney FW Jr, Martin WE. Effects of continuous positive-pressure ventilation on gas exchange in acute pulmonary edema. *J Appl Physiol*. 1971;30(3):378–381. [PubMed: 5544117]

332. Cook CD, Mead J, Schreiner GL, et al. Pulmonary mechanics during induced pulmonary edema in anesthetized dogs. *J Appl Physiol*. 1959;14(2):177–186. [PubMed: 13641138]

333. Grossman RF, Jones JG, Murray JF. Effects of oleic acid-induced pulmonary edema on lung mechanics. *J Appl Physiol*. 1980;48(6):1045–1051. [PubMed: 7380701]

334. Slutsky AS, Scharf SM, Brown R, Ingram RH Jr. The effect of oleic acid-induced pulmonary edema on pulmonary and chest wall mechanics in dogs. *Am Rev Respir Dis*. 1980;121(1):91–96. [PubMed: 7352717]

335. Lemaire F, Harf A, Simonneau G, et al. Echanges gazeux, courbe statique pression-volume et ventilation en pression positive de fin d'expiration. Etude dans seize cas d'insuffisance respiratoire aigue de l'adulte. *Ann Anesthesiol Fr*. 1981;22(5):435–441. [PubMed: 6122395]

336. Benito S, Lemaire F. Pulmonary pressure-volume relationship in acute respiratory distress syndrome in adults: role of positive end expiratory pressure. *J Crit Care*. 1990(1):27–34.

337. Dall'ava-Santucci J, Armaganidis A, Brunet F, et al. Causes of error of respiratory pressure-volume curves in paralyzed subjects. *J Appl Physiol*. 1988;64(1):42–49. [PubMed: 3356659]

338. Muscedere JG, Mullen JB, Gan K, Slutsky AS. Tidal ventilation at low airway pressures can augment lung injury. *Am J Respir Crit Care Med*. 1994;149(5):1327–1334. [PubMed: 8173774]

339. Hartog A, Vazquez de Anda GF, Gommers D, et al. At surfactant deficiency, application of “the open lung concept” prevents protein leakage and attenuates changes in lung mechanics. *Crit Care Med.* 2000;28(5):1450–1454. [[PubMed: 10834694](#)]

340. Grasso S, Terragni P, Mascia L, et al. Airway pressure-time curve profile (stress index) detects tidal recruitment/hyperinflation in experimental acute lung injury. *Crit Care Med.* 2004;32(4):1018–1027. [[PubMed: 15071395](#)]

341. Glaister DH, Schroter RC, Sudlow MF, Milic-Emili J. Bulk elastic properties of excised lungs and the effect of a transpulmonary pressure gradient. *Respir Physiol.* 1973;17(3):347–364. [[PubMed: 4634023](#)]

342. Vieillard-Baron A, Prin S, Schmitt JM, et al. Pressure-volume curves in acute respiratory distress syndrome: clinical demonstration of the influence of expiratory flow limitation on the initial slope. *Am J Respir Crit Care Med.* 2002;165(8):1107–1112. [[PubMed: 11956053](#)]

343. Glaister DH, Schroter RC, Sudlow MF, Milic-Emili J. Transpulmonary pressure gradient and ventilation distribution in excised lungs. *Respir Physiol.* 1973;17(3):365–385. [[PubMed: 4634024](#)]

344. Caramaz MP, Kacmarek RM, Helmy M, et al. A comparison of methods to identify open-lung PEEP. *Intensive Care Med.* 2009;35(4):740–747. [[PubMed: 19183951](#)]

345. Thille AW, Richard JC, Maggiore SM, et al. Alveolar recruitment in pulmonary and extrapulmonary acute respiratory distress syndrome: comparison using pressure-volume curve or static compliance. *Anesthesiology.* 2007;106(2):212–217. [[PubMed: 17264713](#)]

346. Mancebo J, Benito S, Martin M, Net A. Value of static pulmonary compliance in predicting mortality in patients with acute respiratory failure. *Intensive Care Med.* 1988;14(2):110–114. [[PubMed: 3361017](#)]

347. Suter PM, Fairley HB, Isenberg MD. Effect of tidal volume and positive end-expiratory pressure on compliance during mechanical ventilation. *Chest.* 1978;73:158–162. [[PubMed: 340159](#)]

348. Downie JM, Nam AJ, Simon BA. Pressure-volume curve does not predict steady-state lung volume in canine lavage lung injury. *Am J Respir Crit Care Med.* 2004;169(8):957–962. [[PubMed: 14764430](#)]

349. Carvalho AR, Spieth PM, Pelosi P, et al. Ability of dynamic airway pressure curve profile and elastance for positive end-expiratory pressure titration. *Intensive Care Med.* 2008;34(12):2291–2299. [[PubMed: 18825365](#)]

350. Stahl CA, Moller K, Schumann S, et al. Dynamic versus static respiratory mechanics in acute lung injury and acute respiratory distress syndrome. *Crit Care Med.* 2006;34(8):2090–2098. [[PubMed: 16755254](#)]

351. Suarez-Sipmann F, Bohm SH, Tusman G, et al. Use of dynamic compliance for open lung positive end-expiratory pressure titration in an experimental study. *Crit Care Med.* 2007;35(1):214–221. [[PubMed: 17110872](#)]

352. Eissa NT, Ranieri VM, Corbeil C, et al. Effect of PEEP on the mechanics of the respiratory system in ARDS patients. *J Appl Physiol.* 1992;73(5):1728–1735. [[PubMed: 1474044](#)]

353. Pesenti A, Pelosi P, Rossi N, et al. Respiratory mechanics and bronchodilator responsiveness in patients with the adult respiratory distress syndrome. *Crit Care Med.* 1993;21(1):78–83. [[PubMed: 8420734](#)]

354. Guerin C, Coussa ML, Eissa NT, et al. Lung and chest wall mechanics in mechanically ventilated COPD patients. *J Appl Physiol.* 1993;74(4):1570–1580. [[PubMed: 8514671](#)]

355. Jolliet P, Watremez C, Roeseler J, et al. Comparative effects of helium-oxygen and external positive end-expiratory pressure on respiratory mechanics, gas exchange, and ventilation-perfusion relationships in mechanically ventilated patients with chronic obstructive pulmonary disease. *Intensive Care Med.* 2003;29(9):1442–1450. [[PubMed: 12851764](#)]

356. Guerin C, LeMasson S, de Varax R, et al. Small airway closure and positive end-expiratory pressure in mechanically ventilated patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 1997;155(6):1949–1956. [[PubMed: 9196101](#)]

357. Kondili E, Alexopoulou C, Prinianakis G, et al. Pattern of lung emptying and expiratory resistance in mechanically ventilated patients with chronic obstructive pulmonary disease. *Intensive Care Med.* 2004;30(7):1311–1318. [[PubMed: 15054570](#)]

358. Fernandez R, Mancebo J, Blanch L, et al. Intrinsic PEEP on static pressure-volume curves. *Intensive Care Med.* 1990;16(4):233–236. [[PubMed: 2193042](#)]

359. Georgopoulos D, Giannouli E, Patakas D. Effects of extrinsic positive end-expiratory pressure on mechanically ventilated patients with chronic obstructive pulmonary disease and dynamic hyperinflation. *Intensive Care Med.* 1993;19(4):197–203. [[PubMed: 8366227](#)]

360. Lourens MS, Berg BV, Hoogsteden HC, Bogaard JM. Detection of flow limitation in mechanically ventilated patients. *Intensive Care Med.* 2001;27(8):1312–1320. [[PubMed: 11511944](#)]

361. Aerts JG, van den Berg B, Bogaard JM. Controlled expiration in mechanically-ventilated patients with chronic obstructive pulmonary disease (COPD). *Eur Respir J.* 1997;10(3):550–556. [[PubMed: 9072983](#)]

362. Kaczka DW, Ingenito EP, Body SC, et al. Inspiratory lung impedance in COPD: effects of PEEP and immediate impact of lung volume reduction surgery. *J Appl Physiol.* 2001;90(5):1833–1841. [[PubMed: 11299274](#)]

363. Allo JC, Beck JC, Brander L, et al. Influence of neurally adjusted ventilatory assist and positive end-expiratory pressure on breathing pattern in rabbits with acute lung injury. *Crit Care Med.* 2006;34(12):2997–3004. [[PubMed: 16957635](#)]

364. Beck J, Brander L, Slutsky AS, et al. Non-invasive neurally adjusted ventilatory assist in rabbits with acute lung injury. *Intensive Care Med.* 2008;34(2):316–323. [[PubMed: 17960364](#)]

365. Passath C, Takala J, Tuchscherer D, et al. Physiologic response to changing positive end-expiratory pressure during neurally adjusted ventilatory assist in sedated, critically ill adults. *Chest.* 2010;138(3):578–587. [[PubMed: 20435654](#)]

366. Parker JC, Hernandez LA, Peevy KJ. Mechanisms of ventilator-induced lung injury. *Crit Care Med.* 1993;21(1):131–143. [[PubMed: 8420720](#)]

367. Dos Santos CC, Slutsky AS. Invited review: mechanisms of ventilator-induced lung injury: a perspective. *J Appl Physiol.* 2000;89(4):1645–1655.

368. Dreyfuss D, Basset G, Soler P, Saumon G. Intermittent positive-pressure hyperventilation with high inflation pressures produces pulmonary microvascular injury in rats. *Am Rev Respir Dis.* 1985;132(4):880–884. [[PubMed: 3901844](#)]

369. Ranieri VM, Suter PM, Tortorella C, et al. Effect of mechanical ventilation on inflammatory mediators in patients with acute respiratory distress syndrome: a randomized controlled trial. *JAMA.* 1999;282(1):54–61. [[PubMed: 10404912](#)]

370. The Acute Respiratory Distress Syndrome Network. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med.* 2000;342(18):1301–1308.

371. Goldstein I, Bughalo MT, Marquette CH, et al. Mechanical ventilation-induced air-space enlargement during experimental pneumonia in piglets. *Am J Respir Crit Care Med.* 2001;163(4):958–964. [[PubMed: 11282773](#)]

372. Rouby JJ, Lherm T, Martin de Lassale E, et al. Histologic aspects of pulmonary barotrauma in critically ill patients with acute respiratory failure. *Intensive Care Med.* 1993;19(7):383–389. [[PubMed: 8270717](#)]

373. Argiras EP, Blakeley CR, Dunnill MS, et al. High PEEP decreases hyaline membrane formation in surfactant deficient lungs. *Br J Anaesth.* 1987;59(10):1278–1285. [[PubMed: 3314957](#)]

374. Mead J, Takishima T, Leith D. Stress distribution in lungs: a model of pulmonary elasticity. *J Appl Physiol.* 1970;28(5):596–608. [[PubMed: 5442255](#)]

375. Marini JJ. Ventilator-induced airway dysfunction? *Am J Respir Crit Care Med.* 2001;163(4):806–807. [[PubMed: 11282741](#)]

376. Sinclair SE, Molthen RC, Haworth ST, et al. Airway strain during mechanical ventilation in an intact animal model. *Am J Respir Crit Care Med.* 2007;176(8):786–794. [[PubMed: 17626911](#)]

377. Pierson DJ. Barotrauma and bronchopleural fistula. In: Tobin MJ, ed. *Principles and Practice of Mechanical Ventilation.* New York, NY: McGraw-Hill; 1994:813–836.

378. Villar J, Herrera-Abreu MT, Valladares F, et al. Experimental ventilator-induced lung injury: exacerbation by positive end-expiratory pressure. *Anesthesiology.* 2009;110(6):1341–1347. [[PubMed: 19417614](#)]

379. Carlton DP, Cummings JJ, Scheerer RG, et al. Lung overexpansion increases pulmonary microvascular protein permeability in young lambs. *J Appl Physiol.* 1990;69(2):577–583. [[PubMed: 2228868](#)]

380. Hernandez LA, Peevy KJ, Moise AA, Parker JC. Chest wall restriction limits high airway pressure-induced lung injury in young rabbits. *J Appl Physiol.* 1989;66(5):2364–2368. [PubMed: 2745302]
381. Meier T, Lange A, Papenberg H, et al. Pulmonary cytokine responses during mechanical ventilation of noninjured lungs with and without end-expiratory pressure. *Anesth Analg.* 2008;107(4):1265–1275. [PubMed: 18806039]
382. Caldini P, Leith JD, Brennan MJ. Effect of continuous positive-pressure ventilation (CPPV) on edema formation in dog lung. *J Appl Physiol.* 1975;39(4):672–679. [PubMed: 1104563]
383. Tounge T, Saharia P, Permutt S, et al. Aspiration pneumonia: beneficial and harmful effects of positive end-expiratory pressure. *Surgery.* 1977;82(2):279–283. [PubMed: 327600]
384. Hopewell PC. Failure of positive end-expiratory pressure to decrease lung water content in alloxan-induced pulmonary edema. *Am Rev Respir Dis.* 1979;120(4):813–819. [PubMed: 389115]
385. Hopewell PC, Murray JF. Effects of continuous positive-pressure ventilation in experimental pulmonary edema. *J Appl Physiol.* 1976;40(4):568–574. [PubMed: 6421]
386. Luce JM, Huang TW, Robertson HT, et al. The effects of prophylactic expiratory positive airway pressure on the resolution of oleic acid-induced lung injury in dogs. *Ann Surg.* 1983;197(3):327–336. [PubMed: 6338844]
387. Luce JM. The cardiovascular effects of mechanical ventilation and positive end-expiratory pressure. *JAMA.* 1984;252(6):807–811. [PubMed: 6379209]
388. Garcia-Delgado M, Tuma-Fernandez A, Chamorro-Marin V, et al. Alveolar fluid clearance in healthy pigs and influence of positive end-expiratory pressure. *Crit Care.* 2010;14(2):R36.
389. Colmenero-Ruiz M, Fernandez-Mondejar E, Fernandez-Sacristan MA, et al. PEEP and low tidal volume ventilation reduce lung water in porcine pulmonary edema. *Am J Respir Crit Care Med.* 1997;155(3):964–970. [PubMed: 9117033]
390. de Prost N, Roux D, Dreyfuss D, et al. Alveolar edema dispersion and alveolar protein permeability during high volume ventilation: effect of positive end-expiratory pressure. *Intensive Care Med.* 2007;33(4):711–717.
391. Schreiber T, Hueter L, Gaser E, et al. PEEP has beneficial effects on inflammation in the injured and no deleterious effects on the noninjured lung after unilateral lung acid instillation. *Intensive Care Med.* 2006;32(5):740–749. [PubMed: 16534568]
392. Dreyfuss D, Saumon G. Barotrauma is volutrauma, but which volume is the one responsible? *Intensive Care Med.* 1992;18(3):139–141. [PubMed: 1644960]
393. Tremblay L, Valenza F, Ribeiro SP, et al. Injurious ventilatory strategies increase cytokines and c-fos m-RNA expression in an isolated rat lung model. *J Clin Invest.* 1997;99(5):944–952. [PubMed: 9062352]
394. Tobin MJ. Advances in mechanical ventilation. *N Engl J Med.* 2001;344(26):1986–1996. [PubMed: 11430329]
395. McCann UG 2nd, Schiller HJ, Carney DE, et al. Visual validation of the mechanical stabilizing effects of positive end-expiratory pressure at the alveolar level. *J Surg Res.* 2001;99(2):335–342. [PubMed: 11469907]
396. Fanelli V, Mascia L, Puntorieri V, et al. Pulmonary atelectasis during low stretch ventilation: “open lung” versus “lung rest” strategy. *Crit Care Med.* 2009;37(3):1046–1053. [PubMed: 19237916]
397. Farias LL, Faffe DS, Xisto DG, et al. Positive end-expiratory pressure prevents lung mechanical stress caused by recruitment/derecruitment. *J Appl Physiol.* 2005;98(1):53–61. [PubMed: 15377644]
398. Pavone LA, Albert S, Carney D, et al. Injurious mechanical ventilation in the normal lung causes a progressive pathologic change in dynamic alveolar mechanics. *Crit Care.* 2007;11(3):R64.
399. Sandhar BK, Niblett DJ, Argiras EP, et al. Effects of positive end-expiratory pressure on hyaline membrane formation in a rabbit model of the neonatal respiratory distress syndrome. *Intensive Care Med.* 1988;14(5):538–546. [PubMed: 3146593]
400. Taskar V, John J, Evander E, et al. Surfactant dysfunction makes lungs vulnerable to repetitive collapse and reexpansion. *Am J Respir Crit Care Med.* 1997;155(1):313–320. [PubMed: 9001330]

401. Woo SW, Hedley-Whyte J. Macrophage accumulation and pulmonary edema due to thoracotomy and lung over inflation. *J Appl Physiol.* 1972;33(1):14–21. [[PubMed: 5037402](#)]

402. Lachmann B. Open up the lung and keep the lung open. *Intensive Care Med.* 1992;18(6):319–321. [[PubMed: 1469157](#)]

403. Faridy EE. Effect of ventilation on movement of surfactant in airways. *Respir Physiol.* 1976;27(3):323–334. [[PubMed: 989609](#)]

404. Wyszogrodski I, Kyei-Aboagye K, Taeusch HW Jr, Avery ME. Surfactant inactivation by hyperventilation: conservation by end-expiratory pressure. *J Appl Physiol.* 1975;38(3):461–466. [[PubMed: 1097385](#)]

405. Faridy EE, Permutt S, Riley RL. Effect of ventilation on surface forces in excised dogs' lungs. *J Appl Physiol.* 1966;21(5):1453–1462. [[PubMed: 5923215](#)]

406. Verbrugge SJ, Bohm SH, Gommers D, et al. Surfactant impairment after mechanical ventilation with large alveolar surface area changes and effects of positive end-expiratory pressure. *Br J Anaesth.* 1998;80(3):360–364. [[PubMed: 9623439](#)]

407. Ko SC, Zhang H, Haitisma JJ, et al. Effects of PEEP levels following repeated recruitment maneuvers on ventilator-induced lung injury. *Acta Anaesthesiol Scand.* 2008;52(4):514–521. [[PubMed: 18261196](#)]

408. Sugiura M, McCulloch PR, Wren S, et al. Ventilator pattern influences neutrophil influx and activation in atelectasis-prone rabbit lung. *J Appl Physiol.* 1994;77(3):1355–1365. [[PubMed: 7836140](#)]

409. Nahum A, Hoyt J, Schmitz L, et al. Effect of mechanical ventilation strategy on dissemination of intratracheally instilled *Escherichia coli* in dogs. *Crit Care Med.* 1997;25(10):1733–1743. [[PubMed: 9377891](#)]

410. Verbrugge SJ, Sorm V, van 't Veen A, et al. Lung overinflation without positive end-expiratory pressure promotes bacteremia after experimental *Klebsiella pneumoniae* inoculation. *Intensive Care Med.* 1998;24(2):172–177. [[PubMed: 9539077](#)]

411. van Kaam AH, Lachmann RA, Herting E, et al. Reducing atelectasis attenuates bacterial growth and translocation in experimental pneumonia. *Am J Respir Crit Care Med.* 2004;169(9):1046–1053.

412. Slutsky AS, Tremblay LN. Multiple system organ failure. Is mechanical ventilation a contributing factor? *Am J Respir Crit Care Med.* 1998;157(6 Pt 1):1721–1725.

413. Chiumello D, Pristine G, Slutsky AS. Mechanical ventilation affects local and systemic cytokines in an animal model of acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1999;160(1):109–116. [[PubMed: 10390387](#)]

414. Imai Y, Parodo J, Kajikawa O, et al. Injurious mechanical ventilation and end-organ epithelial cell apoptosis and organ dysfunction in an experimental model of acute respiratory distress syndrome. *JAMA.* 2003;289(16):2104–2112. [[PubMed: 12709468](#)]

415. Lachmann RA, van Kaam AH, Haitisma JJ, Lachmann B. High positive end-expiratory pressure levels promote bacterial translocation in experimental pneumonia. *Intensive Care Med.* 2007;33(10):1800–1804. [[PubMed: 17576531](#)]

416. Passaro CP, Silva PL, Rzezinski AF, et al. Pulmonary lesion induced by low and high positive end-expiratory pressure levels during protective ventilation in experimental acute lung injury. *Crit Care Med.* 2009;37(3):1011–1017. [[PubMed: 19237911](#)]

417. Ranieri VM, Giunta F, Suter PM, Slutsky AS. Mechanical ventilation as a mediator of multisystem organ failure in acute respiratory distress syndrome. *JAMA.* 2000;284(1):43–44. [[PubMed: 10872010](#)]

418. Cassidy SS, Schwiep F. Cardiovascular effects of positive end-expiratory pressure. In: Scharf SM, Cassidy SS, eds. *Heart-Lung Interactions in Health and Disease*. New York, NY: Marcel Dekker; 1989:463–506.

419. Dhainaut JF, Aouate P, Brunet FP. Circulatory effects of positive end-expiratory pressure in patients with acute lung injury. In: Scharf SM CS, ed. *Heart-Lung Interactions in Health and Disease*. New York, NY: Marcel Dekker; 1989:809–838.

420. Cassidy SS. Heart-lung interactions in health and disease. *Am J Med Sci.* 1987;294(6):451–461. [[PubMed: 3321986](#)]

421. Fessler HE, Permutt S. Interactions between the circulatory and ventilatory pumps. The thorax. In: Roussos C, Macklem PT, eds. *Lung Biology in Health and Disease*. 2nd ed. New York, NY: Marcel Dekker; 1995:1621–1640.

422. Fessler HE. Heart-lung interactions: applications in the critically ill. *Eur Respir J.* 1997;10(1):226–237. [[PubMed: 9032519](#)]

423. Michard F, Teboul JL. Using heart-lung interactions to assess fluid responsiveness during mechanical ventilation. *Crit Care*. 2000;4(5):282–289. [PubMed: 11094507]

424. Pinsky MR. Recent advances in the clinical application of heart-lung interactions. *Curr Opin Crit Care*. 2002;8(1):26–31. [PubMed: 12205403]

425. Vieillard-Baron A, Prin S, Chergui K, et al. Echo-Doppler demonstration of acute cor pulmonale at the bedside in the medical intensive care unit. *Am J Respir Crit Care Med*. 2002;166(10):1310–1319. [PubMed: 12421740]

426. Jardin F, Vieillard-Baron A. Right ventricular function and positive pressure ventilation in clinical practice: from hemodynamic subsets to respirator settings. *Intensive Care Med*. 2003;29(9):1426–1434. [PubMed: 12910335]

427. Cournand A, Motley H, Werko L, Richards D. Physiologic studies of the effects of intermittent positive pressure breathing on cardiac output in man. *Am J Physiol*. 1948;125:261–273.

428. Liebman PR, Patten MT, Manny J, et al. The mechanism of depressed cardiac output on positive end-expiratory pressure (PEEP). *Surgery*. 1978;83(5):594–598. [PubMed: 347615]

429. Guyton AC, Lindsey AW, Abernathy B, Richardson T. Venous return at various right atrial pressures and the normal venous return curve. *Am J Physiol*. 1957;189(3):609–615. [PubMed: 13458395]

430. Powers SR Jr, Mannal R, Neclerio M, et al. Physiologic consequences of positive end-expiratory pressure (PEEP) ventilation. *Ann Surg*. 1973;178(3):265–272. [PubMed: 4580970]

431. Scharf SM, Caldini P, Ingram RH Jr. Cardiovascular effects of increasing airway pressure in the dog. *Am J Physiol*. 1977;232(1):H35–H43.

432. Jellinek H, Krenn H, Oczenski W, et al. Influence of positive airway pressure on the pressure gradient for venous return in humans. *J Appl Physiol*. 2000;88(3):926–932. [PubMed: 10710387]

433. Fessler HE, Brower RG, Wise RA, Permutt S. Effects of positive end-expiratory pressure on the gradient for venous return. *Am Rev Respir Dis*. 1991;143(1):19–24. [PubMed: 1986678]

434. Peters J, Hecker B, Neuser D, Schaden W. Regional blood volume distribution during positive and negative airway pressure breathing in supine humans. *J Appl Physiol*. 1993;75(4):1740–1747. [PubMed: 8282628]

435. van den Berg PC, Jansen JR, Pinsky MR. Effect of positive pressure on venous return in volume-loaded cardiac surgical patients. *J Appl Physiol*. 2002;92(3):1223–1231.

436. Nanas S, Magder S. Adaptations of the peripheral circulation to PEEP. *Am Rev Respir Dis*. 1992;146(3):688–693. [PubMed: 1519849]

437. Sykes MK, Adams AP, Finlay WE, et al. The effects of variations in end-expiratory inflation pressure on cardiorespiratory function in normo-, hypo- and hypervolaemic dogs. *Br J Anaesth*. 1970;42(8):669–677. [PubMed: 5485130]

438. Schreuder JJ, Jansen JR, Versprille A. Hemodynamic effects of PEEP applied as a ramp in normo-, hyper-, and hypovolemia. *J Appl Physiol*. 1985;59(4):1178–1184. [PubMed: 3902776]

439. Haynes JB, Carson SD, Whitney WP, et al. Positive end-expiratory pressure shifts left ventricular diastolic pressure-area curves. *J Appl Physiol*. 1980;48(4):670–676. [PubMed: 6991464]

440. Dhainaut JF, Devaux JY, Monsallier JF, et al. Mechanisms of decreased left ventricular preload during continuous positive pressure ventilation in ARDS. *Chest*. 1986;90(1):74–80. [PubMed: 3522122]

441. Lemaire F, Teboul JL, Cinotti L, et al. Acute left ventricular dysfunction during unsuccessful weaning from mechanical ventilation. *Anesthesiology*. 1988;69(2):171–179. [PubMed: 3044189]

442. Taylor RR, Covell JW, Sonnenblick EH, Ross J Jr. Dependence of ventricular distensibility on filling of the opposite ventricle. *Am J Physiol*. 1967;213(3):711–718. [PubMed: 6036789]

443. Scharf SM, Brown R. Influence of the right ventricle on canine left ventricular function with PEEP. *J Appl Physiol*. 1982;52(1):254–259. [PubMed: 7037711]

444. Cassidy SS, Mitchell JH, Johnson RL Jr. Dimensional analysis of right and left ventricles during positive-pressure ventilation in dogs. *Am J Physiol.* 1982;242(4):H549–H556.
445. Whittenberger JL, Mc GM, Berglund E, Borst HG. Influence of state of inflation of the lung on pulmonary vascular resistance. *J Appl Physiol.* 1960;15:878–882. [[PubMed: 13784949](#)]
446. Permutt S, Howell JB, Proctor DF, Riley RL. Effect of lung inflation on static pressure-volume characteristics of pulmonary vessels. *J Appl Physiol.* 1961;16:64–70. [[PubMed: 13734447](#)]
447. Brower R, Wise RA, Hassapoyannes C, et al. Effect of lung inflation on lung blood volume and pulmonary venous flow. *J Appl Physiol.* 1985;58(3):954–963. [[PubMed: 3884583](#)]
448. Vieillard-Baron A, Loubieres Y, Schmitt JM, et al. Cyclic changes in right ventricular output impedance during mechanical ventilation. *J Appl Physiol.* 1999;87(5):1644–1650. [[PubMed: 10562603](#)]
449. Artucio H, Hurtado J, Zimet L, et al. PEEP-induced tricuspid regurgitation. *Intensive Care Med.* 1997;23(8):836–840. [[PubMed: 9310800](#)]
450. Culver BH, Marini JJ, Butler J. Lung volume and pleural pressure effects on ventricular function. *J Appl Physiol.* 1981;50(3):630–635. [[PubMed: 7019172](#)]
451. Gernoth C, Wagner G, Pelosi P, Luecke T. Respiratory and haemodynamic changes during decremental open lung positive end-expiratory pressure titration in patients with acute respiratory distress syndrome. *Crit Care.* 2009;13(2):R59.
452. Mekontso Dessap A, Charron C, Devaquet J, et al. Impact of acute hypercapnia and augmented positive end-expiratory pressure on right ventricle function in severe acute respiratory distress syndrome. *Intensive Care Med.* 2009;35(11):1850–1858.
453. Cabrera MR, Nakamura GE, Montague DA, Cole RP. Effect of airway pressure on pericardial pressure. *Am Rev Respir Dis.* 1989;140(3):659–667. [[PubMed: 2675704](#)]
454. Chapin JC, Downs JB, Douglas ME, et al. Lung expansion, airway pressure transmission, and positive end-expiratory pressure. *Arch Surg.* 1979;114(10):1193–1197. [[PubMed: 384964](#)]
455. O'Quin RJ, Marini JJ, Culver BH, Butler J. Transmission of airway pressure to pleural space during lung edema and chest wall restriction. *J Appl Physiol.* 1985;59(4):1171–1177. [[PubMed: 3902775](#)]
456. Venus B, Cohen LE, Smith RA. Hemodynamics and intrathoracic pressure transmission during controlled mechanical ventilation and positive end-expiratory pressure in normal and low compliant lungs. *Crit Care Med.* 1988;16(7):686–690. [[PubMed: 3286121](#)]
457. Miller RR, Vismara LA, Zelis R, et al. Clinical use of sodium nitroprusside in chronic ischemic heart disease. Effects on peripheral vascular resistance and venous tone and on ventricular volume, pump and mechanical performance. *Circulation.* 1975;51(2):328–336. [[PubMed: 1112013](#)]
458. Summer WR, Permutt S, Sagawa K, et al. Effects of spontaneous respiration on canine left ventricular function. *Circ Res.* 1979;45(6):719–728. [[PubMed: 498435](#)]
459. Hall MJ, Ando S, Floras JS, Bradley TD. Magnitude and time course of hemodynamic responses to Mueller maneuvers in patients with congestive heart failure. *J Appl Physiol.* 1998;85(4):1476–1484. [[PubMed: 9760344](#)]
460. Noble WH, Kay JC, Obdrzalek J. Lung mechanics in hypervolemic pulmonary edema. *J Appl Physiol.* 1975;38(4):681–687. [[PubMed: 1141099](#)]
461. Bernard GR, Pou NA, Coggeshall JW, et al. Comparison of the pulmonary dysfunction caused by cardiogenic and noncardiogenic pulmonary edema. *Chest.* 1995;108(3):798–803. [[PubMed: 7656636](#)]
462. Hogg JC, Agarawal JB, Gardiner AJ, et al. Distribution of airway resistance with developing pulmonary edema in dogs. *J Appl Physiol.* 1972;32(1):20–24. [[PubMed: 5007012](#)]
463. Tang GJ, Freed AN. The role of submucosal oedema in increased peripheral airway resistance by intravenous volume loading in dogs. *Eur Respir J.* 1994;7(2):311–317. [[PubMed: 7512932](#)]
464. Pellegrino R, Dellaca R, Macklem PT, et al. Effects of rapid saline infusion on lung mechanics and airway responsiveness in humans. *J Appl Physiol.* 2003;95(2):728–734. [[PubMed: 12730149](#)]

465. Aubier M, Trippebach T, Roussos C. Respiratory muscle fatigue during cardiogenic shock. *J Appl Physiol*. 1981;51(2):499–508. [PubMed: 6790504]
466. Rutledge FS, Hussain SN, Roussos C, Magder S. Diaphragmatic energetics and blood flow during pulmonary edema and hypotension. *J Appl Physiol*. 1988;64(5):1908–1915. [PubMed: 3391891]
467. Lockhat D, Roussos C, Ianuzzo CD. Metabolite changes in the loaded hypoperfused and failing diaphragm. *J Appl Physiol*. 1988;65(4):1563–1571. [PubMed: 3141359]
468. Supinski GS. Respiratory muscle blood flow. *Clin Chest Med*. 1988;9(2):211–223. [PubMed: 3292124]
469. Schumacker PT, Cain SM. The concept of a critical oxygen delivery. *Intensive Care Med*. 1987;13(4):223–229. [PubMed: 3301969]
470. Gilston A. PEEP and oxygen balance: where are the emperor's clothes? *Intensive Crit Care Dig*. 1990;9:7–13.
471. Danek SJ, Lynch JP, Weg JG, Dantzker DR. The dependence of oxygen uptake on oxygen delivery in the adult respiratory distress syndrome. *Am Rev Respir Dis*. 1980;122(3):387–395. [PubMed: 7416614]
472. Mohsenifar Z, Goldbach P, Tashkin DP, Campisi DJ. Relationship between O₂ delivery and O₂ consumption in the adult respiratory distress syndrome. *Chest*. 1983;84(3):267–271. [PubMed: 6349945]
473. Thorens JB, Joliet P, Ritz M, Chevrolet JC. Effects of rapid permissive hypercapnia on hemodynamics, gas exchange, and oxygen transport and consumption during mechanical ventilation for the acute respiratory distress syndrome. *Intensive Care Med*. 1996;22(3):182–191. [PubMed: 8727430]
474. Carvalho CR, Barbas CS, Medeiros DM, et al. Temporal hemodynamic effects of permissive hypercapnia associated with ideal PEEP in ARDS. *Am J Respir Crit Care Med*. 1997;156(5):1458–1466. [PubMed: 9372661]
475. Puybasset L, Stewart T, Rouby JJ, et al. Inhaled nitric oxide reverses the increase in pulmonary vascular resistance induced by permissive hypercapnia in patients with acute respiratory distress syndrome. *Anesthesiology*. 1994;80(6):1254–1267. [PubMed: 8010472]
476. Venus B, Mathru M, Smith RA, et al. Renal function during application of positive end-expiratory pressure in swine: effects of hydration. *Anesthesiology*. 1985;62(6):765–769. [PubMed: 3890619]
477. Annat G, Viale JP, Bui Xuan B, et al. Effect of PEEP ventilation on renal function, plasma renin, aldosterone, neurophysins and urinary ADH, and prostaglandins. *Anesthesiology*. 1983;58(2):136–141. [PubMed: 6337527]
478. Andrivet P, Adnot S, Sanker S, et al. Hormonal interactions and renal function during mechanical ventilation and ANF infusion in humans. *J Appl Physiol*. 1991;70(1):287–292. [PubMed: 1826291]
479. Gammanpila S, Bevan DR, Bhudu R. Effect of positive and negative expiratory pressure on renal function. *Br J Anaesth*. 1977;49(3):199–205. [PubMed: 334205]
480. Hemmer M, Suter PM. Treatment of cardiac and renal effects of PEEP with dopamine in patients with acute respiratory failure. *Anesthesiology*. 1979;50(5):399–403. [PubMed: 378029]
481. Jacob LP, Chazalet JJ, Payen DM, et al. Renal hemodynamic and functional effect of PEEP ventilation in human renal transplantations. *Am J Respir Crit Care Med*. 1995;152(1):103–107. [PubMed: 7599806]
482. Jarnberg PO, de Villota ED, Eklund J, Granberg PO. Effects of positive end-expiratory pressure on renal function. *Acta Anaesthesiol Scand*. 1978;22(5):508–514. [PubMed: 358723]
483. Kaczmarczyk G, Vogel S, Krebs M, et al. Vasopressin and renin-angiotensin maintain arterial pressure during PEEP in nonexpanded, conscious dogs. *Am J Physiol Regul Integr Comp Physiol*. 1996;271(5):R1396–R1402.
484. Kaukinen S, Eerola R. Positive end expiratory pressure ventilation, renal function and renin. *Ann Clin Res*. 1979;11(2):58–62. [PubMed: 378092]
485. Kharasch ED, Yeo KT, Kenny MA, Buffington CW. Atrial natriuretic factor may mediate the renal effects of PEEP ventilation. *Anesthesiology*. 1988;69(6):862–869. [PubMed: 2973759]
486. Andrivet P AS, Brun-Buisson C, Chabrier PE, et al. Involvement of ANF in the acute antidiuresis during PEEP ventilation. *J Appl Physiol*. 1988;65(5):1967–1974. [PubMed: 2974845]

487. Payen DM, Brun-Buisson CJ, Carli PA, et al. Hemodynamic, gas exchange, and hormonal consequences of LBPP during PEEP ventilation. *J Appl Physiol.* 1987;62(1):61–70. [[PubMed: 3549670](#)]

488. Shirakami G, Magaribuchi T, Shingu K, et al. Positive end-expiratory pressure ventilation decreases plasma atrial and brain natriuretic peptide levels in humans. *Anesth Analg.* 1993;77(6):1116–1121. [[PubMed: 8250300](#)]

489. Ramamoorthy C, Rooney MW, Dries DJ, Mathru M. Aggressive hydration during continuous positive-pressure ventilation restores atrial transmural pressure, plasma atrial natriuretic peptide concentrations, and renal function. *Crit Care Med.* 1992;20(7):1014–1019. [[PubMed: 1535581](#)]

490. Leithner C, Frass M, Pacher R, et al. Mechanical ventilation with positive end-expiratory pressure decreases release of alpha-atrial natriuretic peptide. *Crit Care Med.* 1987;15(5):484–488. [[PubMed: 2952461](#)]

491. Payen DM, Farge D, Beloucif S, et al. No involvement of antidiuretic hormone in acute antidiuresis during PEEP ventilation in humans. *Anesthesiology.* 1987;66(1):17–23. [[PubMed: 3541690](#)]

492. Boemke W, Krebs MO, Djalali K, et al. Renal nerves are not involved in sodium and water retention during mechanical ventilation in awake dogs. *Anesthesiology.* 1998;89(4):942–953. [[PubMed: 9778012](#)]

493. Christensen G, Bugge JF, Ostensen J, Kiil F. Atrial natriuretic factor and renal sodium excretion during ventilation with PEEP in hypervolemic dogs. *J Appl Physiol.* 1992;72(3):993–997. [[PubMed: 1533215](#)]

494. Fewell JE, Bond GC. Renal denervation eliminates the renal response to continuous positive-pressure ventilation. *Proc Soc Exp Biol Med.* 1979;161(4):574–578. [[PubMed: 384401](#)]

495. Mullins RJ, Dawe EJ, Lucas CE, et al. Mechanisms of impaired renal function with PEEP. *J Surg Res.* 1984;37(3):189–196. [[PubMed: 6379297](#)]

496. Rossaint R, Jorres D, Nienhaus M, et al. Positive end-expiratory pressure reduces renal excretion without hormonal activation after volume expansion in dogs. *Anesthesiology.* 1992;77(4):700–708. [[PubMed: 1416167](#)]

497. Aibiki M, Koyama S, Ogi K, Shirakawa Y. Reflex mechanisms for changes in renal nerve activity during positive end-expiratory pressure. *J Appl Physiol.* 1988;65(1):109–115. [[PubMed: 3042737](#)]

498. Kaczmarczyk G, Rossaint R, Altmann C, et al. ACE inhibition facilitates sodium and water excretion during PEEP in conscious volume-expanded dogs. *J Appl Physiol.* 1992;73(3):962–967. [[PubMed: 1400063](#)]

499. Farge D, De la Coussaye JE, Beloucif S, et al. Interactions between hemodynamic and hormonal modifications during PEEP-induced antidiuresis and antinatriuresis. *Chest.* 1995;107(4):1095–1100. [[PubMed: 7705122](#)]

500. Rossaint R, Krebs M, Forther J, et al. Inferior vena caval pressure increase contributes to sodium and water retention during PEEP in awake dogs. *J Appl Physiol.* 1993;75(6):2484–2492. [[PubMed: 8125866](#)]

501. Chernow B, Soldano S, Cook D, et al. Positive end-expiratory pressure increases plasma catecholamine levels in non-volume loaded dogs. *Anaesth Intensive Care.* 1986;14(4):421–425. [[PubMed: 3551676](#)]

502. Sha M, Saito Y, Yokoyama K, et al. Effects of continuous positive-pressure ventilation on hepatic blood flow and intrahepatic oxygen delivery in dogs. *Crit Care Med.* 1987;15(11):1040–1043. [[PubMed: 3315466](#)]

503. Fujita Y. Effects of PEEP on splanchnic hemodynamics and blood volume. *Acta Anaesthesiol Scand.* 1993;37(4):427–431. [[PubMed: 8322573](#)]

504. Steinberg S, Azar G, Love R, et al. Dopexamine prevents depression of mesenteric blood flow caused by positive end-expiratory pressure in rats. *Surgery.* 1996;120(4):597–601; discussion 601–602. [[PubMed: 8862366](#)]

505. Kiefer P, Nunes S, Kosonen P, Takala J. Effect of positive end-expiratory pressure on splanchnic perfusion in acute lung injury. *Intensive Care Med.* 2000;26(4):376–383. [[PubMed: 10872128](#)]

506. Brienza N, Revelly JP, Ayuse T, Robotham JL. Effects of PEEP on liver arterial and venous blood flows. *Am J Respir Crit Care Med.* 1995;152(2):504–510. [[PubMed: 7633699](#)]

507. Aneman A, Ponten J, Fandriks L, et al. Hemodynamic, sympathetic and angiotensin II responses to PEEP ventilation before and during administration of isoflurane. *Acta Anaesthesiol Scand.* 1997;41(1 Pt 1):41–48.

508. Fournell A, Schwarte LA, Kindgen-Milles D, et al. Assessment of microvascular [oxygen](#) saturation in gastric mucosa in volunteers breathing continuous positive airway pressure. *Crit Care Med*. 2003;31(6):1705–1710. [[PubMed: 12794408](#)]

509. Trager K, Radermacher P, Georgieff M. PEEP and hepatic metabolic performance in septic shock. *Intensive Care Med*. 1996;22(11):1274–1275. [[PubMed: 9120128](#)]

510. Brienza N, Dalfino L, Cinnella G, et al. Jaundice in critical illness: promoting factors of a concealed reality. *Intensive Care Med*. 2006;32(2):267–274. [[PubMed: 16450099](#)]

511. Saner FH, Pavlakovic G, Gu Y, et al. Does PEEP impair the hepatic outflow in patients following liver transplantation? *Intensive Care Med*. 2006;32(10):1584–1590. [[PubMed: 16941166](#)]

512. Kocabayoglu P, Treckmann JW, Beckebaum S, et al. Does long-term ventilation with PEEP > or = 10 mbar influence graft-function in patients following liver transplantation? *Hepatogastroenterology*. 2010;57(101):839–844. [[PubMed: 21033239](#)]

513. Cooper KR, Boswell PA. Reduced functional residual capacity and abnormal oxygenation in patients with severe head injury. *Chest*. 1983;84(1):29–35. [[PubMed: 6345100](#)]

514. Holt J. The effect of positive and negative intrathoracic pressure on peripheral venous pressure in man. *Am J Physiol*. 1943;139:209–211.

515. Luce JM, Huseby JS, Kirk W, Butler J. Mechanism by which positive end-expiratory pressure increases cerebrospinal fluid pressure in dogs. *J Appl Physiol*. 1982;52(1):231–235. [[PubMed: 7037710](#)]

516. Hormann C, Mohsenipour I, Gottardis M, Benzer A. Response of cerebrospinal fluid pressure to continuous positive airway pressure in volunteers. *Anesth Analg*. 1994;78(1):54–57. [[PubMed: 8267181](#)]

517. Lodrini S, Montolivo M, Pluchino F, Borroni V. Positive end-expiratory pressure in supine and sitting positions: its effects on intrathoracic and intracranial pressures. *Neurosurgery*. 1989;24(6):873–877. [[PubMed: 2664547](#)]

518. Muench E, Bauhuf C, Roth H, et al. Effects of positive end-expiratory pressure on regional cerebral blood flow, intracranial pressure, and brain tissue oxygenation. *Crit Care Med*. 2005;33(10):2367–2372. [[PubMed: 16215394](#)]

519. Caricato A, Conti G, Della Corte F, et al. Effects of PEEP on the intracranial system of patients with head injury and subarachnoid hemorrhage: the role of respiratory system compliance. *J Trauma*. 2005;58(3):571–576. [[PubMed: 15761353](#)]

520. Mascia L, Grasso S, Fiore T, et al. Cerebro-pulmonary interactions during the application of low levels of positive end-expiratory pressure. *Intensive Care Med*. 2005;31(3):373–379. [[PubMed: 15668765](#)]

521. Cooper KR, Boswell PA, Choi SC. Safe use of PEEP in patients with severe head injury. *J Neurosurg*. 1985;63(4):552–555. [[PubMed: 3897477](#)]

522. McGuire G, Crossley D, Richards J, Wong D. Effects of varying levels of positive end-expiratory pressure on intracranial pressure and cerebral perfusion pressure. *Crit Care Med*. 1997;25(6):1059–1062. [[PubMed: 9201061](#)]

523. Videtta W, Villarejo F, Cohen M, et al. Effects of positive end-expiratory pressure on intracranial pressure and cerebral perfusion pressure. *Acta Neurochir Suppl*. 2002;81:93–97. [[PubMed: 12168368](#)]

524. Huynh T, Messer M, Sing RF, et al. Positive end-expiratory pressure alters intracranial and cerebral perfusion pressure in severe traumatic brain injury. *J Trauma*. 2002;53(3):488–492; discussion 492–493. [[PubMed: 12352486](#)]

525. Georgiadis D, Schwarz S, Baumgartner RW, et al. Influence of positive end-expiratory pressure on intracranial pressure and cerebral perfusion pressure in patients with acute stroke. *Stroke*. 2001;32(9):2088–2092. [[PubMed: 11546901](#)]

526. Frost EA. Effects of positive end-expiratory pressure on intracranial pressure and compliance in brain-injured patients. *J Neurosurg*. 1977;47(2):195–200. [[PubMed: 327031](#)]

527. Mascia L. Acute lung injury in patients with severe brain injury: a double hit model. *Neurocrit Care*. 2009;11(3):417–426. [[PubMed: 19548120](#)]

528. Baile EM, Albert RK, Kirk W, et al. Positive end-expiratory pressure decreases bronchial blood flow in the dog. *J Appl Physiol*. 1984;56(5):1289–1293. [[PubMed: 6373694](#)]

529. Cassidy SS, Haynes MS. The effects of ventilation with positive end-expiratory pressure on the bronchial circulation. *Respir Physiol.* 1986;66(3):269–278. [\[PubMed: 3541095\]](#)
530. Agostoni P, Arena V, Biglioli P, et al. Increase of alveolar pressure reduces systemic-to-pulmonary bronchial blood flow in humans. *Chest.* 1989;96(5):1081–1085. [\[PubMed: 2680319\]](#)
531. Haider M, Schad H, Mendler N. Thoracic duct lymph and PEEP studies in anaesthetized dogs. II. Effect of a thoracic duct fistula on the development of a hyponcotic-hydrostatic pulmonary oedema. *Intensive Care Med.* 1987;13(4):278–283. [\[PubMed: 3301971\]](#)
532. Lakshminarayan S, Agostoni PG, Kirk W. Vagal cooling and positive end-expiratory pressure reduce systemic to pulmonary bronchial blood flow in dogs. *Respiration.* 1990;57(2):85–89. [\[PubMed: 2236937\]](#)
533. van der Zee H, Cooper JA, Hakim TS, Malik AB. Alterations in pulmonary fluid balance induced by positive end-expiratory pressure. *Respir Physiol.* 1986;64(2):125–133.
534. Navalesi P, Hernandez P, Laporta D, et al. Influence of site of tracheal pressure measurement on in situ estimation of endotracheal tube resistance. *J Appl Physiol.* 1994;77(6):2899–2906. [\[PubMed: 7896638\]](#)
535. Valta P, Corbeil C, Lavoie A, et al. Detection of expiratory flow limitation during mechanical ventilation. *Am J Respir Crit Care Med.* 1994;150(5 Pt 1):1311–1317.
536. Gay PC, Rodarte JR, Hubmayr RD. The effects of positive expiratory pressure on isovolume flow and dynamic hyperinflation in patients receiving mechanical ventilation. *Am Rev Respir Dis.* 1989;139(3):621–626. [\[PubMed: 2647006\]](#)
537. Alvisi V, Romanello A, Badet M, et al. Time course of expiratory flow limitation in COPD patients during acute respiratory failure requiring mechanical ventilation. *Chest.* 2003;123(5):1625–1632. [\[PubMed: 12740283\]](#)
538. Ninane V, Leduc D, Kafi SA, et al. Detection of expiratory flow limitation by manual compression of the abdominal wall. *Am J Respir Crit Care Med.* 2001;163(6):1326–1330. [\[PubMed: 11371396\]](#)
539. Maltais F, Sovilj M, Goldberg P, Gottfried SB. Respiratory mechanics in status asthmaticus. Effects of inhalational anesthesia. *Chest.* 1994;106(5):1401–1406. [\[PubMed: 7956391\]](#)
540. Pellegrino R, Violante B, Nava S, et al. Expiratory airflow limitation and hyperinflation during methacholine-induced bronchoconstriction. *J Appl Physiol.* 1993;75(4):1720–1727. [\[PubMed: 8282625\]](#)
541. Ferretti A, Giampiccolo P, Cavalli A, et al. Expiratory flow limitation and orthopnea in massively obese subjects. *Chest.* 2001;119(5):1401–1408. [\[PubMed: 11348945\]](#)
542. Marini JJ. Auto-positive end-expiratory pressure and flow limitation in adult respiratory distress syndrome—intrinsically different? *Crit Care Med.* 2002;30(9):2140–2141. [\[PubMed: 12352054\]](#)
543. Gottfried SB, Reissman H, Ranieri VM. A simple method for the measurement of intrinsic positive end-expiratory pressure during controlled and assisted modes of mechanical ventilation. *Crit Care Med.* 1992;20(5):621–629. [\[PubMed: 1572187\]](#)
544. Leatherman JW, Ravenscraft SA. Low measured auto-positive end-expiratory pressure during mechanical ventilation of patients with severe asthma: hidden auto-positive end-expiratory pressure. *Crit Care Med.* 1996;24(3):541–546. [\[PubMed: 8625647\]](#)
545. Haluszka J, Chartrand DA, Grassino AE, Milic-Emili J. Intrinsic PEEP and arterial PCO₂ in stable patients with chronic obstructive pulmonary disease. *Am Rev Respir Dis.* 1990;141(5 Pt 1):1194–1197.
546. Hernandez P, Navalesi P, Maltais F, et al. Comparison of static and dynamic measurements of intrinsic PEEP in anesthetized cats. *J Appl Physiol.* 1994;76(6):2437–2442. [\[PubMed: 7928868\]](#)
547. Maltais F, Reissmann H, Navalesi P, et al. Comparison of static and dynamic measurements of intrinsic PEEP in mechanically ventilated patients. *Am J Respir Crit Care Med.* 1994;150(5 Pt 1):1318–1324.
548. Jubran A, Laghi F, Mazur M, et al. Partitioning of lung and chest-wall mechanics before and after lung-volume-reduction surgery. *Am J Respir Crit Care Med.* 1998;158(1):306–310. [\[PubMed: 9655744\]](#)

549. Yan S, Kayser B, Tobiasz M, Sliwinski P. Comparison of static and dynamic intrinsic positive end-expiratory pressure using the Campbell diagram. *Am J Respir Crit Care Med*. 1996;154(4 Pt 1):938–944.
550. Fleury B, Murciano D, Talamo C, et al. Work of breathing in patients with chronic obstructive pulmonary disease in acute respiratory failure. *Am Rev Respir Dis*. 1985;131(6):822–827. [[PubMed: 4003930](#)]
551. MacIntyre NR, Cheng KC, McConnell R. Applied PEEP during pressure support reduces the inspiratory threshold load of intrinsic PEEP. *Chest*. 1997;111(1):188–193. [[PubMed: 8996015](#)]
552. Ninane V, Rypens F, Yernault JC, De Troyer A. Abdominal muscle use during breathing in patients with chronic airflow obstruction. *Am Rev Respir Dis*. 1992;146(1):16–21. [[PubMed: 1385684](#)]
553. Ninane V, Yernault JC, de Troyer A. Intrinsic PEEP in patients with chronic obstructive pulmonary disease. Role of expiratory muscles. *Am Rev Respir Dis*. 1993;148(4 Pt 1):1037–1042.
554. Parthasarathy S, Jubran A, Tobin MJ. Cycling of inspiratory and expiratory muscle groups with the ventilator in airflow limitation. *Am J Respir Crit Care Med*. 1998;158(5 Pt 1):1471–1478.
555. Gorini M, Misuri G, Duranti R, et al. Abdominal muscle recruitment and PEEPi during bronchoconstriction in chronic obstructive pulmonary disease. *Thorax*. 1997;52(4):355–361. [[PubMed: 9196519](#)]
556. Lessard MR, Lofaso F, Brochard L. Expiratory muscle activity increases intrinsic positive end-expiratory pressure independently of dynamic hyperinflation in mechanically ventilated patients. *Am J Respir Crit Care Med*. 1995;151(2 Pt 1):562–569.
557. Zakyntinos SG, Vassilakopoulos T, Zakyntinos E, Roussos C. Accurate measurement of intrinsic positive end-expiratory pressure: how to detect and correct for expiratory muscle activity. *Eur Respir J*. 1997;10(3):522–529. [[PubMed: 9072979](#)]
558. Zakyntinos SG, Vassilakopoulos T, Zakyntinos E, et al. Contribution of expiratory muscle pressure to dynamic intrinsic positive end-expiratory pressure: validation using the Campbell diagram. *Am J Respir Crit Care Med*. 2000;162(5):1633–1640. [[PubMed: 11069788](#)]
559. Zakyntinos SG, Vassilakopoulos T, Zakyntinos E, et al. Correcting static intrinsic positive end-expiratory pressure for expiratory muscle contraction. Validation of a new method. *Am J Respir Crit Care Med*. 1999;160(3):785–790. [[PubMed: 10471597](#)]
560. Cole AG, Weller SF, Sykes MK. Inverse ratio ventilation compared with PEEP in adult respiratory failure. *Intensive Care Med*. 1984;10(5):227–232. [[PubMed: 6386921](#)]
561. Connery LE, Deignan MJ, Gujer MW, Richardson MG. Cardiovascular collapse associated with extreme iatrogenic PEEPi in patients with obstructive airways disease. *Br J Anaesth*. 1999;83(3):493–495. [[PubMed: 10655931](#)]
562. Williams TJ, Tuxen DV, Scheinkestel CD, et al. Risk factors for morbidity in mechanically ventilated patients with acute severe asthma. *Am Rev Respir Dis*. 1992;146(3):607–615. [[PubMed: 1519836](#)]
563. Dal Vecchio L, Polese G, Poggi R, Rossi A. “Intrinsic” positive end-expiratory pressure in stable patients with chronic obstructive pulmonary disease. *Eur Respir J*. 1990;3(1):74–80.
564. Chen Rong C, Yan S. Perceived inspiratory difficulty during inspiratory threshold and hyperinflationary loadings. *Am J Respir Crit Care Med*. 1999;159(3):720–727. [[PubMed: 10051242](#)]
565. Appendini L. About the relevance of dynamic intrinsic PEEP (PEEPi, dyn) measurement. *Intensive Care Med*. 1999;25(3):252–254. [[PubMed: 10229157](#)]
566. Appendini L, Purro A, Gudjonsdottir M, et al. Physiologic response of ventilator-dependent patients with chronic obstructive pulmonary disease to proportional assist ventilation and continuous positive airway pressure. *Am J Respir Crit Care Med*. 1999;159(5 Pt 1):1510–1517.
567. Brochard L. Intrinsic (or auto-) positive end-expiratory pressure during spontaneous or assisted ventilation. *Intensive Care Med*. 2002;28(11):1552–1554. [[PubMed: 12583374](#)]
568. Sydow M, Golisch W, Buscher H, et al. Effect of low-level PEEP on inspiratory work of breathing in intubated patients, both with healthy lungs and with COPD. *Intensive Care Med*. 1995;21(11):887–895. [[PubMed: 8636520](#)]

569. Fernandez R, Benito S, Blanch L, Net A. Intrinsic PEEP: a cause of inspiratory muscle ineffectivity. *Intensive Care Med.* 1988;15(1):51–52. [[PubMed: 3068269](#)]
570. Leung P, Jubran A, Tobin MJ. Comparison of assisted ventilator modes on triggering, patient effort, and dyspnea. *Am J Respir Crit Care Med.* 1997;155(6):1940–1948. [[PubMed: 9196100](#)]
571. Nava S, Ambrosino N, Bruschi C, et al. Physiological effects of flow and pressure triggering during non-invasive mechanical ventilation in patients with chronic obstructive pulmonary disease. *Thorax.* 1997;52(3):249–254. [[PubMed: 9093341](#)]
572. Kim H, Bach JR. Central alveolar hypoventilation in neurosarcoidosis. *Arch Phys Med Rehabil.* 1998;79(11):1467–1468. [[PubMed: 9821911](#)]
573. Smith J, Bellemare F. Effect of lung volume on in vivo contraction characteristics of human diaphragm. *J Appl Physiol.* 1987;62(5):1893–1900. [[PubMed: 3597263](#)]
574. Macklem PT. Hyperinflation. *Am Rev Respir Dis.* 1984;129(1):1–2. [[PubMed: 6703469](#)]
575. Collett PW, Engel LA. Influence of lung volume on oxygen cost of resistive breathing. *J Appl Physiol.* 1986;61(1):16–24. [[PubMed: 3733600](#)]
576. Beck J, Sinderby C, Lindstrom L, Grassino A. Effects of lung volume on diaphragm EMG signal strength during voluntary contractions. *J Appl Physiol.* 1998;85(3):1123–1134. [[PubMed: 9729591](#)]
577. Kawagoe Y, Permutt S, Fessler HE. Hyperinflation with intrinsic PEEP and respiratory muscle blood flow. *J Appl Physiol.* 1994;77(5):2440–2448. [[PubMed: 7868467](#)]
578. Orozco-Levi M, Lloreta J, Minguella J, et al. Injury of the human diaphragm associated with exertion and chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2001;164(9):1734–1739. [[PubMed: 11719318](#)]
579. Ranieri VM, Dambrosio M, Brienza N. Intrinsic PEEP and cardiopulmonary interaction in patients with COPD and acute ventilatory failure. *Eur Respir J.* 1996;9(6):1283–1292. [[PubMed: 8804950](#)]
580. Coussa ML, Guerin C, Eissa NT, et al. Partitioning of work of breathing in mechanically ventilated COPD patients. *J Appl Physiol.* 1993;75(4):1711–1719. [[PubMed: 8282624](#)]
581. Jubran A, Tobin MJ. Pathophysiologic basis of acute respiratory distress in patients who fail a trial of weaning from mechanical ventilation. *Am J Respir Crit Care Med.* 1997;155(3):906–915. [[PubMed: 9117025](#)]
582. Vassilakopoulos T, Zakynthinos S, Roussos C. The tension-time index and the frequency/tidal volume ratio are the major pathophysiologic determinants of weaning failure and success. *Am J Respir Crit Care Med.* 1998;158(2):378–385. [[PubMed: 9700110](#)]
583. Marini JJ. Should PEEP be used in airflow obstruction? *Am Rev Respir Dis.* 1989;140(1):1–3. [[PubMed: 2665582](#)]
584. Pride NB, Permutt S, Riley RL, Bromberger-Barnea B. Determinants of maximal expiratory flow from the lungs. *J Appl Physiol.* 1967;23(5):646–662. [[PubMed: 6061379](#)]
585. Tobin MJ, Lodato RF. PEEP, auto-PEEP, and waterfalls. *Chest.* 1989;96(3):449–451. [[PubMed: 2670461](#)]
586. Caramaz MP, Borges JB, Tucci MR, et al. Paradoxical responses to positive end-expiratory pressure in patients with airway obstruction during controlled ventilation. *Crit Care Med.* 2005;33(7):1519–1528. [[PubMed: 16003057](#)]
587. de Durante G, del Turco M, Rustichini L, et al. ARDSNet lower tidal volume ventilatory strategy may generate intrinsic positive end-expiratory pressure in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2002;165(9):1271–1274.
588. Richard JC, Brochard L, Breton L, et al. Influence of respiratory rate on gas trapping during low volume ventilation of patients with acute lung injury. *Intensive Care Med.* 2002;28(8):1078–1083. [[PubMed: 12185428](#)]
589. Hough CL, Kallet RH, Ranieri VM, et al. Intrinsic positive end-expiratory pressure in Acute Respiratory Distress Syndrome (ARDS) Network subjects. *Crit Care Med.* 2005;33(3):527–532. [[PubMed: 15753743](#)]
590. Maggiore SM, Iacobone E, Toto R, et al. High vs low PEEP in patients with ARDS: role of intrinsic PEEP (abstract). *Am J Respir Crit Care Med.* 2007;175:A34.

591. van den Berg B, Aerts JG, Bogaard JM. Effect of continuous positive airway pressure (CPAP) in patients with chronic obstructive pulmonary disease (COPD) depending on intrinsic PEEP levels. *Acta Anaesthesiol Scand*. 1995;39(8):1097–1102.

592. Reissmann HK, Ranieri VM, Goldberg P, Gottfried SB. Continuous positive airway pressure facilitates spontaneous breathing in weaning chronic obstructive pulmonary disease patients by improving breathing pattern and gas exchange. *Intensive Care Med*. 2000;26(12):1764–1772. [[PubMed: 11271083](#)]

593. Baigorri F, de Monte A, Blanch L, et al. Hemodynamic responses to external counterbalancing of auto-positive end-expiratory pressure in mechanically ventilated patients with chronic obstructive pulmonary disease. *Crit Care Med*. 1994;22(11):1782–1791. [[PubMed: 7956282](#)]

594. Tan IK, Bhatt SB, Tam YH, Oh TE. Effects of PEEP on dynamic hyperinflation in patients with airflow limitation. *Br J Anaesth*. 1993;70(3):267–272. [[PubMed: 8471367](#)]

595. Mancini M, Zavala E, Mancebo J, et al. Mechanisms of pulmonary gas exchange improvement during a protective ventilatory strategy in acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2001;164(8 Pt 1):1448–1453.

596. Halter JM, Steinberg JM, Schiller HJ, et al. Positive end-expiratory pressure after a recruitment maneuver prevents both alveolar collapse and recruitment/derecruitment. *Am J Respir Crit Care Med*. 2003;167(12):1620–1626. [[PubMed: 12615628](#)]

597. Artigas A, Bernard GR, Carlet J, et al. The American-European Consensus Conference on ARDS, part 2: ventilatory, pharmacologic, supportive therapy, study design strategies, and issues related to recovery and remodeling. Acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 1998;157(4 Pt 1):1332–1347.

598. Pepe PE, Hudson LD, Carrico CJ. Early application of positive end-expiratory pressure in patients at risk for the adult respiratory-distress syndrome. *N Engl J Med*. 1984;311(5):281–286. [[PubMed: 6377071](#)]

599. Carroll GC, Tuman KJ, Braverman B, et al. Minimal positive end-expiratory pressure (PEEP) may be “best PEEP.” *Chest*. 1988;93(5):1020–1025. [[PubMed: 3282814](#)]

600. Springer RR, Stevens PM. The influence of PEEP on survival of patients in respiratory failure: a retrospective analysis. *Am J Med*. 1979;66(2):196–200. [[PubMed: 371394](#)]

601. Brower RG, Lanken PN, MacIntyre N, et al. Higher versus lower positive end-expiratory pressures in patients with the acute respiratory distress syndrome. *N Engl J Med*. 2004;351(4):327–336. [[PubMed: 15269312](#)]

602. Meade MO, Cook DJ, Guyatt GH, et al. Ventilation strategy using low tidal volumes, recruitment maneuvers, and high positive end-expiratory pressure for acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA*. 2008;299(6):637–645. [[PubMed: 18270352](#)]

603. Mercat A, Richard JC, Vielle B, et al. Positive end-expiratory pressure setting in adults with acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA*. 2008;299(6):646–655. [[PubMed: 18270353](#)]

604. Villar J, Kacmarek RM, Perez-Mendez L, Aguirre-Jaime A. A high positive end-expiratory pressure, low tidal volume ventilatory strategy improves outcome in persistent acute respiratory distress syndrome: a randomized, controlled trial. *Crit Care Med*. 2006;34(5):1311–1318. [[PubMed: 16557151](#)]

605. Tobin MJ. Culmination of an era in research on the acute respiratory distress syndrome. *N Engl J Med*. 2000;342(18):1360–1361. [[PubMed: 10793169](#)]

606. Gattinoni L, Caironi P, Carlesso E. How to ventilate patients with acute lung injury and acute respiratory distress syndrome. *Curr Opin Crit Care*. 2005;11(1):69–76. [[PubMed: 15659948](#)]

607. Levy MM. PEEP in ARDS—how much is enough? *N Engl J Med*. 2004;351(4):389–391. [[PubMed: 15269320](#)]

608. Briel M, Meade M, Mercat A, et al. Higher vs lower positive end-expiratory pressure in patients with acute lung injury and acute respiratory distress syndrome: systematic review and meta-analysis. *JAMA*. 2010;303(9):865–873. [[PubMed: 20197533](#)]

609. Oba Y, Thameem DM, Zaza T. High levels of PEEP may improve survival in acute respiratory distress syndrome: A meta-analysis. *Respir Med*. 2009;103(8):1174–1181. [[PubMed: 19269800](#)]

610. Phoenix SI, Paravastu S, Columb M, et al. Does a higher positive end expiratory pressure decrease mortality in acute respiratory distress syndrome? A systematic review and meta-analysis. *Anesthesiology*. 2009;110(5):1098–1105. [[PubMed: 19352160](#)]

611. Putensen C, Theuerkauf N, Zinserling J, et al. Meta-analysis: ventilation strategies and outcomes of the acute respiratory distress syndrome and acute lung injury. *Ann Intern Med.* 2009;151(8):566–576. [[PubMed: 19841457](#)]

612. Bernard GR, Artigas A, Brigham KL, et al. The American-European Consensus Conference on ARDS. Definitions, mechanisms, relevant outcomes, and clinical trial coordination. *Am J Respir Crit Care Med.* 1994;149(3 Pt 1):818–824.

613. Villar J, Perez-Mendez L, Lopez J, et al. An early PEEP/FiO₂ trial identifies different degrees of lung injury in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 2007;176(8):795–804. [[PubMed: 17585106](#)]

614. Downs JB, Klein EF Jr, Modell JH. The effect of incremental PEEP on PaO₂ in patients with respiratory failure. *Anesth Analg.* 1973;52(2):210–215. [[PubMed: 4572339](#)]

615. Gallagher TJ, Civetta JM, Kirby RR. Terminology update: optimal PEEP. *Crit Care Med.* 1978;6(5):323–326. [[PubMed: 363358](#)]

616. Hartmann M, Rosberg B, Jonsson K. The influence of different levels of PEEP on peripheral tissue perfusion measured by subcutaneous and transcutaneous oxygen tension. *Intensive Care Med.* 1992;18(8):474–478. [[PubMed: 1289372](#)]

617. Hurewitz A, Bergofsky EH. Adult respiratory distress syndrome: physiologic basis of treatment. *Med Clin North Am.* 1981;65(1):33–51. [[PubMed: 7010022](#)]

618. Weisman IM, Rinaldo JE, Rogers RM. Current concepts: positive end-expiratory pressure in adult respiratory failure. *N Engl J Med.* 1982;307(22):1381–1384. [[PubMed: 6752716](#)]

619. Murray IP, Modell JH, Gallagher TJ, Banner MJ. Titration of PEEP by the arterial minus end-tidal carbon dioxide gradient. *Chest.* 1984;85(1):100–104. [[PubMed: 6360567](#)]

620. Albert RK. Least PEEP: primum non nocere. *Chest.* 1985;87(1):2–4. [[PubMed: 3880692](#)]

621. Badet M, Bayle F, Richard JC, Guerin C. Comparison of optimal positive end-expiratory pressure and recruitment maneuvers during lung-protective mechanical ventilation in patients with acute lung injury/acute respiratory distress syndrome. *Respir Care.* 2009;54(7):847–854. [[PubMed: 19558735](#)]

622. Girgis K, Hamed H, Khater Y, Kacmarek RM. A decremental PEEP trial identifies the PEEP level that maintains oxygenation after lung recruitment. *Respir Care.* 2006;51(10):1132–1139. [[PubMed: 17005058](#)]

623. Tusman G, Suarez-Sipmann F, Bohm SH, et al. Monitoring dead space during recruitment and PEEP titration in an experimental model. *Intensive Care Med.* 2006;32(11):1863–1871. [[PubMed: 17047925](#)]

624. Gattinoni L, Caironi P. Refining ventilatory treatment for acute lung injury and acute respiratory distress syndrome. *JAMA.* 2008;299(6):691–693. [[PubMed: 18270359](#)]

625. Ranieri VM, Zhang H, Mascia L, et al. Pressure-time curve predicts minimally injurious ventilatory strategy in an isolated rat lung model. *Anesthesiology.* 2000;93(5):1320–1328. [[PubMed: 11046222](#)]

626. Talmor D, Sarge T, Malhotra A, et al. Mechanical ventilation guided by esophageal pressure in acute lung injury. *N Engl J Med.* 2008;359(20):2095–2104. [[PubMed: 19001507](#)]

627. Lindberg P, Gunnarsson L, Tokics L, et al. Atelectasis and lung function in the postoperative period. *Acta Anaesthesiol Scand.* 1992;36(6):546–553. [[PubMed: 1514340](#)]

628. Arozullah AM, Daley J, Henderson WG, Khuri SF. Multifactorial risk index for predicting postoperative respiratory failure in men after major noncardiac surgery. The National Veterans Administration Surgical Quality Improvement Program. *Ann Surg.* 2000;232(2):242–253. [[PubMed: 10903604](#)]

629. O'Donohue WJ Jr. National survey of the usage of lung expansion modalities for the prevention and treatment of postoperative atelectasis following abdominal and thoracic surgery. *Chest.* 1985;87(1):76–80. [[PubMed: 3880695](#)]

630. Thompson JS, Baxter BT, Allison JG, et al. Temporal patterns of postoperative complications. *Arch Surg.* 2003;138(6):596–602; discussion 602–603. [[PubMed: 12799329](#)]

631. Lang M, Niskanen M, Miettinen P, et al. Outcome and resource utilization in gastroenterological surgery. *Br J Surg.* 2001;88(7):1006–1014. [[PubMed: 11442536](#)]

632. Pasquina P, Merlani P, Granier JM, Ricou B. Continuous positive airway pressure versus noninvasive pressure support ventilation to treat atelectasis after cardiac surgery. *Anesth Analg*. 2004;99(4):1001–1008, table of contents. [PubMed: 15385340]
633. Stock MC, Downs JB, Gauer PK, et al. Prevention of postoperative pulmonary complications with CPAP, incentive spirometry, and conservative therapy. *Chest*. 1985;87(2):151–157. [PubMed: 3881226]
634. Dehaven CB Jr, Hurst JM, Branson RD. Postextubation hypoxemia treated with a continuous positive airway pressure mask. *Crit Care Med*. 1985;13(1):46–48. [PubMed: 3880690]
635. Lindner KH, Lotz P, Ahnefeld FW. Continuous positive airway pressure effect on functional residual capacity, vital capacity and its subdivisions. *Chest*. 1987;92(1):66–70. [PubMed: 3297521]
636. Ricksten SE, Bengtsson A, Soderberg C, et al. Effects of periodic positive airway pressure by mask on postoperative pulmonary function. *Chest*. 1986;89(6):774–781. [PubMed: 3519107]
637. Antonelli M, Conti G, Bui M, et al. Noninvasive ventilation for treatment of acute respiratory failure in patients undergoing solid organ transplantation: a randomized trial. *JAMA*. 2000;283(2):235–241. [PubMed: 10634340]
638. Huerta S, DeShields S, Shpiner R, et al. Safety and efficacy of postoperative continuous positive airway pressure to prevent pulmonary complications after Roux-en-Y gastric bypass. *J Gastrointest Surg*. 2002;6(3):354–358. [PubMed: 12022987]
639. Duggan M, McCaul CL, McNamara PJ, et al. Atelectasis causes vascular leak and lethal right ventricular failure in uninjured rat lungs. *Am J Respir Crit Care Med*. 2003;167(12):1633–1640. [PubMed: 12663325]
640. Drummond GB, Stedul K, Kingshott R, et al. Automatic CPAP compared with conventional treatment for episodic hypoxemia and sleep disturbance after major abdominal surgery. *Anesthesiology*. 2002;96(4):817–826. [PubMed: 11964587]
641. Carlsson C, Sonden B, Thylen U. Can postoperative continuous positive airway pressure (CPAP) prevent pulmonary complications after abdominal surgery? *Intensive Care Med*. 1981;7(5):225–229. [PubMed: 7024383]
642. Pinilla JC, Oleniuk FH, Tan L, et al. Use of a nasal continuous positive airway pressure mask in the treatment of postoperative atelectasis in aortocoronary bypass surgery. *Crit Care Med*. 1990;18(8):836–840. [PubMed: 2199148]
643. Jousela I, Rasanen J, Verkkala K, et al. Continuous positive airway pressure by mask in patients after coronary surgery. *Acta Anaesthesiol Scand*. 1994;38(4):311–316. [PubMed: 8067215]
644. Fagevik Olsen M, Wennberg E, Johnsson E, et al. Randomized clinical study of the prevention of pulmonary complications after thoracoabdominal resection by two different breathing techniques. *Br J Surg*. 2002;89(10):1228–1234.
645. Squadrone V, Coxa M, Cerutti E, et al. Continuous positive airway pressure for treatment of postoperative hypoxemia: a randomized controlled trial. *JAMA*. 2005;293(5):589–595. [PubMed: 15687314]
646. Ferreyra GP, Baussano I, Squadrone V, et al. Continuous positive airway pressure for treatment of respiratory complications after abdominal surgery: a systematic review and meta-analysis. *Ann Surg*. 2008;247(4):617–626. [PubMed: 18362624]
647. Similowski T, Derenne JP, Milic-Emili J. Respiratory mechanics during acute respiratory failure of chronic obstructive pulmonary disease. In: Derenne JP, Whitelaw WA, Similowski T, eds. *Acute Respiratory Failure in Chronic Obstructive Pulmonary Disease*. New York, NY: Marcel Dekker; 1996:23–46.
648. Nava S, Navalesi P. Bronchodilators and mechanical ventilation in COPD patients. Emptying, pumping or both? *Intensive Care Med*. 1999;25(11):1206–1208. [PubMed: 10654201]
649. Soto FJ, Varkey B. Evidence-based approach to acute exacerbations of COPD. *Curr Opin Pulm Med*. 2003;9(2):117–124. [PubMed: 12574691]
650. Sherk PA, Grossman RF. The chronic obstructive pulmonary disease exacerbation. *Clin Chest Med*. 2000;21(4):705–721. [PubMed: 11194781]
651. Lu CC. Bronchodilator therapy for chronic obstructive pulmonary disease. *Respirology*. 1997;2(4):317–322. [PubMed: 9525304]
652. Bach PB, Brown C, Gelfand SE, McCrory DC. Management of acute exacerbations of chronic obstructive pulmonary disease: a summary and appraisal of published evidence. *Ann Intern Med*. 2001;134(7):600–620. [PubMed: 11281745]

653. Niewoehner DE, Erbland ML, Deupree RH, et al. Effect of systemic glucocorticoids on exacerbations of chronic obstructive pulmonary disease. Department of Veterans Affairs Cooperative Study Group. *N Engl J Med*. 1999;340(25):1941–1947. [PubMed: 10379017]
654. Dhand R, Duarte AG, Jubran A, et al. Dose-response to bronchodilator delivered by metered-dose inhaler in ventilator-supported patients. *Am J Respir Crit Care Med*. 1996;154(2 Pt 1):388–393.
655. Mouloudi E, Katsanoulas K, Anastasaki M, et al. Bronchodilator delivery by metered-dose inhaler in mechanically ventilated COPD patients: influence of tidal volume. *Intensive Care Med*. 1999;25(11):1215–1221. [PubMed: 10654204]
656. Mouloudi E, Maliotakis C, Kondili E, et al. Duration of salbutamol-induced bronchodilation delivered by metered-dose inhaler in mechanically ventilated COPD patients. *Monaldi Arch Chest Dis*. 2001;56(3):189–194. [PubMed: 11665496]
657. Mouloudi E, Prinianakis G, Kondili E, Georgopoulos D. Effect of inspiratory flow rate on beta2-agonist induced bronchodilation in mechanically ventilated COPD patients. *Intensive Care Med*. 2001;27(1):42–46. [PubMed: 11280671]
658. Rubini F, Rampulla C, Nava S. Acute effect of corticosteroids on respiratory mechanics in mechanically ventilated patients with chronic airflow obstruction and acute respiratory failure. *Am J Respir Crit Care Med*. 1994;149(2 Pt 1):306–310.
659. Guerin C, Chevre A, Dessirier P, et al. Inhaled fenoterol-ipratropium bromide in mechanically ventilated patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 1999;159(4 Pt 1):1036–1042.
660. Fink JB, Dhand R, Duarte AG, et al. Aerosol delivery from a metered-dose inhaler during mechanical ventilation. An in vitro model. *Am J Respir Crit Care Med*. 1996;154(2 Pt 1):382–387.
661. Celikel T, Sungur M, Ceyhan B, Karakurt S. Comparison of noninvasive positive pressure ventilation with standard medical therapy in hypercapnic acute respiratory failure. *Chest*. 1998;114(6):1636–1642. [PubMed: 9872200]
662. Plant PK, Owen JL, Elliott MW. Early use of non-invasive ventilation for acute exacerbations of chronic obstructive pulmonary disease on general respiratory wards: a multicentre randomised controlled trial. *Lancet*. 2000;355(9219):1931–1935. [PubMed: 10859037]
663. Conti G, Antonelli M, Navalesi P, et al. Noninvasive vs. conventional mechanical ventilation in patients with chronic obstructive pulmonary disease after failure of medical treatment in the ward: a randomized trial. *Intensive Care Med*. 2002;28(12):1701–1707. [PubMed: 12447511]
664. Sinderby C, Navalesi P, Beck J, et al. Neural control of mechanical ventilation in respiratory failure. *Nat Med*. 1999;5(12):1433–1436. [PubMed: 10581089]
665. Colombo D, Cammarota G, Bergamaschi V, et al. Physiologic response to varying levels of pressure support and neurally adjusted ventilatory assist in patients with acute respiratory failure. *Intensive Care Med*. 2008;34(11):2010–2018. [PubMed: 18629471]
666. Hill AR. Respiratory muscle function in asthma. *J Assoc Acad Minor Phys*. 1991;2(3):100–108. [PubMed: 1809452]
667. Peigang Y, Marini JJ. Ventilation of patients with asthma and chronic obstructive pulmonary disease. *Curr Opin Crit Care*. 2002;8(1):70–76. [PubMed: 12205409]
668. Shivaram U, Donath J, Khan FA, Juliano J. Effects of continuous positive airway pressure in acute asthma. *Respiration*. 1987;52(3):157–162. [PubMed: 3326082]
669. Shivaram U, Miro AM, Cash ME, et al. Cardiopulmonary responses to continuous positive airway pressure in acute asthma. *J Crit Care*. 1993;8(2):87–92. [PubMed: 8343867]
670. Loughheed DM, Webb KA, O'Donnell DE. Breathlessness during induced lung hyperinflation in asthma: the role of the inspiratory threshold load. *Am J Respir Crit Care Med*. 1995;152(3):911–920. [PubMed: 7663804]
671. Lin HC, Wang CH, Yang CT, et al. Effect of nasal continuous positive airway pressure on methacholine-induced bronchoconstriction. *Respir Med*. 1995;89(2):121–128. [PubMed: 7708996]
672. Martin JG, Shore S, Engel LA. Effect of continuous positive airway pressure on respiratory mechanics and pattern of breathing in induced asthma. *Am Rev Respir Dis*. 1982;126(5):812–817. [PubMed: 6216839]
673. Meduri GU, Cook TR, Turner RE, et al. Noninvasive positive pressure ventilation in status asthmaticus. *Chest*. 1996;110(3):767–774. [PubMed: 8797425]

674. Soroksky A, Stav D, Shpirer I. A pilot prospective, randomized, placebo-controlled trial of bilevel positive airway pressure in acute asthmatic attack. *Chest*. 2003;123(4):1018–1025. [[PubMed: 12684289](#)]
675. Moritz F, Benichou J, Vanheste M, et al. Boussignac continuous positive airway pressure device in the emergency care of acute cardiogenic pulmonary oedema: a randomized pilot study. *Eur J Emerg Med*. 2003;10(3):204–208. [[PubMed: 12972896](#)]
676. Bellone A, Monari A, Cortellaro F, et al. Myocardial infarction rate in acute pulmonary edema: noninvasive pressure support ventilation versus continuous positive airway pressure. *Crit Care Med*. 2004;32(9):1860–1865. [[PubMed: 15343013](#)]
677. Ferrari G, Milan A, Groff P, et al. Continuous positive airway pressure vs. pressure support ventilation in acute cardiogenic pulmonary edema: a randomized trial. *J Emerg Med*. 2010;39(5):676–684. [[PubMed: 19818574](#)]
678. Ferrari G, Olliveri F, De Filippi G, et al. Noninvasive positive airway pressure and risk of myocardial infarction in acute cardiogenic pulmonary edema: continuous positive airway pressure vs noninvasive positive pressure ventilation. *Chest*. 2007;132(6):1804–1809. [[PubMed: 17908705](#)]
679. Gray A, Goodacre S, Newby DE, et al. Noninvasive ventilation in acute cardiogenic pulmonary edema. *N Engl J Med*. 2008;359(2):142–151. [[PubMed: 18614781](#)]
680. Moritz F, Brousse B, Gellee B, et al. Continuous positive airway pressure versus bilevel noninvasive ventilation in acute cardiogenic pulmonary edema: a randomized multicenter trial. *Ann Emerg Med*. 2007;50(6):666–675, 675.e1. [[PubMed: 17764785](#)]
681. Noura S, Boukef R, Bouida W, et al. Non-invasive pressure support ventilation and CPAP in cardiogenic pulmonary edema: a multicenter randomized study in the emergency department. *Intensive Care Med*. 2011;37(2):249–256. [[PubMed: 21136039](#)]
682. Crane SD, Elliott MW, Gilligan P, et al. Randomised controlled comparison of continuous positive airways pressure, bilevel non-invasive ventilation, and standard treatment in emergency department patients with acute cardiogenic pulmonary oedema. *Emerg Med J*. 2004;21(2):155–161. [[PubMed: 14988338](#)]
683. Park M, Sangean MC, Volpe Mde S, et al. Randomized, prospective trial of oxygen, continuous positive airway pressure, and bilevel positive airway pressure by face mask in acute cardiogenic pulmonary edema. *Crit Care Med*. 2004;32(12):2407–2415. [[PubMed: 15599144](#)]
684. Masip J, Roque M, Sanchez B, et al. Noninvasive ventilation in acute cardiogenic pulmonary edema: systematic review and meta-analysis. *JAMA*. 2005;294(24):3124–3130. [[PubMed: 16380593](#)]
685. Vital FM, Saconato H, Ladeira MT, et al. Non-invasive positive pressure ventilation (CPAP or bilevel NPPV) for cardiogenic pulmonary edema. *Cochrane Database Syst Rev*. 2008(3):CD005351.
686. Winck JC, Azevedo LF, Costa-Pereira A, et al. Efficacy and safety of non-invasive ventilation in the treatment of acute cardiogenic pulmonary edema—a systematic review and meta-analysis. *Crit Care*. 2006;10(2):R69.
687. Hubble MW, Richards ME, Jarvis R, et al. Effectiveness of prehospital continuous positive airway pressure in the management of acute pulmonary edema. *Prehosp Emerg Care*. 2006;10(4):430–439. [[PubMed: 16997770](#)]
688. Hubble MW, Richards ME, Wilfong DA. Estimates of cost-effectiveness of prehospital continuous positive airway pressure in the management of acute pulmonary edema. *Prehosp Emerg Care*. 2008;12(3):277–285. [[PubMed: 18584492](#)]
689. Foti G, Sangalli F, Berra L, et al. Is helmet CPAP first line pre-hospital treatment of presumed severe acute pulmonary edema? *Intensive Care Med*. 2009;35(4):656–662. [[PubMed: 19030843](#)]
690. Frontin P, Bounes V, Houze-Cerfon CH, et al. Continuous positive airway pressure for cardiogenic pulmonary edema: a randomized study. *Am J Emerg Med*. 2011;29(7):775–781. [[PubMed: 20825901](#)]
691. Thompson J, Petrie DA, Ackroyd-Stolarz S, Bardua DJ. Out-of-hospital continuous positive airway pressure ventilation versus usual care in acute respiratory failure: a randomized controlled trial. *Ann Emerg Med*. 2008;52(3):232–241, 241e1. [[PubMed: 18387700](#)]
692. Nava S, Navalesi P, Conti G. Time of non-invasive ventilation. *Intensive Care Med*. 2006;32(3):361–370. [[PubMed: 16477416](#)]
693. Schmidt GB, O'Neill WW, Kotb K, et al. Continuous positive airway pressure in the prophylaxis of the adult respiratory distress syndrome. *Surg Gynecol Obstet*. 1976;143(4):613–618. [[PubMed: 785646](#)]
694. Shah DM, Powers SR, Jr. Prevention of pulmonary complications in high risk patients. *Surg Clin North Am*. 1980;60(6):1359–1372. [[PubMed: 7006122](#)]

695. Weigelt JA, Mitchell RA, Snyder WH 3rd. Early positive end-expiratory pressure in the adult respiratory distress syndrome. *Arch Surg.* 1979;114(4):497–501. [\[PubMed: 373705\]](#)
696. McAslan TC, Cowley RA. The preventive use of PEEP in major trauma. *Am Surg.* 1979;45(3):159–167. [\[PubMed: 373532\]](#)
697. Demling RH. Improved survival after massive burns. *J Trauma.* 1983;23(3):179–184. [\[PubMed: 6403712\]](#)
698. Goris RJ, Gimbere JS, van Niekerk JL, et al. Early osteosynthesis and prophylactic mechanical ventilation in the multitrauma patient. *J Trauma.* 1982;22(11):895–903. [\[PubMed: 6754962\]](#)
699. Askanazi J, Wax SD, Neville JF Jr, et al. Prevention of pulmonary insufficiency through prophylactic use of PEEP and rapid respiratory rates. *J Thorac Cardiovasc Surg.* 1978;75(2):267–272. [\[PubMed: 342835\]](#)
700. Barash PG, Bunke MC, Tilson MD, et al. The salutary effects of positive end expiratory pressure (PEEP) in experimentally induced pseudomonas pneumonia. *Anesth Analg.* 1979;58(3):208–215. [\[PubMed: 36814\]](#)
701. Luce JM, Robertson HT, Huang T, et al. The effects of expiratory positive airway pressure on the resolution of oleic acid-induced lung injury in dogs. *Am Rev Respir Dis.* 1982;125(6):716–722. [\[PubMed: 6807153\]](#)
702. Valdes ME, Powers SR Jr, Shah DM, et al. Continuous positive airway pressure in prophylaxis of adult respiratory distress syndrome in trauma patients. *Surg Forum.* 1978;29:187–189. [\[PubMed: 401406\]](#)
703. Manzano F, Fernandez-Mondejar E, Colmenero M, et al. Positive-end expiratory pressure reduces incidence of ventilator-associated pneumonia in nonhypoxemic patients. *Crit Care Med.* 2008;36(8):2225–2231. [\[PubMed: 18664777\]](#)
704. Lucangelo U, Zin WA, Antonaglia V, et al. Effect of positive expiratory pressure and type of tracheal cuff on the incidence of aspiration in mechanically ventilated patients in an intensive care unit. *Crit Care Med.* 2008;36(2):409–413. [\[PubMed: 18007268\]](#)
705. Pitts R, Fisher D, Sulemanji D, et al. Variables affecting leakage past endotracheal tube cuffs: a bench study. *Intensive Care Med.* 2010;36(12):2066–2073. [\[PubMed: 20852839\]](#)
706. Zanella A, Scaravilli V, Isgro S, et al. Fluid leakage across tracheal tube cuff, effect of different cuff material, shape, and positive expiratory pressure: a bench-top study. *Intensive Care Med.* 2011;37(2):343–347. [\[PubMed: 21152894\]](#)
707. Metnitz PG, Metnitz B, Moreno RP, et al. Epidemiology of mechanical ventilation: analysis of the SAPS 3 database. *Intensive Care Med.* 2009;35(5):816–825. [\[PubMed: 19288079\]](#)
708. Maitre B, Jaber S, Maggiore SM, et al. Continuous positive airway pressure during fiberoptic bronchoscopy in hypoxemic patients. A randomized double-blind study using a new device. *Am J Respir Crit Care Med.* 2000;162(3 Pt 1):1063–1067.
709. Gachot B, Clair B, Wolff M, et al. Continuous positive airway pressure by face mask or mechanical ventilation in patients with human immunodeficiency virus infection and severe *Pneumocystis carinii* pneumonia. *Intensive Care Med.* 1992;18(3):155–159. [\[PubMed: 1644963\]](#)
710. Hilbert G, Gruson D, Vargas F, et al. Noninvasive continuous positive airway pressure in neutropenic patients with acute respiratory failure requiring intensive care unit admission. *Crit Care Med.* 2000;28(9):3185–3190. [\[PubMed: 11008980\]](#)
711. Conti G, Marino P, Cogliati A, et al. Noninvasive ventilation for the treatment of acute respiratory failure in patients with hematologic malignancies: a pilot study. *Intensive Care Med.* 1998;24(12):1283–1288. [\[PubMed: 9885881\]](#)
712. Joris JL, Sottiaux TM, Chiche JD, et al. Effect of bi-level positive airway pressure (BiPAP) nasal ventilation on the postoperative pulmonary restrictive syndrome in obese patients undergoing gastropasty. *Chest.* 1997;111(3):665–670. [\[PubMed: 9118706\]](#)
713. Lindholm CE, Ollman B, Snyder JV, et al. Cardiorespiratory effects of flexible fiberoptic bronchoscopy in critically ill patients. *Chest.* 1978;74(4):362–368. [\[PubMed: 699643\]](#)
714. Matsushima Y, Jones RL, King EG, et al. Alterations in pulmonary mechanics and gas exchange during routine fiberoptic bronchoscopy. *Chest.* 1984;86(2):184–188. [\[PubMed: 6744959\]](#)
715. Verra F, Hmouda H, Rauss A, et al. Bronchoalveolar lavage in immunocompromised patients. Clinical and functional consequences. *Chest.* 1992;101(5):1215–1220. [\[PubMed: 1582274\]](#)

716. Valenza F, Guglielmi M, Irace M, et al. Positive end-expiratory pressure delays the progression of lung injury during ventilator strategies involving high airway pressure and lung overdistention. *Crit Care Med*. 2003;31(7):1993–1998. [[PubMed: 12847394](#)]

717. Gajic O, Dara SI, Mendez JL, et al. Ventilator-associated lung injury in patients without acute lung injury at the onset of mechanical ventilation. *Crit Care Med*. 2004;32(9):1817–1824. [[PubMed: 15343007](#)]

718. Wolthuis EK, Choi G, Dessing MC, et al. Mechanical ventilation with lower tidal volumes and positive end-expiratory pressure prevents pulmonary inflammation in patients without preexisting lung injury. *Anesthesiology*. 2008;108(1):46–54. [[PubMed: 18156881](#)]

719. Hong CM, Xu DZ, Lu Q, et al. Low tidal volume and high positive end-expiratory pressure mechanical ventilation results in increased inflammation and ventilator-associated lung injury in normal lungs. *Anesth Analg*. 2010;110(6):1652–1660. [[PubMed: 20103541](#)]

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